Real Analysis

Lecture Notes at Nanjing University
Translated by ChatGPT

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Preface

Back in 2011, I wrote a set of real analysis notes in Chinese at Nanjing University. They grew out of a course I took with Prof. Chengkui Zhong. It was my first exposure to real analysis, and the experience was formative. The course's level of detail and rigor strongly influenced my later academic interests. Around the same time, I had just started using LaTeX; these notes became a practical way for me to learn mathematical typesetting while working through the material carefully.

The notes circulated on the university forum and on a Sina knowledge-sharing platform (iask), where they accumulated a few thousand downloads. That site has long since disappeared (it eventually became an online medical Q&A portal). For years I have felt that the notes deserve a wider audience, and I have periodically considered translating them into English.

Recently, I revisited the manuscript and produced a complete English translation with ChatGPT. With the current generation of LLM-assisted tooling—especially the ability to edit files and iterate quickly through compilation and correction—the conversion process was far smoother than I expected. I ended up with a near-perfect translation in about two hours. I hope the rise of LLMs can truly break down barriers to knowledge (and language) and make it accessible to everyone.

Real Analysis Real analysis begins as a continuation of calculus, but it matures the moment one recognizes that "well-behaved" functions are not the default. The subject is, in a precise sense, the art of building a calculus that survives contact with pathological examples: functions that oscillate too violently, sets too irregular to be captured by elementary geometry, and limiting processes that erase the properties one would like to preserve.

These notes take a classical route through this landscape while remaining modern in spirit. They do not treat measure theory as an isolated technical appendix; rather, they develop Lebesgue measure and the Lebesgue integral as a structural repair of the weaknesses of the Riemann integral and of pointwise calculus. Along the way, the language of "almost everywhere" becomes not a convenience but the correct resolution at which analytic stability is restored.

Why another integral? The opening overview is organized around three recurring themes.

First, **limits**. A sequence of continuous functions can converge pointwise to a function that is no longer continuous and may fail to be Riemann integrable. Yet the integrals of the approximating functions can converge, suggesting that the integral itself—not merely the class of functions one integrates—should be redesigned to interact well with limits under natural hypotheses.

Second, **geometry at scale**. The length of a curve, defined as the supremum of polygonal approximations, is a geometric quantity that can exist even when classical differentiability fails. The correct analytic condition governing rectifiability is bounded variation, and the familiar arc-length formula generally requires attention to parameterization.

Third, **differentiation versus integration**. The fundamental theorem of calculus is not wrong; it is incomplete. Its sharp form requires the right notion of integrability (Lebesgue) and the right regularity class (absolute continuity). The final chapter returns to this theme from a measure-theoretic perspective, where differentiation is defined not only for functions but for measures.

A small collection of principles guides the development.

- Replace pointwise structure by *measure-theoretic structure*: work "up to null sets," where analytic objects behave robustly.
- Replace rigid descriptions by *approximation*: measurable sets by simple geometric sets, measurable functions by simple functions and by continuous functions on large subsets, and a.e. convergence by almost uniform convergence outside small exceptional sets.
- Replace ad hoc arguments by *closure principles*: identify the smallest class closed under the operations at hand and containing the desired generators.

These principles are not merely pedagogical; they are the mechanism by which the modern theory becomes usable.

Chapter 1: Sets, topology, and descriptive foundations. This chapter develops the set-theoretic and topological language in which measure theory naturally lives: Borel sets, compactness (Heine–Borel), and category arguments (via the Baire theorem). Two ideas recur throughout: inverse images preserve structure (the basis for continuity and measurability), and "largeness" can be formalized in more than one way (measure versus category).

Chapter 2: Constructing Lebesgue measure. Lebesgue measure is built from first principles: begin with a pre-measure on a concrete ring of boxes, extend to an outer measure by countable coverings, and then select the measurable sets via Carathéodory's criterion. This chapter also develops regularity (open/closed approximation) and explains why nonmeasurable sets are unavoidable.

Chapter 3: Measurable functions and convergence. Measurability for functions is defined through measurability of level sets. The chapter emphasizes approximation theorems (simple functions; Lusin) and convergence modes (a.e. and in measure), including Egorov's almost uniform convergence on finite-measure sets and the Riesz subsequence principle.

Chapter 4: Lebesgue integration. Integration is developed in stages: bounded functions on finite-measure sets, then nonnegative functions via truncation and exhaustion, then general integrable functions via positive and negative parts. The central limit theorems (Fatou, monotone convergence, dominated convergence) and Tonelli/Fubini's theorems then become natural consequences of the construction.

Chapter 5: Differentiation of measures and the Newton–Leibniz formula. The final chapter connects integration back to differentiation. Vitali covering arguments yield differentiation theorems (Lebesgue points), which in turn support Radon–Nikodym and the modern form of Newton–Leibniz for absolutely continuous functions. In this way, the notes close the loop: the Lebesgue integral is not a replacement for calculus, but its completion.

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Overview

From a narrow perspective, real analysis studies real-valued functions of one (or several) real variables and the limiting processes that govern them. Unlike elementary calculus, one quickly encounters functions and sets whose behavior is "pathological" from the classical point of view. From a broader perspective, many of the central ideas extend naturally to abstract measure spaces. In that setting, Lebesgue measure theory and Lebesgue integration are the core organizing tools: they provide a stable framework for limits, for geometric notions such as size and length, and for the modern relationship between integration and differentiation.

Real analysis demands careful logical reasoning, but it is also driven by geometric intuition. Many standard exercises are deceptively difficult: they require one to decide which notion of convergence is relevant, which approximation principle to invoke, and which hypotheses are genuinely necessary.

0.1 Limits of Continuous Functions

Let $\{f_n\}$ be a sequence of continuous functions on $[0,1]$, and suppose $f_n(x) \to f(x)$ for every $x \in [0,1]$. If the convergence is uniform, then f is continuous; this is one of the first stability results learned in analysis. Without uniform convergence, however, the situation changes drastically. One can construct continuous functions f_n such that

- $0 \le f_n(x) \le 1$ for all $x \in [0,1]$,
- $f_{n+1}(x) \le f_n(x)$ for all $x \in [0,1]$ (pointwise monotone decrease),
- the pointwise limit f is not Riemann integrable.

Despite this, the sequence of classical integrals $\int_0^1 f_n(x)\,dx$ can still converge. This tension motivates a central question: can one define an integral that interacts better with limits, so that

$$\int_{[0,1]} f(x)\,dx \;=\; \lim_{n\to\infty} \int_0^1 f_n(x)\,dx$$

holds under natural, checkable hypotheses? Related tools appear later in Section 3.3 (convergence) and Section 4.3 (limit theorems for the Lebesgue integral).

0.2 Length of Curves

Let Γ be a continuous curve in the plane given parametrically by

$$\Gamma = \{(x(t), y(t)) : \; a \le t \le b\},$$

where $x(t)$ and $y(t)$ are continuous. The length of Γ is usually defined as the supremum of polygonal lengths obtained by joining finitely many points on Γ in increasing order of t. When this supremum is finite, Γ is called *rectifiable*. If x and y are continuously differentiable, then the length is given by the classical formula

$$L = \int_a^b \sqrt{(x'(t))^2 + (y'(t))^2}\, dt. \tag{1}$$

For a general curve, two questions arise:

1. Under what conditions on $x(t)$ and $y(t)$ is Γ rectifiable?
2. When those conditions hold, in what sense is (1) valid?

The first question admits a precise answer: Γ is rectifiable if and only if x and y are of bounded variation. Under bounded variation, the integral in (1) is always meaningful, but the equality need not hold as written for an arbitrary parameter; it can, however, be restored after a suitable reparameterization. For bounded variation and the analytic content behind this discussion (differentiability a.e. and integrability of derivatives), see Section 5.5.2.

0.3 Differentiation and Integration

The fundamental theorem of calculus asserts that differentiation and integration are inverse operations. It is commonly written in the two forms

$$F(b) - F(a) = \int_a^b F'(x)\, dx, \tag{2}$$

$$\frac{d}{dx} \int_0^x f(t)\, dt = f(x). \tag{3}$$

Both identities are subtle once one leaves the classical setting. There exist continuous functions F that are nowhere differentiable, and there exist functions with $F'(x)$ defined everywhere but not Riemann integrable. Thus, a natural goal is to identify a broad and robust class of functions for which (2) remains valid.

For (3), the question becomes: for which integrable functions f does the differentiation of the integral recover f (almost everywhere)? Answering this requires a covering theory and the notion of absolute continuity; see Section 5.1 and Section 5.6.

0.4 Limitations of the Riemann Integral

We recall the classical definition.

Definition 0.4.1 (Riemann Integral)**.** *Let f be bounded on $[a, b]$, and let*

$$\Delta : \quad a = x_0 < x_1 < \cdots < x_n = b$$

be a partition. Choose tags $\xi_i \in [x_{i-1}, x_i]$ and form the Riemann sum

$$S(f, \Delta) = \sum_{i=1}^{n} f(\xi_i)(x_i - x_{i-1}).$$

If there exists $I \in \mathbb{R}$ such that for every $\varepsilon > 0$ there is $\delta > 0$ with

$$|\Delta| \overset{\text{def}}{=} \max_{1 \le i \le n} (x_i - x_{i-1}) < \delta \quad \Longrightarrow \quad |S(f, \Delta) - I| < \varepsilon, \tag{4}$$

then f is called Riemann integrable on $[a, b]$ and I is its Riemann integral, denoted

$$(R) \int_a^b f(x)\, dx.$$

When no confusion can arise, the (R) on the left is omitted.

In topological language, the limit in (4) is a limit over a directed set of partitions (a net).

The integrability problem is central in Riemann integration. While the Darboux criterion characterizes Riemann integrability, it is often difficult to classify functions directly from that criterion.

Example 0.4.1 (Dirichlet Function)**.** *The function*

$$D(x) = \begin{cases} 1, & x \in \mathbb{Q} \cap [0, 1], \\ 0, & x \in [0, 1] \setminus \mathbb{Q}, \end{cases} \tag{5}$$

is not Riemann integrable.

Moreover, the class of Riemann integrable functions is not closed under pointwise limits even under uniform boundedness. For instance, if $\{f_n\}$ are Riemann integrable, uniformly bounded, and converge pointwise to f, the limit f need not be Riemann integrable. These limitations are closely tied to the fact that the Riemann integral is not designed to commute with limits or to support clean exchange-of-limit operations.

Example 0.4.2. *Let $f_n(x)$ be a sequence of Riemann integrable functions on $[a, b]$, and suppose*

(1) *there exists $M > 0$ such that $|f_n(x)| \leqslant M$ for all $x \in [a, b]$;*

(2) *there exists a real-valued function $f(x)$ on $[a, b]$ such that for all $x \in [a, b]$,*

$$\lim_{n \to \infty} f_n(x) = f(x).$$

Question: is $f(x)$ Riemann integrable on $[a, b]$?

Remark 0.4.1. *If $f(x)$ is Riemann integrable, then*

$$\lim_{n \to \infty} (R) \int_a^b f_n(x)\, dx = (R) \int_a^b f(x)\, dx.$$

0.5 Lebesgue Integral and Lebesgue Measure

The Lebesgue integral reverses the Riemann viewpoint: it partitions the *range* of the function and measures the size of level sets in the *domain*.

Definition 0.5.1 (Lebesgue Integral)**.** *Let $f(x)$ be a real-valued function on $[a, b]$ satisfying*

$$m \leqslant f(x) < M.$$

Choose a partition of $[m, M)$,

$$m = y_0 < y_1 < \cdots < y_n = M,$$

and define the level sets

$$E_i = \{x \in [a, b] : \ y_{i-1} \leq f(x) < y_i\}.$$

A Lebesgue-type sum takes the form

$$S_L(f, \Delta) = \sum_{i=1}^{n} \xi_i \, m(E_i), \qquad \xi_i \in [y_{i-1}, y_i),$$

where $\xi_i \in [y_{i-1}, y_i)$ *is arbitrary and* $m(E_i)$ *denotes the* measure *(length) of* E_i. *If there exists a constant* I *such that for every* $\varepsilon > 0$, *there exists* $\delta > 0$ *and whenever*

$$|\Delta| \overset{\text{def.}}{=} \max_{1 \leqslant i \leqslant n} \{y_i - y_{i-1}\} < \delta,$$

we have

$$|S_L(f, \Delta) - I| < \varepsilon,$$

then $f(x)$ *is called Lebesgue integrable on* $[a, b]$, *and* I *is called the Lebesgue integral of* $f(x)$ *on* $[a, b]$, *denoted by*

$$(L) \int_{[a,b]} f(x) \, dx.$$

When no confusion can arise, the (L) *on the left is omitted.*

Two questions are immediate and decisive:

- What is $m(E)$? It is a generalization of length, called *Lebesgue measure.*
- For which sets E is $m(E)$ defined, and which properties does it satisfy?

Example 0.5.1 (Lebesgue Integral of the Dirichlet Function)**.** *Since the Dirichlet function* (5) *takes only the values* 0 *and* 1, *the Lebesgue integral of* D *is determined by the measure of* $\mathbb{Q} \cap [0, 1]$

$$S_L^*(D, \Delta) = m(\mathbb{Q} \cap [0, 1]).$$

Then what is the "length" of the set of all rational numbers in $[0, 1]$*? The decisive fact is that* $\mathbb{Q} \cap [0, 1]$ *has Lebesgue measure zero, so* $\int_{[0,1]} D(x) \, dx = 0$.

Finally, the Lebesgue integral extends the Riemann integral in the expected way.

Proposition 0.5.1 (Relation Between the Lebesgue and Riemann Integrals)**.** *if* f *is Riemann integrable on* $[a, b]$, *then* f *is Lebesgue integrable and both integrals coincide.*

$$(L) \int_{[a,b]} f(x) \, dx = (R) \int_a^b f(x) \, dx.$$

Measure theory rests on Cantor's set theory and grew through the work of Peano, Jordan, Borel, Lebesgue, and later Riesz and Carathéodory. Jordan's approach yields a natural integration theory on Jordan measurable sets but is too restrictive: there exist open sets that are not Jordan measurable. Later, Borel further developed measure theory and established Borel measure on the class generated from open and closed sets by basic operations such as intersections, unions, and differences (the Borel class, a σ-ring).

Lebesgue's construction produces a much larger class of measurable sets (a σ-ring) and yields an integral designed to support the limit operations that arise throughout modern analysis. Further development of measure theory relied on work by Riesz, Caratheodory, and others.

Chapter 1

Set Theory

Set theory, initiated by Georg Cantor (1845–1918), provides the basic language of modern analysis. Its concepts—sets, mappings, cardinality, limit points, and the operations that build complicated objects from simple ones—will be used throughout these notes, especially in the construction of measurable sets and the Lebesgue integral.

A *set* is a collection of distinct objects, called its elements. In everyday mathematics it is natural to describe a set by a property,

$$A = \{x \mid P(x)\},$$

as in Cantor's descriptive definition of sets. However, if taken literally, this "naive" principle leads to contradictions. The most famous is Russell's paradox (1903).

Definition 1.0.1 (Paradox)**.** *An argument is called a paradox if it leads to a conclusion opposite to common judgment, while it is difficult to provide a justified refutation.*

Definition 1.0.2 (Fallacy)**.** *A statement is called a fallacy if both the statement and its negation can be proved by seemingly logically equivalent reasoning, and no error in the derivation can be identified.*

Example 1.0.1 (Russell paradox)**.** *Consider the set*

$$E = \{x \mid x \notin x\}.$$

If $E \in E$, then by definition $E \notin E$. Conversely, if $E \notin E$, then E satisfies the defining property and hence $E \in E$. Thus

$$E \in E \iff E \notin E,$$

a contradiction.

Remark 1.0.1. *Modern set theory avoids such paradoxes by working within an axiomatic framework (e.g. ZFC), in which not every property determines a set. In these notes we use the standard set-theoretic toolkit informally, as is customary in real analysis, without emphasizing the axioms.*

1.1 Set Operations

The basic operations on sets are well known; we give only a simple example.

Example 1.1.1. *Let $f : [a, b] \to \mathbb{R}^1$. Then*

$$[a, b] = \bigcup_{n=0}^{\infty} \{x||f(x)| \leqslant n\},$$

$$\{x||f(x)| > 0\} = \bigcup_{n=1}^{\infty} \{x||f(x)| \geqslant \frac{1}{n}\}.$$

Besides intersection, union, and difference, we also define symmetric difference and limit operations for sets.

Definition 1.1.1 (Symmetric Difference)**.** *Let A, B be sets. Define*

$$C = (A\backslash B) \cup (B\backslash A),$$

called the symmetric difference of A and B, denoted by $A\triangle B$.

To motivate the definition of set limits, let us recall limsup and liminf for sequences. Let $\{a_n\}$ be a real sequence and define

$$b_k = \sup_{i \geqslant k} a_i,$$

then $\{b_k\}$ is monotone decreasing and hence convergent. We define

$$\limsup_{n\to\infty} a_n = \inf_k b_k = \inf_k \sup_{i\geqslant k} a_i.$$

Similarly, we define upper and lower limits of a sequence of sets.

Definition 1.1.2 (Limit Set)**.** *Let $\{A_k\}$ be a sequence of sets. The set*

$$\bigcap_{j=1}^{\infty} \bigcup_{k=j}^{\infty} A_k$$

is called the upper limit set of $\{A_k\}$, denoted by $\overline{\lim_{k\to\infty}} A_k$ or $\limsup_{k\to\infty} A_k$. The set

$$\bigcup_{j=1}^{\infty} \bigcap_{k=j}^{\infty} A_k$$

is called the lower limit set of $\{A_k\}$, denoted by $\underline{\lim_{k\to\infty}} A_k$ or $\liminf_{k\to\infty} A_k$. If

$$\overline{\lim_{k\to\infty}} A_k = \underline{\lim_{k\to\infty}} A_k,$$

then this common set is called the limit set of $\{A_k\}$, denoted by $\lim_{k\to\infty} A_k$.

Remark 1.1.1. *In fact, set inclusion forms a partial order. For any partially ordered set, one can define suprema and infima, and therefore limsup and liminf analogously. Here, $\bigcup_{k=j}^{\infty} A_k$ is the supremum of $\{A_k\}_{k=j}^{\infty}$.*

Now let us look at a basic example.

Example 1.1.2. *Let*

$$A_k = \begin{cases} [-1,1] \cup [1,2], & k \text{ even;} \\ [-1,1] \cup [-2,-1], & k \text{ odd.} \end{cases}$$

Then

$$\overline{\lim_{k\to\infty}} A_k = [-2,2], \ \underline{\lim_{k\to\infty}} A_k = [-1,1].$$

Think: 1. For an arbitrary sequence of sets $\{A_k\}$, do the upper and lower limit sets always exist?

2. Assume $\lim_{n\to\infty} x_n = x$ and $\lim_{n\to\infty} r_n = r$. Study limsup and liminf for the family of n-dimensional balls $\{B(x_n, r_n)\}$.

Definition 1.1.3 (Monotone sequences of sets)**.** *Let $\{A_k\}$ be a sequence of sets.*

- *If $A_k \subseteq A_{k+1}$ for every $k \in \mathbb{N}$, then $\{A_k\}$ is called* monotone increasing*.*
- *If $A_{k+1} \subseteq A_k$ for every $k \in \mathbb{N}$, then $\{A_k\}$ is called* monotone decreasing*.*

Theorem 1.1.1 (Limit of a monotone sequence of sets)**.** *If $\{A_k\}$ is monotone, then $\limsup_{k\to\infty} A_k = \liminf_{k\to\infty} A_k$, so the limit set exists. More precisely,*

$$\lim_{k\to\infty} A_k = \begin{cases} \bigcup_{k=1}^{\infty} A_k, & \text{if } \{A_k\} \text{ is increasing,} \\ \bigcap_{k=1}^{\infty} A_k, & \text{if } \{A_k\} \text{ is decreasing.} \end{cases}$$

Proof. Assume first that $A_k \subseteq A_{k+1}$ for all k. For fixed j, the monotonicity implies

$$\bigcap_{k=j}^{\infty} A_k = A_j,$$

and therefore

$$\liminf_{k\to\infty} A_k = \bigcup_{j=1}^{\infty} \bigcap_{k=j}^{\infty} A_k = \bigcup_{j=1}^{\infty} A_j.$$

On the other hand, the sets $\bigcup_{k=j}^{\infty} A_k$ decrease as j increases, and their intersection equals the union of the whole sequence. Hence

$$\limsup_{k\to\infty} A_k = \bigcap_{j=1}^{\infty} \bigcup_{k=j}^{\infty} A_k = \bigcup_{k=1}^{\infty} A_k.$$

Thus $\limsup = \liminf = \bigcup_{k=1}^{\infty} A_k$.

If instead $A_{k+1} \subseteq A_k$ for all k, then $\bigcup_{k=j}^{\infty} A_k = A_j$, while $\bigcap_{k=j}^{\infty} A_k$ increases with j and

$$\bigcup_{j=1}^{\infty} \bigcap_{k=j}^{\infty} A_k = \bigcap_{k=1}^{\infty} A_k.$$

The same calculation yields $\limsup = \liminf = \bigcap_{k=1}^{\infty} A_k$. □

Recall the monotone convergence theorem in classical analysis.

Proposition 1.1.1. *Let $\{A_k\}$ be a sequence of subsets of E. Then*

1. $E\backslash\Big(\overline{\lim\limits_{k\to\infty}} A_k\Big) = \underline{\lim\limits_{k\to\infty}} (E\backslash A_k)$,
2. $E\backslash\Big(\underline{\lim\limits_{k\to\infty}} A_k\Big) = \overline{\lim\limits_{k\to\infty}} (E\backslash A_k)$.

Proof. Take the first result as an example, by De Morgan's law,

$$\begin{aligned} E\backslash\Big(\overline{\lim_{k\to\infty}} A_k\Big) &= E\backslash\Big(\bigcap_{j=1}^{\infty}\bigcup_{k=j}^{\infty} A_k\Big) \\ &= \bigcup_{j=1}^{\infty} (E\backslash(\bigcup_{k=j}^{\infty} A_k)) \\ &= \bigcup_{j=1}^{\infty}\bigcap_{k=j}^{\infty}(E\backslash A_k) \\ &= \underline{\lim_{k\to\infty}} (E\backslash A_k). \end{aligned}$$

□

Theorem 1.1.2. *Let $\{A_k\}$ be a sequence of sets. Then*

(1) $\overline{\lim\limits_{k\to\infty}} A_k = \{x|\forall j \in \mathbb{N}, \exists k \geqslant j, s.t. x \in A_k\}$, *i.e., there exists a subsequence $\{A_{k_i}\}$ such that $x \in A_{k_i}$.*

(2) $\underline{\lim\limits_{k\to\infty}} A_k = \{x|\exists j_0 \in \mathbb{N}, s.t. \forall k \geqslant j_0, x \in A_k\}$, *i.e., from some index j_0 onward, x belongs to all A_k.*

Proof. This descriptive formulation follows directly from the definition of limsup/liminf. □

Example 1.1.3 (Structure of Non-convergence Points)**.** *Let $\{f_n(x)\}$ and $f(x)$ be real-valued functions on $\mathbb{R}^1$. Denote by D the set of points where $\{f_n(x)\}$ does not converge to $f(x)$. Then*

$$D = \bigcup_{k=1}^{\infty}\Big(\bigcap_{N=1}^{\infty}\bigcup_{n=N}^{\infty}\{x||f_n(x) - f(x)| \geqslant \frac{1}{k}\}\Big).$$

Remark 1.1.2. *We will use this in the proof of Egorov's theorem; see Theorem 3.3.2.*

At the end of this section, we define Cartesian products. Let X, Y be nonempty sets. The Cartesian product is

$$X \times Y = \{(x, y)\big|x \in X, y \in Y\}.$$

The definition of infinite products is more subtle and is formulated via maps. Let $\{X_\alpha\}_{\alpha\in I}$ be a family of sets. Define

$$\prod_{\alpha\in I} X_\alpha = \Big\{x\Big|x : I \to \bigcup_{\alpha\in I} X_\alpha, s.t., x(\alpha) \in X_\alpha, \forall\alpha \in I\Big\}.$$

Conversely, a map can be viewed as an element of a product space:

$$\{f|f : X \to Y\} \cong Y^X.$$

As a simple example, when $X = \{1, 2, \cdots, n\}$, elements of $\{X \to Y\}$ correspond one-to-one to elements of $Y \times Y \times \cdots \times Y$.

Exercises 1.1

[Unless otherwise noted, page and problem numbers in exercises refer to the corresponding page/problem in Zhou Mingqiang's *Real Analysis*, 2nd edition.]

1. Prove Theorem 1.1.2.
2. Characterize convergence of a function sequence $\{f_n(x)\}$ to $f(x)$.
3. P11 1,2,3
4. P63 1,2

1.2 Mappings and Cardinality

1.2.1 Mappings

Sets are a central object of study in mathematics. For a given set, one may introduce a topology to obtain a topological space, or an algebraic structure (usually natural) to obtain a group, ring, or field, or study both structures simultaneously. These are studies of individual sets in relative isolation. A more effective approach is to classify a family of sets by grouping together sets with the same properties (equivalent sets). A key way to realize this classification is to build relations between sets, and one major tool is mappings between sets.

Definition 1.2.1 (Mapping)**.** *Let X, Y be sets. If there is a rule f such that for each $x \in X$ there exists a unique $y \in Y$ corresponding to x, then this relation is called a mapping, also called a function or transformation.*

Let $f : X \to Y$ be a mapping, with $A \subset X$, $B \subset Y$. Define

$$f(A) = \{f(x)|x \in A\}$$

as the image of A under f, and

$$f^{-1}(B) = \{x|f(x) \in B\}$$

as the preimage of B under f. Question: is $f^{-1}(B)$ unique?

Definition 1.2.2. *Let $f : X \to Y$ be a mapping. If for all $x_1, x_2 \in X$ with $x_1 \neq x_2$, one has*

$$f(x_1) \neq f(x_2),$$

then f is called injective (one-to-one). If $f(X) = Y$, then f is called surjective (onto). If f is both injective and surjective, then f is called bijective.

Let $f : X \to Y$ be bijective. Define $g : Y \to X$ by

$$g(y) = x,$$

where $y = f(x)$. Then g is called the inverse mapping of f, denoted by f^{-1}. Clearly, $f^{-1}(f(x)) = x$ and $f(f^{-1}(y)) = y$.

Proposition 1.2.1. *Let $f : X \to Y$ be a mapping. Let $\{A_i\}_{i\in I}$ be a family of subsets of X and $\{B_j\}_{j\in J}$ a family of subsets of Y. Then*

1. $f(\bigcup_{i\in I} A_i) = \bigcup_{i\in I} f(A_i)$.
2. $f(\bigcap_{i\in I} A_i) \subseteq \bigcap_{i\in I} f(A_i)$.
3. $f^{-1}(\bigcup_{j\in J} B_j) = \bigcup_{j\in J} f^{-1}(B_j)$.
4. $f^{-1}(\bigcap_{j\in J} B_j) = \bigcap_{j\in J} f^{-1}(B_j)$.
5. $f^{-1}(B^c) = (f^{-1}(B))^c$ *for every* $B \subseteq Y$.

Proof. Items (1)–(4) follow directly from the definitions of image and inverse image. For (5), note that

$$x \in f^{-1}(B^c) \iff f(x) \in B^c \iff f(x) \notin B \iff x \notin f^{-1}(B) \iff x \in (f^{-1}(B))^c. \qquad \square$$

Remark 1.2.1. *For a general mapping f, the direct image $A \mapsto f(A)$ behaves well with unions but can fail to preserve intersections and complements. In contrast, the inverse image $B \mapsto f^{-1}(B)$ preserves the Boolean operations (unions, intersections, complements, and differences). For this reason, structural notions in analysis are naturally phrased in terms of inverse images:*

1. *A function is continuous if the inverse image of every open set is open.*
2. *A function is Borel measurable if the inverse image of every Borel set is Borel (or, more generally, measurable).*

Definition 1.2.3 (Composite Mapping)**.** *Let $f : X \to Y$ and $g : Y \to Z$ be mappings. Define $h : X \to Z$ by*

$$h(x) = g(f(x)), \forall x \in X,$$

called the composition of f and g, denoted by $g \circ f$.

1.2.2 Characteristic Functions and Power Sets

Definition 1.2.4 (Power Set)**.** *Let X be a set. The set*

$$\mathcal{P}(X) = \{A | A \subset X\}$$

is called the power set of X.

Definition 1.2.5 (Characteristic Function)**.** *Let X be a set. For any subset $A \subset X$, define*

$$\chi_A(x) = \begin{cases} 1, & x \in A; \\ 0, & x \notin A. \end{cases}$$

This map is called the characteristic function of A.

Let X be a set and define

$$f : \mathcal{P}(X) \to \chi(X)$$
$$A \mapsto \chi_A,$$

where $\chi(X)$ denotes the set of all characteristic functions on X. Then f is bijective.

Some basic properties of characteristic functions are listed below; their proofs are straightforward.

Proposition 1.2.2. *Let A, B be sets. Then*

(1) *If $A \cap B = \varnothing$, then $\chi_{A\cup B} = \chi_A + \chi_B$;*

(2) $\chi_{A\cup B} = \chi_{A\setminus B} + \chi_{B\setminus A} + \chi_{A\cap B} = \chi_A + \chi_B - \chi_{A\cap B}$;

(3) $\chi_{A\cap B} = \chi_A \times \chi_B$;

(4) $\chi_{A\setminus B} = \chi_A - \chi_{A\cap B} = \chi_A(1 - \chi_B)$;

(5) $\chi_{A\triangle B} = |\chi_A - \chi_B| = \begin{cases} \chi_A - \chi_B, & x \in A\setminus B; \\ \chi_B - \chi_A, & x \in B\setminus A. \end{cases}$

These identities connect operations on sets with operations on functions. Furthermore, if one introduces an order structure on the function space and considers lattice operations

$$\chi_A \vee \chi_B = \chi_{A\cup B}, \chi_A \wedge \chi_B = \chi_{A\cap B},$$

one obtains an isomorphism between the two structures in the lattice sense.

Think: Prove

$$\limsup_{n\to\infty} \chi_{A_n} = \chi_{\limsup_{n\to\infty} A_n}, \liminf_{n\to\infty} \chi_{A_n} = \chi_{\liminf_{n\to\infty} A_n}.$$

In fact, since the definitions of lim sup and lim inf only involve order structure and χ gives an isomorphism of the two order relations, the conclusion should hold.

Proposition 1.2.3 (Fixed Point Problem for Monotone Set Maps)**.** *Let X be nonempty and let $f : \mathcal{P}(X) \to \mathcal{P}(X)$ satisfy the monotonicity condition*

$$A \subset B \Rightarrow f(A) \subset f(B).$$

Then there exists $T \in \mathcal{P}(X)$ such that $f(T) = T$.

Proof. Let

$$S = \{A \subseteq X \mid A \subseteq f(A)\}.$$

Since $\varnothing \subseteq f(\varnothing)$, the family S is nonempty. Define

$$T = \bigcup_{A\in S} A.$$

For every $A \in S$ we have $A \subseteq T$, hence by monotonicity $f(A) \subseteq f(T)$. Combining this with $A \subseteq f(A)$ gives $A \subseteq f(T)$. Taking the union over all $A \in S$ yields $T \subseteq f(T)$.

Now $T \subseteq f(T)$ implies $f(T) \subseteq f(f(T))$, so $f(T) \in S$. By definition of T as the union of all sets in S, this gives $f(T) \subseteq T$. Therefore $f(T) = T$. □

Example 1.2.1. *Let $f : X \to X$ be a mapping. Define $F : \mathcal{P}(X) \to \mathcal{P}(X)$ by*

$$F(A) = f(A).$$

Call F the map induced by f. Clearly F is monotone and has the trivial fixed point $\varnothing$. Question: does F have nontrivial fixed points?

1.2.3 Cardinality

One major question in set theory is how to describe and compare the number of elements in two sets; this is essentially a classification problem. For a finite set A with n elements, we say A has cardinality n, written $\overline{\overline{A}} = n$ or Card$A = n$. What about infinite sets?

- The algebraic viewpoint classifies by isomorphism.
- The topological viewpoint classifies by homotopy/homeomorphism.
- The differential-geometric viewpoint classifies by diffeomorphism.
- The set-theoretic viewpoint classifies by equipotence.

Definition 1.2.6 (Equipotence)**.** *Let A, B be sets. If there exists a bijection from A to B, then A and B are called equipotent, denoted $A \sim B$.*

Ignoring the Russell paradox issue, equipotence is an equivalence relation between sets.[1] If $A \sim B$, then A and B have the same cardinal number, written

$$\overline{\overline{A}} = \overline{\overline{B}}.$$

Thus cardinality is a concept on equivalence classes of sets.

Intuitively, if A is equipotent to a subset of B, define

$$\overline{\overline{A}} \leqslant \overline{\overline{B}}.$$

Conversely, if B is equipotent to a subset of A, define

$$\overline{\overline{A}} \geqslant \overline{\overline{B}}.$$

Then we ask:

$$\left.\begin{array}{l} \overline{\overline{A}} \leqslant \overline{\overline{B}} \\ \overline{\overline{A}} \geqslant \overline{\overline{B}} \end{array}\right\} \Rightarrow \overline{\overline{A}} = \overline{\overline{B}}?$$

Theorem 1.2.1 (Cantor–Bernstein)**.** *If X is equipotent to a subset of Y, and Y is equipotent to a subset of X, then $X \sim Y$.*

Using the following lemmas, we obtain a proof of Theorem 1.2.1.

Lemma 1.2.1. *Let $X_1 \subset X_2$, $Y_1 \subset Y_2$ be sets, and $\varphi : X \to Y$ a bijection, with*

$$X_i \overset{\varphi}{\sim} Y_i, i = 1, 2.$$

Then

$$(X_2 \backslash X_1) \overset{\varphi}{\sim} (Y_2 \backslash Y_1).$$

Proof. Omitted. □

[1] An equivalence relation can be viewed as the diagonal subset in the product space of two sets.

Lemma 1.2.2. *Let $\{A_i\}_{i\in I}$ and $\{B_i\}_{i\in I}$ be two families indexed by the same set I. Assume members of $\{A_i\}$ are pairwise disjoint and members of $\{B_i\}$ are pairwise disjoint, and*

$$A_i \sim B_i.$$

Then

$$\bigcup_{i\in I} A_i \sim \bigcup_{i\in I} B_i.$$

Proof. Omitted. □

Remark 1.2.2. *If the two families use different index sets that are themselves equipotent, the lemma still holds.*

Proof. (Proof of Theorem 1.2.1). By assumption there exist subsets $Y_1 \subset Y$ and $X_1 \subset X$ together with bijections

$$f : X \xrightarrow{\sim} Y_1, \qquad g : Y \xrightarrow{\sim} X_1.$$

Write $Y_1 = f(X)$ and $X_1 = g(Y)$, and set

$$\varphi = g \circ f : X \longrightarrow X_2, \qquad X_2 = \varphi(X) = g(Y_1) \subset X_1.$$

Then φ is a bijection from X onto X_2.

Define $X_0 = X$ and recursively

$$X_{n+2} = \varphi(X_n), \qquad n \geqslant 0.$$

Since $X_2 \subset X_1 \subset X_0$ and φ is bijective, we obtain a decreasing chain

$$X_0 \supset X_1 \supset X_2 \supset X_3 \supset \cdots \supset X_n \supset \cdots .$$

Let

$$X_\infty = \bigcap_{n=1}^{\infty} X_n.$$

Then we have disjoint decompositions

$$X = \Big(\bigsqcup_{n=1}^{\infty} (X_{n-1} \setminus X_n) \Big) \sqcup X_\infty,$$
$$X_1 = \Big(\bigsqcup_{n=1}^{\infty} (X_n \setminus X_{n+1}) \Big) \sqcup X_\infty.$$

For each $n \geqslant 0$, the restriction of φ gives a bijection

$$\varphi : X_{2n} \setminus X_{2n+1} \xrightarrow{\sim} X_{2n+2} \setminus X_{2n+3},$$

which follows from Lemma 1.2.1 applied to the inclusions $X_{2n+1} \subset X_{2n}$ and $X_{2n+3} \subset X_{2n+2}$. On the other hand, the identity map gives

$$(X_{2n+1} \setminus X_{2n+2}) \xrightarrow{\mathrm{id}} (X_{2n+1} \setminus X_{2n+2}).$$

Now match the pieces of the two disjoint decompositions as follows:

- map $X_{2n} \setminus X_{2n+1}$ to $X_{2n+2} \setminus X_{2n+3}$ via φ;
- map $X_{2n+1} \setminus X_{2n+2}$ to itself via the identity;
- map X_∞ to itself via the identity.

By Lemma 1.2.2 (disjoint unions preserve equipotence), this produces a bijection $X \sim X_1$. Since $X_1 \sim Y$ via g, we conclude $X \sim Y$. □

There is another proof of Theorem 1.2.1. We list it below, first requiring one lemma.

Lemma 1.2.3 (Decomposition Theorem Under Set Mappings)**.** *Let X, Y be sets. If $f : X \to Y$ and $g : Y \to X$, then there are decompositions*

$$X = A \cup A_1, A \cap A_1 = \varnothing,$$

$$Y = B \cup B_1, B \cap B_1 = \varnothing,$$

such that

$$f(A) = B, g(B_1) = A_1.$$

Proof. Without loss of generality assume

$$f(X) \subsetneqq Y, g(Y) \subsetneqq X.$$

Call a subset $E \subset X$ *separated* if

$$E \cap g(Y \setminus f(E)) = \varnothing.$$

Let Γ be the family of all separated sets. Since $\varnothing \in \Gamma$, Γ is nonempty. We claim Γ is closed under arbitrary unions. Indeed, if A is a union of members of Γ, then

$$\begin{aligned} & E \in \Gamma, (Y \setminus f(A)) \subset (Y \setminus f(E)) \\ \Rightarrow \quad & \forall E \in \Gamma, E \cap g(Y \setminus f(A)) = \varnothing \\ \Rightarrow \quad & A \cap g(Y \setminus f(A)) = \varnothing \\ \Rightarrow \quad & A \text{ is separated.} \end{aligned}$$

With this property, define[2]

$$A = \bigcup_{E \in \Gamma} E.$$

Then:

(1) By definition, A is maximal among separated sets under inclusion.

(2) Set $B = f(A)$, $B_1 = Y \setminus B$, $A_1 = g(B_1)$. We prove

$$A \cap A_1 = \varnothing, \text{and} A_1 = X \setminus A.$$

First,

$$A \cap g(Y \setminus f(A)) = \varnothing \Rightarrow A \cap A_1 = \varnothing.$$

[2]This avoids explicitly invoking the well-ordering principle.

Next prove $A_1 = X\backslash A$ by contradiction. Suppose $\exists x_0 \in X\backslash A$ and $x_0 \notin A_1$. Let $A^* = A \cup \{x_0\}$. Then

$$\begin{aligned} & A^* = A \cup \{x_0\} \\ \Rightarrow \quad & f(A) \subset f(A^*) \\ \Rightarrow \quad & Y\backslash f(A^*) \subset Y\backslash f(A) \\ \Rightarrow \quad & \Big(g(Y\backslash f(A^*)) \subset g(Y\backslash f(A))\Big) + \Big(A \cap g(Y\backslash f(A)) = \varnothing\Big) \\ \Rightarrow \quad & A \cap g(Y\backslash f(A^*)) = \varnothing. \end{aligned}$$

Since $x_0 \notin A_1$, we have $x_0 \notin g(Y\backslash f(A^*)) \subset g(Y\backslash f(A))$. Hence $A^* \in \Gamma$, but $A \subsetneqq A^*$, contradicting maximality. □

Using this lemma, Theorem 1.2.1 can be proved (left as an exercise).

Let us also see a corollary.

Corollary 1.2.1. *Let $C \subset A \subset B$ and $B \sim C$. Then $A \sim B$.*

Proof. Take $f : B \to C \subsetneqq A$ and $g : A \to A \subsetneqq B$. □

Theorem 1.2.1 not only solves

$$\left.\begin{array}{l} \overline{\overline{A}} \leqslant \overline{\overline{B}} \\ \overline{\overline{A}} \geqslant \overline{\overline{B}} \end{array}\right\} \Rightarrow \overline{\overline{A}} = \overline{\overline{B}}?$$

but also provides a convenient tool for proving equipotence. In the next part, we focus on the following cardinalities:

1. finite sets;
2. countable sets;
3. $\overline{\overline{[0,1]}}$;
4. $\overline{\overline{\mathcal{P}(X)}} = \overline{\overline{2^X}} > \overline{\overline{X}}$.

Definition 1.2.7 (Finite Set)**.** *Let A be nonempty. Write $M_n = \{1, 2, \cdots, n\}$. If $A \sim M_n$ for some n, then A is called finite and n is its cardinality. If $A \nsim M_n$ for every natural number n, then A is called infinite.*

Definition 1.2.8 (Countable Set)**.** *A set equipotent to $\mathbb{N}$ is called countable. The cardinality of a countable set is denoted by $\aleph_0$.*

Remark 1.2.3. *The cardinality of $\mathbb{R}^1$ (equivalently of the interval $(0,1)$) is called the* continuum*. In these notes we denote it by $\aleph$, so*

$$\overline{\overline{\mathbb{R}^1}} = \aleph, \qquad \overline{\overline{\mathcal{P}(\mathbb{R}^1)}} = 2^{\aleph}.$$

The next two examples are standard countability examples.

Example 1.2.2. *Any family of pairwise disjoint open intervals in $\mathbb{R}^1$ is at most countable.*

Proof. It suffices to build a one-to-one correspondence with a subset of $\mathbb{Q}$. □

Remark 1.2.4. *This property will be used in describing the structure of open sets in* $\mathbb{R}^1$.

Example 1.2.3. *The set of discontinuities of a monotone function* f *is at most countable.*

Proof. Omitted. □

Theorem 1.2.2. *Every infinite set* A *contains a countable subset.*

Example 1.2.4 (Characterization of Function Discontinuity Points)**.** *Let* $f : \mathbb{R} \to \mathbb{R}$ *be an arbitrary function. For* $r > 0$ *define the oscillation of* f *on the interval* $(x - r, x + r)$ *by*

$$\omega(f; x, r) = \sup\{f(y) \mid |y - x| < r\} - \inf\{f(y) \mid |y - x| < r\}.$$

The (pointwise) oscillation of f *at* x *is*

$$\omega(f; x) = \inf_{r>0} \omega(f; x, r) = \lim_{r \downarrow 0} \omega(f; x, r).$$

Then

$$f \text{ is continuous at } x \quad \Longleftrightarrow \quad \omega(f; x) = 0.$$

Moreover, if we set

$$\Omega_{k,m} = \Big\{ x \in \mathbb{R} \;\Big|\; \omega(f; x, \tfrac{1}{m}) \geqslant \tfrac{1}{k} \Big\}, \qquad k, m \in \mathbb{N},$$

then each $\Omega_{k,m}$ *is closed and the set of discontinuity points of* f *is*

$$\Omega^{(+)} = \bigcup_{k=1}^{\infty} \bigcup_{m=1}^{\infty} \Omega_{k,m}.$$

In particular, the discontinuity set of an arbitrary real function is always an F_σ *set.*

Theorem 1.2.3. *For any set* A*,*

$$\overline{\overline{\mathcal{P}(A)}} = \overline{\overline{2^A}} > \overline{\overline{A}}.$$

Proof. First define $\varphi : A \to \mathcal{P}(A)$ by

$$x \mapsto \{x\},$$

so clearly $\overline{\overline{\mathcal{P}(A)}} \geqslant \overline{\overline{A}}$. We prove strict inequality by contradiction. Assume $\overline{\overline{\mathcal{P}(A)}} = \overline{\overline{A}}$. Then there is a bijection $\psi : A \to \mathcal{P}(A)$. Let

$$B = \{x \in A | x \notin \psi(x)\}.$$

Then $B \neq \varnothing$. Since ψ is onto, there exists $x^* \in A$ with

$$\psi(x^*) = B.$$

Thus

$$x^* \in B \Leftrightarrow x^* \notin B,$$

a contradiction. Hence $\overline{\overline{\mathcal{P}(A)}} \neq \overline{\overline{A}}$, and therefore $\overline{\overline{\mathcal{P}(A)}} > \overline{\overline{A}}$. □

Now let us look at two examples on cardinality.

Example 1.2.5. *The cardinality of the set* $\mathcal{F}$ *of all real-valued functions on* $\mathbb{R}^1$ *is* $2^{\aleph}$.

Proof. For each subset $A \subseteq \mathbb{R}^1$, the characteristic function χ_A is an element of $\mathcal{F}$. The map

$$A \mapsto \chi_A$$

is injective from $\mathcal{P}(\mathbb{R}^1)$ into $\mathcal{F}$, hence

$$\overline{\overline{\mathcal{F}}} \geqslant \overline{\overline{\mathcal{P}(\mathbb{R}^1)}} = 2^{\aleph}.$$

Conversely, associate to each $f \in \mathcal{F}$ its graph

$$\mathrm{Graph}(f) = \{(x, f(x)) \mid x \in \mathbb{R}^1\} \subset \mathbb{R}^2.$$

The map $f \mapsto \mathrm{Graph}(f)$ is injective from $\mathcal{F}$ into $\mathcal{P}(\mathbb{R}^2)$, so

$$\overline{\overline{\mathcal{F}}} \leqslant \overline{\overline{\mathcal{P}(\mathbb{R}^2)}} = 2^{\aleph}.$$

Therefore $\overline{\overline{\mathcal{F}}} = 2^{\aleph}$. □

Remark 1.2.5. *If one regards maps as elements of infinite product spaces, then $\mathcal{F} \sim \mathbb{R}^{\mathbb{R}}$. Hence*

$$\mathrm{Card}\mathcal{F} = 2^{\aleph}.$$

Example 1.2.6. *The cardinality of $C(\mathbb{R}^1)$, the set of all continuous functions on $\mathbb{R}^1$, is $\aleph$.*

Proof. Write $\mathbb{Q} = \{r_1, r_2, \cdots\}$. Define

$$T : C(\mathbb{R}^1) \longrightarrow (\mathbb{R}^1)^{\mathbb{N}}, \qquad T(f) = \big(f(r_1), f(r_2), \cdots\big).$$

If $T(f) = T(g)$, then f and g agree on the dense set $\mathbb{Q}$, hence by continuity $f \equiv g$ on $\mathbb{R}^1$. Therefore T is injective and

$$\overline{\overline{C(\mathbb{R}^1)}} \leqslant \overline{\overline{(\mathbb{R}^1)^{\mathbb{N}}}} = \aleph.$$

On the other hand, the map $a \mapsto f_a$, where $f_a(x) \equiv a$ is the constant function, is an injection $\mathbb{R}^1 \hookrightarrow C(\mathbb{R}^1)$. Hence $\overline{\overline{C(\mathbb{R}^1)}} \geqslant \overline{\overline{\mathbb{R}^1}} = \aleph$, and the conclusion follows. □

Exercises 1.2

1. Let $f(x)$ be continuous on $[a, b]$. Then for any $\varepsilon > 0$, there exist finitely many pairwise disjoint intervals $\{[a_i, b_i)\}_{i=1}^n$ and $\xi_i \in \mathbb{R}$, $i = 1, 2, \cdots, n$, such that

$$\Big| f(x) - \sum_{i=1}^{n} \xi_i \chi_{[a_i, b_i)}(x) \Big| < \varepsilon, \forall x \in [a, b].$$

2. Prove that the definition of cardinality for finite sets is well-defined.

3. If A is infinite, then A is equipotent to a proper subset of itself. [3]

4. Give a bijection between $(0, 1)$ and $[0, 1]$, and prove that no continuous bijection exists between them.

[3] This can also be used as a definition of infinite sets.

5. Find the cardinality of $C(\mathbb{R}^1)$.
6. Find the cardinality of monotone functions on $\mathbb{R}^1$.[4]
7. Find the cardinality of all maps from $[0,1]$ to $\mathbb{N}$.
8. P18 1,2,3
9. P24 5,6,7,9,10
10. P28 13,14,15,16,17
11. P64 5,7,9,10
12. P69 27

1.3 Sets in $\mathbb{R}^n$

1.3.1 Metrics in $\mathbb{R}^n$

The n-dimensional Euclidean space $\mathbb{R}^n$ naturally carries the Euclidean distance. For any

$$x = (x_1, x_2, \cdots, x_n) \in \mathbb{R}^n, y = (y_1, y_2, \cdots, y_n) \in \mathbb{R}^n,$$

define

$$\mathrm{d}(x,y) = \|x-y\|_{\mathbb{R}^n} = \Big(\sum_{i=1}^{n} |x_i - y_i|^2\Big)^{\frac{1}{2}}.$$

It is easy to prove that $\mathrm{d}(x,y)$ satisfies:

(1) [Positive definiteness] $\mathrm{d}(x,y) \geqslant 0$, with equality iff $x = y$;
(2) [Symmetry] $\mathrm{d}(x,y) = \mathrm{d}(y,x)$;
(3) [Triangle inequality] $\mathrm{d}(x,y) \leqslant \mathrm{d}(x,z) + \mathrm{d}(z,y)$.

Definition 1.3.1. *Let $\{x^k\}_{k=1}^{\infty} \subset \mathbb{R}^n$, $x \in \mathbb{R}^n$. We say x is the limit of $\{x^k\}$ as $k \to \infty$ if*

$$\lim_{k\to\infty} \mathrm{d}(x^k, x) = 0.$$

Theorem 1.3.1. *Let $\{x^k\}_{k=1}^{\infty} \subset \mathbb{R}^n$, $x \in \mathbb{R}^n$. Then*

$$x^k \to x \Leftrightarrow x_i^k \to x_i, i = 1, 2, \cdots.$$

Proof. Omitted. □

Definition 1.3.2. *Let $A \subset \mathbb{R}^n$. If there exists $M > 0$ such that for all $x \in A$,*

$$\|x\| \overset{\text{def.}}{=} d(x,0) \leqslant M,$$

then A is called bounded.

Theorem 1.3.2 (Bolzano–Weierstrass)**.** *If $\{x^k\} \subset \mathbb{R}^n$ is bounded, then it has a convergent subsequence in $\mathbb{R}^n$.*

Theorem 1.3.1 and Theorem 1.3.2 are standard results from analysis, so we omit proofs.

[4] 2011 midterm.

1.3.2 Neighborhoods, Interior Points, Open Sets, Closed Sets

Let $x_0 \in \mathbb{R}^n$. Define

$$B_\delta(x_0) = \{x \in \mathbb{R}^n | d(x, x_0) < \delta\}$$

as the δ-neighborhood of x_0, also denoted by $N_\delta(x_0)$ or $B(x_0, \delta)$.

Definition 1.3.3 (Open Set)**.** *Let $A \subset \mathbb{R}^n$, $x_0 \in A$. We call x_0 an interior point of A if there exists $\delta > 0$ such that*

$$N_\delta(x_0) \subset A.$$

The set of all interior points is called the interior of A, denoted by $\overset{\circ}{A}$. A is called open in $\mathbb{R}^n$ if $A = \overset{\circ}{A}$.

Remark 1.3.1. *The definition of openness depends on the chosen ambient space.*

Theorem 1.3.3. *Open sets in $\mathbb{R}^n$ satisfy:*

(1) *$\varnothing$ and $\mathbb{R}^n$ are open;*

(2) *if $\{G_i\}_{i \in I}$ is a family of open sets, then $\bigcup\limits_{i \in I} G_i$ is open;*

(3) *finite intersections of open sets are open.*

Proof. For (3), let G_i, $i = 1, 2, \cdots, n$, be open in $\mathbb{R}^n$. Take $x_0 \in \bigcap\limits_{i=1}^{n} G_i$. For each $1 \leqslant i \leqslant n$, since $x_0 \in G_i$, there exists $\delta_i > 0$ such that $B_{\delta_i}(x_0) \subset G_i$. Let $\delta = \min\{\delta_1, \delta_2, \cdots, \delta_n\}$, then

$$B_\delta(x_0) \subset G_i, i = 1, 2, \cdots, n,$$

so

$$B_\delta(x_0) \subset \Big(\bigcap_{i=1}^{n} G_i\Big).$$

Hence x_0 is an interior point of $\bigcap\limits_{i=1}^{n} G_i$. □

Definition 1.3.4 (Path Connectedness)**.** *Let $A \subset \mathbb{R}^n$, $x, y \in A$. We say x, y are path connected in A if there exists a continuous map $\varphi : [0, 1] \to A$ such that*

$$\varphi(0) = x, \varphi(1) = y.$$

A is called path connected if every pair $x, y \in A$ is path connected. A subset $B \subset A$ is called a connected component of A if

(1) *B is path connected;*

(2) *B is maximal: for every path-connected $B_1 \subset A$, one has $B_1 \subset B$.*

Theorem 1.3.4 (Structure Theorem of Open Sets in $\mathbb{R}^n$)**.** *Let G be open in $\mathbb{R}^n$. Then G can be written as the union of at most countably many path-connected components, each of which is open:*

$$G = \bigcup_{i \in I} G_i,$$

where I is at most countable. Each component is called a component interval of G.

Proof. See [3], Theorem 4.2. □

Remark 1.3.2. *For any $x, y \in G$, define $x \sim y$ if x and y are path connected. This is an equivalence relation. For each $x \in G$, let*

$$G_x = \{y \in G | y \sim x\},$$

then G_x is a path-connected component of G, and is open.

1.3.3 Limit Points and Closure

Definition 1.3.5 (Limit Point)**.** *Let $A \subset \mathbb{R}^n$, $x_0 \in \mathbb{R}^n$. We call x_0 a limit point (accumulation point) of A if for every δ-neighborhood $N_\delta(x_0)$,*

$$\Big(N_\delta(x_0) \backslash \{x_0\}\Big) \cap A \neq \varnothing.$$

The set of all limit points is called the derived set, denoted by A'.

x_0 is called an isolated point of A if there exists $\delta > 0$ such that

$$\Big(N_\delta(x_0) \backslash \{x_0\}\Big) \cap A = \varnothing.$$

x_0 is called a boundary point of A if for every δ-neighborhood $N_\delta(x_0)$,

$$\Big(N_\delta(x_0) \backslash \{x_0\}\Big) \cap A \neq \varnothing,$$

and

$$\Big(N_\delta(x_0) \backslash \{x_0\}\Big) \cap A^c \neq \varnothing.$$

The set of all boundary points is called the boundary of A, denoted by $\mathrm{Bd}A$ or ∂A.

Definition 1.3.6 (Closed Set)**.** *$A \subset \mathbb{R}^n$ is called closed if $A' \subset A$. Its closure is $\overline{A} = A' \cup A$.*

Remark 1.3.3. *The definition of closedness depends on the ambient space.*

We have the following elementary fact.

Theorem 1.3.5. *$A \subset \mathbb{R}^n$ is closed in $\mathbb{R}^n$ iff A^c is open in $\mathbb{R}^n$.*

Proof. Omitted. □

From the definitions of closed set and derived set, one easily obtains:

Example 1.3.1. *For $E_1, E_2 \subset \mathbb{R}^n$,*

$$(E_1 \cup E_2)' = E_1' \cup E_2'.$$

Hence

$$\overline{E_1 \cup E_2} = \overline{E_1} \cup \overline{E_2}.$$

Corollary 1.3.1. *For $A \subset \mathbb{R}^n$, the closure $\overline{A}$ is closed in $\mathbb{R}^n$.*

Proof. This follows by contradiction. □

The next theorem corresponds exactly to Theorem 1.3.3.

Theorem 1.3.6. *Closed sets in $\mathbb{R}^n$ satisfy:*

(1) *$\varnothing$ and $\mathbb{R}^n$ are closed;*

(2) *if $\{G_i\}_{i\in I}$ is a family of closed sets, then $\bigcap_{i\in I} G_i$ is closed;*

(3) *finite unions of closed sets are closed.*

Proof. Consider complements, and apply De Morgan's law with Theorem 1.3.3. □

We now prove the famous Heine–Borel theorem (finite open cover principle). We first define refinement covers.

Definition 1.3.7 (Refinement Cover)**.** *Let $A \subset \mathbb{R}^n$ and let $\{G_i\}_{i\in I}$ be an open cover of A. We call $\{U_j\}_{j\in J}$ a refinement of $\{G_i\}_{i\in I}$ if $\{U_j\}_{j\in J}$ is also an open cover of A, and for each $j \in J$ there exists $i_j \in I$ such that*

$$U_j \subset G_{i_j}.$$

Density is another frequent concept.

Definition 1.3.8 (Dense)**.** *Let A be nonempty. A set E is dense in A if $A \subset \overline{E}$. If additionally $E \subset A$, then E is called a dense subset of A.*

With density, we define separability.

Definition 1.3.9 (Separable)**.** *A set A is called separable if it has a countable dense subset.*

Second countability is also important.

Definition 1.3.10 (Second Countable)**.** *A topological space (Ω, τ) is called second countable if τ has a countable base.*

Remark 1.3.4. *1. The proof of the next lemma uses that $\mathbb{R}^n$ is second countable.*[5]

2. For $\mathbb{R}^n$, separability, second countability, and the Lindelöff property below are equivalent. In general topological spaces, they need not be equivalent.

Lemma 1.3.1 (Lindelöff Property)**.** *Let $A \subset \mathbb{R}^n$ and $\{G_i\}_{i\in I}$ be any open cover of A. Then $\{G_i\}_{i\in I}$ has a countable subcover of A.*

Proof. For each $x \in A$, since $A \subset \bigcup_{i\in I} G_i$, there exists $i_x \in I$ such that $x \in G_{i_x}$. Since G_{i_x} is open, there exists $\delta_x > 0$ with

$$N_{\delta_x}(x) \subset G_{i_x}.$$

Because $\mathbb{Q}^n$ is dense in $\mathbb{R}^n$, there exists $y_x \in \mathbb{Q}^n$ with $\mathrm{d}(y_x, x) < \delta_x/4$. Choose $\varepsilon_x \in \mathbb{Q}$ with $\delta_x/4 \leqslant \varepsilon_x \leqslant \delta_x/2$, then $x \in N_{\delta_x}(y_x) \subset G_{i_x}$. Let $X = \{N_{\delta_x}(y_x) | x \in A\}$. Then X is at most countable and is a refinement of $\{G_i\}$. Hence

$$A \subset \bigcup_{i=1}^{\infty} N_{\delta_{x_j}}(x_j) \subset \bigcup_{i=1}^{\infty} G_{i_j},$$

a countable cover of A. □

[5]The proof does not need explicit topological language; readers unfamiliar with point-set topology may ignore this remark.

This lemma itself is also very useful.

Theorem 1.3.7 (Heine–Borel)**.** *$A \subset \mathbb{R}^n$ is bounded and closed iff every open cover of A has a finite subcover.*

Proof. Necessity. Let $\{G_i\}_{i\in I}$ be an open cover of A. By Lemma 1.3.1, assume it is at most countable. Suppose no finite subcover exists. Then for each k we can pick

$$x_{k+1} \in A\backslash\Big(\bigcup_{i=1}^{k} G_i\Big).$$

By Theorem 1.3.2, there is a subsequence (still denoted x_k) with $x_k \to x_0 \in A$. Then $x_0 \in G_{k_0}$ for some k_0. Since G_{k_0} is open, there exists $\delta > 0$ such that $N_\delta(x_0) \subset G_{k_0}$. From $x_k \to x_0$, for sufficiently large k we have $\mathrm{d}(x_k, x_0) < \delta$, i.e.

$$x_k \in N_\delta(x_0) \subset G_{k_0},$$

contradicting the construction of x_k.

Sufficiency. First, boundedness is easy. Next prove closedness by contradiction. Assume A is not closed. Then $\exists x_0 \in A'$ with $x_0 \notin A$. Let

$$G_n = \{y \in \mathbb{R}^n | \mathrm{d}(y, x_0) > \frac{1}{n}\},$$

then

$$A = \bigcup_{n=1}^{\infty} G_n,$$

but this cover has no finite subcover, contradiction. □

Remark 1.3.5. *1. If every open cover of $A \subset \mathbb{R}^n$ has a finite subcover, then A is called compact in $\mathbb{R}^n$.*[6] *2. The necessity part fails in infinite-dimensional spaces. In fact, whether necessity holds can serve as a criterion for finite dimensionality. The sufficiency part always holds.*

For compactness, we give a sample problem.

Example 1.3.2. *Let $E \subset \mathbb{R}^n$ be compact and $x \in E^c$. Then x and E can be separated by open sets, i.e., there exist disjoint open sets U, V such that $x \in U$ and $E \subset V$.*

Proof. Sketch: first note any two distinct points in $\mathbb{R}^n$ can be separated by open sets; this yields an open cover of E. Then extract a finite subcover by compactness, and finally use closure of open sets under finite intersections. □

1.3.4 Extension Theorem for Continuous Functions on Closed Sets

In this section we discuss some properties of continuous functions and present an important extension theorem.

[6] More generally, for a topological space (Ω, τ), a subset $A \subset \Omega$ is compact if every open cover has a finite subcover.

Definition 1.3.11 (Continuous Function)**.** *Let $E \subset \mathbb{R}^n$, $f : E \to \mathbb{R}$, $x_0 \in E$. If for every $\varepsilon > 0$ there exists $\delta > 0$ such that whenever $x \in E \cap B(x_0, \delta)$,*

$$|f(x) - f(x_0)| < \varepsilon,$$

then f is continuous at x_0. If f is continuous at every point of E, then f is continuous on E. The set of continuous functions on E is denoted by $C(E, \mathbb{R})$ or $C(E)$.

Continuity has the following equivalent formulation.

Theorem 1.3.8. *f is continuous at x iff for every neighborhood V_y of $y = f(x)$, there exists a neighborhood U_x of x such that $f(U_x) \subset V_y$.*

Continuous functions on bounded closed sets satisfy:

Proposition 1.3.1. *Let $F \subset \mathbb{R}^n$ be bounded closed and $f \in C(F, \mathbb{R})$. Then*

(1) *f is bounded on F;*

(2) *f attains its supremum and infimum on F;*

(3) *f is uniformly continuous on F;*

(4) *if F is path connected, then f has the intermediate value property.*

Proof. Omitted. □

Our goal is to find a method to extend a continuous function on a closed set F to a continuous function on $\mathbb{R}^n$. First define continuous extension.

Definition 1.3.12 (Continuous Extension)**.** *Let $f \in C(E, \mathbb{R})$. A function $g \in C(\mathbb{R}^n, \mathbb{R})$ is called a continuous extension of f to $\mathbb{R}^n$ if*

$$g|_E \equiv f.$$

In general, continuous extension is difficult. To construct extensions on closed sets, we introduce distance to a set.

Definition 1.3.13. *Let $x \in \mathbb{R}^n$, $E \subset \mathbb{R}^n$, $E \neq \varnothing$. Define*

$$\operatorname{dist}(x, E) = \inf_{y \in E} \mathrm{d}(x, y)$$

as the distance from x to E. If $E_1, E_2 \subset \mathbb{R}^n$ are nonempty, define

$$\operatorname{dist}(E_1, E_2) = \inf_{x \in E_1} \operatorname{dist}(x, E_2)$$

as the distance between sets E_1 and E_2.

The next theorem says distance functions are continuous.

Theorem 1.3.9. *Let E be a nonempty subset of $\mathbb{R}^n$. Then*

$$f(x) = \operatorname{dist}(x, E)$$

is uniformly continuous on $\mathbb{R}^n$.

Proof. For every $z \in E$,

$$\mathrm{d}(x,z) \leqslant \mathrm{d}(x,y) + \mathrm{d}(y,z).$$

Taking infimum over z on both sides gives

$$\inf_{z\in E} \mathrm{d}(x,z) \leqslant \mathrm{d}(x,y) + \inf_{z\in E} \mathrm{d}(y,z),$$

i.e.

$$\mathrm{dist}(x,E) \leqslant \mathrm{d}(x,y) + \mathrm{dist}(y,E).$$

Similarly,

$$\mathrm{dist}(y,E) \leqslant \mathrm{d}(y,x) + \mathrm{dist}(x,E).$$

Since $d(x,y) = d(y,x)$,

$$\left|\mathrm{dist}(x,E) - \mathrm{dist}(y,E)\right| \leqslant \mathrm{d}(x,y) = |x-y|.$$

Done. □

Corollary 1.3.2. *Let F be a nonempty closed subset of $\mathbb{R}^n$, $x_0 \in \mathbb{R}^n$. Then there exists $y_0 \in F$ such that*

$$\mathrm{dist}(x_0,F) = \mathrm{d}(x_0,y_0).$$

Proof. Let $r = \mathrm{dist}(x_0,F)$. Choose a sequence $\{y_k\} \subset F$ such that $\mathrm{d}(x_0,y_k) \downarrow r$. Then each y_k lies in the closed ball $\overline{B}(x_0,r+1)$. Hence the set

$$K = F \cap \overline{B}(x_0,r+1)$$

is nonempty, closed, and bounded, so it is compact. The function $y \mapsto \mathrm{d}(x_0,y)$ is continuous, therefore it attains its minimum on K at some $y_0 \in K$. By construction this minimum equals r, i.e. $\mathrm{d}(x_0,y_0) = \mathrm{dist}(x_0,F)$. □

Now we state and prove the extension theorem for continuous functions on closed sets.

Theorem 1.3.10 (Tietze Extension Theorem)**.** *Let $F \subset \mathbb{R}^n$ be closed, $f \in C(F)$, and $|f(x)| < M$ for all $x \in F$. Then there exists $g \in C(\mathbb{R}^n)$ such that*

(1) *$g|_F \equiv f$;*

(2) *$|g(x)| \leqslant M$ for all $x \in \mathbb{R}^n$.*

Proof. Let

$$A = \{x \in F \mid f(x) \geqslant \tfrac{M}{3}\}, \qquad B = \{x \in F \mid f(x) \leqslant -\tfrac{M}{3}\}, \qquad C = \{x \in F \mid |f(x)| < \tfrac{M}{3}\}.$$

Then A and B are disjoint closed subsets of F (hence closed in $\mathbb{R}^n$). We construct a continuous function φ on $\mathbb{R}^n$ satisfying $\varphi = 1$ on A and $\varphi = -1$ on B.

If $A = \varnothing$ and $B = \varnothing$, set $\varphi \equiv 0$. If $A = \varnothing$ and $B \neq \varnothing$, set $\varphi \equiv -1$. If $A \neq \varnothing$ and $B = \varnothing$, set $\varphi \equiv 1$. Otherwise (when both A and B are nonempty) define

$$\varphi(x) = \frac{-\mathrm{dist}(x,A) + \mathrm{dist}(x,B)}{\mathrm{dist}(x,A) + \mathrm{dist}(x,B)}.$$

In the last case the denominator is never zero because $A \cap B = \varnothing$. For all $x \in \mathbb{R}^n$, $|\varphi(x)| \leqslant 1$. Let

$$g_1(x) = \frac{M}{3}\varphi(x),$$

then $|g_1(x)| \leqslant M/3$ for all $x \in \mathbb{R}^n$. Also

$$|f(x) - g_1(x)| \leqslant \frac{2}{3}M, \forall x \in F = A \cup B \cup C.$$

Let $f_1(x) = f(x) - g_1(x)$. Applying the same argument to f_1, there exists a continuous function g_2 on $\mathbb{R}^n$ (constructed via distance functions) such that

$$|g_2(x)| \leqslant \frac{1}{3}(\frac{2}{3}M),$$

$$|f_1(x) - g_2(x)| = \Big|f(x) - \sum_{i=1}^{2} g_i(x)\Big| \leqslant (\frac{2}{3})^2 M, \forall x \in F.$$

Continuing inductively, there exist

$$G_n(x) = \sum_{i=1}^{n} g_i(x),$$

with

$$|g_n(x)| \leqslant \frac{1}{3}(\frac{2}{3})^{n-1} M, \forall x \in \mathbb{R}^n,$$

$$|f(x) - G_n(x)| \leqslant (\frac{2}{3})^n M, \forall x \in F.$$

Now prove $\{G_n\}$ converges uniformly on $\mathbb{R}^n$:

$$\begin{aligned} |G_{n+p}(x) - G_n(x)| = \Big|\sum_{i=n+1}^{n+p} g_i(x)\Big| &\leqslant \sum_{i=n+1}^{n+p} |g_i(x)| \\ &\leqslant \sum_{i=n+1}^{\infty} \frac{1}{3}(\frac{2}{3})^{i-1} M \\ &\leqslant \frac{1}{3}(\frac{2}{3})^n M \sum_{k=0}^{\infty} (\frac{2}{3})^k. \end{aligned}$$

Hence $\{G_n\}$ converges uniformly on $\mathbb{R}^n$. So

$$G(x) = \lim_{n\to\infty} G_n(x) = \sum_{n=1}^{\infty} g_n(x)$$

is continuous on $\mathbb{R}^n$, and

$$|G(x)| = \Big|\sum_{n=1}^{\infty} g_n(x)\Big| \leqslant M.$$

For $x \in F$, we have $f(x) \equiv G(x)$. □

Remark 1.3.6. *In the extension theorem above, we essentially used a special case of Urysohn's lemma in $\mathbb{R}^n$. Urysohn's lemma is one of the deeper theorems in topology, not easy to prove in full generality. But in $\mathbb{R}^n$ (more generally, metric spaces), the distance function makes the proof straightforward.*

Theorem 1.3.11 (Urysohn Lemma)**.** *Two closed sets in a normal space can be separated by a continuous function.*

Theorem 1.3.12 (Urysohn Lemma in $\mathbb{R}^n$)**.** *Let $E, F \subset \mathbb{R}^n$ be disjoint closed sets. Then there exists $f \in C(\mathbb{R}^n)$ such that $0 \leqslant f \leqslant 1$, with $E \subset f^{-1}(\{1\})$ and $F \subset f^{-1}(\{0\})$.*

Proof. Construct directly:

$$f(x) = \frac{\mathrm{d}(x, F)}{\mathrm{d}(x, E) + \mathrm{d}(x, F)}.$$

□

Another frequently used form is:

Theorem 1.3.13 (Urysohn')**.** *Let K be compact and V open with $K \subset V \subset \mathbb{R}^n$. Then there exists a continuous function f with compact support[7] such that $0 \leqslant f \leqslant 1$, $\mathrm{Supp}(f) \subset V$, and $f|_K = 1$. Equivalently, $\chi_K \leqslant f \leqslant \chi_V$.*

Think: (Partition of unity) Let $K \subset \mathbb{R}^n$ be compact and $\{V_i\}_{i=1}^k$ an open cover of K. Then there exists a family of continuous compactly supported functions $\{h_i\}_{i=1}^k$ such that $0 \leqslant h_i \leqslant 1$, $\mathrm{Supp}(h_i) \subset V_i$, and

$$\sum_{i=1}^{k} h_i(x) = 1, x \in K.$$

Finally, we briefly introduce semicontinuity.

Definition 1.3.14 (Semicontinuity)**.** *A function f is upper semicontinuous if for all $\varepsilon > 0$, there exists $\delta > 0$ such that for all $y \in \mathrm{B}(x, \delta)$,*

$$f(y) < f(x) + \varepsilon.$$

A function f is lower semicontinuous if for all $\varepsilon > 0$, there exists $\delta > 0$ such that for all $y \in \mathrm{B}(x, \delta)$,

$$f(x) - \varepsilon < f(y).$$

Remark 1.3.7. *f is upper semicontinuous iff $\{x|f(x) < \lambda\}$ is open for every $\lambda \in \mathbb{R}^1$. f is lower semicontinuous iff $\{x|f(x) > \lambda\}$ is open for every $\lambda \in \mathbb{R}^1$.*

Proposition 1.3.2. *Semicontinuous functions satisfy:*

(1) *If $\{f_\lambda\}_{\lambda\in\Lambda}$ is a family of lower semicontinuous functions and*

$$(\sup_{\lambda\in\Lambda} f_\lambda)(u) = \sup_{\lambda\in\Lambda} \{f_\lambda(u)\},$$

then $\sup_{\lambda\in\Lambda} f_\lambda$ is lower semicontinuous;

(2) *finite sums of lower semicontinuous functions are lower semicontinuous;*

(3) *lower semicontinuous functions attain their minimum on compact sets;*

(4) *if $F \subset \mathbb{R}^1$ is open and $G \subset \mathbb{R}^1$ is closed, then χ_F is lower semicontinuous and χ_G is upper semicontinuous;*

(5) *if f_1 is upper semicontinuous, f_2 is lower semicontinuous, and $f_1 \leqslant f_2$, then there exists a continuous f such that $f_1 \leqslant f \leqslant f_2$.*

[7] The support of f is

$$E = \overline{\{x|f(x) \neq 0\}},$$

denoted by $\mathrm{Supp}(f)$. By definition the support is closed; if also bounded, it is compact support.

Proof. We only prove (3). Let K be compact and f lower semicontinuous, and set

$$\lambda = \inf_{x\in K} f(x).$$

Choose a sequence $\{x_n\} \subset K$ such that $f(x_n) \leqslant \lambda + \frac{1}{n}$. By compactness, there exists a subsequence $x_{n_k} \to x_0 \in K$. Lower semicontinuity is equivalent to the sequential inequality

$$f(x_0) \leqslant \liminf_{k\to\infty} f(x_{n_k}),$$

hence

$$f(x_0) \leqslant \liminf_{k\to\infty} \left(\lambda + \frac{1}{n_k}\right) = \lambda.$$

Since $\lambda \leqslant f(x_0)$ by definition of λ, we conclude $f(x_0) = \lambda$. □

Exercises 1.3

1. Let $X = \{G \subset \mathbb{R}^n | G \text{ open}\}$. Find $\overline{\overline{X}}$.
2. P36 1,2,3,4,5,7
3. P40 2,5
4. P62 1,2,3,4[8],5,6
5. P67 40,41
6. P69 28

1.3.5 Cantor Set

We now recursively define the middle-third Cantor set.

At step 1, set $I_0 = (0,1)$, remove the middle closed interval of length 1/3, namely $I^{1,1} = [\frac{1}{3}, \frac{2}{3}]$, and obtain $I_1 = I_0 \backslash I^{1,1}$, which has two path-connected components $I_{1,1} = (a_{1,1}, b_{1,1})$ and $I_{1,2} = (a_{1,2}, b_{1,2})$. Write $I^1 = I^{1,1}$.

At step 2, for each $I_{1,j}$, $j \in \{1,2\}$, remove the middle closed third,

$$I^{2,j} = [a_{1,j} + \frac{1}{3}(b_{1,j} - a_{1,j}), a_{1,j} + \frac{2}{3}(b_{1,j} - a_{1,j})],$$

and write $I^2 = \bigcup_{j=1}^{2} I^{2,j}$. The remaining set is $I_2 = I_1 \backslash I^2$, with four path-connected components $I_{2,j} = (a_{2,j}, b_{2,j})$, $j \in \{1,2,3,4\}$.

Continue inductively. Suppose step n has been completed, yielding I_n with 2^n path-connected components $I_{n,j} = (a_{n,j}, b_{n,j})$, $j \in \{1, 2, \cdots, 2^n\}$. At step $n+1$, for each $I_{n,j}$ remove the middle closed third,

$$I^{n+1,j} = [a_{n,j} + \frac{1}{3}(b_{n,j} - a_{n,j}), a_{n,j} + \frac{2}{3}(b_{n,j} - a_{n,j})],$$

[8]This problem seems misstated; F should likely be replaced by a bounded closed subset of G.

set $I^{n+1} = \bigcup_{j=1}^{2^n} I^{n+1,j}$, and let $I_{n+1} = I_n \backslash I^{n+1}$. Then I_{n+1} has 2^{n+1} path-connected components $I_{n+1,j} = (a_{n+1,j}, b_{n+1,j})$, $j \in \{1, 2, \cdots, 2^{n+1}\}$.

Thus I_k and I^k are defined recursively, and the Cantor middle-third set is

$$\mathcal{C} = \Big(\bigcap_{k=1}^{\infty} I_k \Big) \cup \{0, 1\} = [0, 1] \backslash \Big(\bigcup_{k=1}^{\infty} I^k \Big).$$

The Cantor set $\mathcal{C}$ has the following properties.

Proposition 1.3.3. *Let $\mathcal{C}$ be the Cantor middle-third set. Then*

(1) *$\mathcal{C}$ is closed;*
(2) $\sum_{k=1}^{\infty} |I^k| = \sum_{k=1}^{\infty} \frac{2^{k-1}}{3^k} = 1$;
(3) $\overline{\overline{\mathcal{C}}} = \aleph$;
(4) $\overset{\circ}{\mathcal{C}} = \varnothing$;
(5) $\mathcal{C} \subset \mathcal{C}' (\Rightarrow \mathcal{C} = \mathcal{C}')$.

The general notion is the Cantor set, namely a nowhere dense perfect set. We first define perfect sets.

Definition 1.3.15 (Perfect Set)**.** *A set E is called perfect if*

$$E = E'.$$

To aid understanding, we give a characterization in one dimension.

Proposition 1.3.4. *$E \subset \mathbb{R}^1$ is perfect iff E^c is open and*

$$E^c = \bigcup_{n=1}^{N} (a_n, b_n),$$

where N may be ∞, and for $n \neq m$, (a_m, b_m) and (a_n, b_n) share no endpoints.

Proof. On one hand, since E is perfect, i.e. $E = E'$, we have $\overline{E} = E \cup E' = E$, so E is closed and E^c is open. By Theorem 1.3.4, E^c is a countable union of disjoint connected open sets, and in $\mathbb{R}^1$ connected open sets are open intervals. Also, perfectness implies no isolated points in E, equivalently component intervals of E^c cannot share endpoints.

On the other hand,

$$E^c \text{ open} \Rightarrow E \text{ closed} \Rightarrow E' \subset E,$$

$$E \text{ has no isolated points} \Rightarrow E \subset E',$$

thus the claim holds. □

Next define nowhere dense sets.

Definition 1.3.16 (Nowhere Dense Set)**.** *Let $A, B \subset \mathbb{R}^n$. If $\overline{A} \supseteq B$, then A is dense in B. If $\overset{\circ}{\overline{A}} = \varnothing$, equivalently for any $x_0 \in \mathbb{R}^n$ and any $\varepsilon > 0$, there exist $x \in B_\varepsilon(x_0)$ and $\delta > 0$ such that $B_\delta(x) \subseteq B_\varepsilon(x_0)$ and*

$$B_\delta(x) \cap A = \varnothing,$$

then A is called nowhere dense (also sparse).

Further, a countable union of nowhere dense sets is called a first category set; sets that are not first category are second category. These notions depend on the ambient space.

Example 1.3.3. *The x-axis is of second category in $\mathbb{R}^1$, but of first category in $\mathbb{R}^2$.*

Now we characterize one-dimensional nowhere dense perfect sets.

Let $E \in \mathbb{R}^n$ be a bounded nowhere dense perfect set, and define

$$m = \inf\{x | x \in E\}, M = \sup\{x | x \in E\}.$$

Since E is closed, $m \in E$ and $M \in E$. By Proposition 1.3.4,

$$E^c = (-\infty, m) \cup (M, \infty) \bigcup_{n=1}^{\infty} (a_n, b_n],$$

and for $n \neq m$, intervals (a_m, b_m) and (a_n, b_n) share no endpoints. Assume $b_n < a_m$. Since E is nowhere dense, there are points of E^c in b_n, a_m. Hence there exists an open interval $(c, d) \subset E^c$, which is another component interval lying between (a_m, b_m) and (a_n, b_n).

This shows that between any two component intervals of E^c, there is another one, and the three have no common endpoints.

The following proposition reveals the structure of complements of one-dimensional nowhere dense perfect sets and also gives the famous Cantor function.

Proposition 1.3.5. *Let E be as above. Then there exists $\varphi : [m, M] \to [m, M]$ with:*

(1) *φ is monotone, continuous, and surjective;*

(2) *$\varphi|_{(a_n, b_n)} \equiv const.$*

Moreover, one can reorder the component intervals of E^c so that their order resembles that of the complement of the Cantor set.

Proof. Step 1. Reorder component intervals of E^c analogously to the Cantor complement order. By Proposition 1.3.4,

$$E^c = (-\infty, m) \cup (M, \infty) \bigcup_{n=1}^{\infty} (a_n, b_n],$$

set $(c_{1,1}, d_{1,1}) = (a_1, b_1)$. Choose the first interval in original order between $(-\infty, m)$ and (a_1, b_1), denote it by $(c_{2,1}, d_{2,1})$, i.e.

$$n_{2,1} = \inf\{n | a_n > m, b_n < a_1\},$$

and set $(c_{2,1}, d_{2,1}) = (a_{n_{2,1}}, b_{n_{2,1}})$. Similarly choose the first interval between (a_1, b_1) and $(M, +\infty)$, denoted $(c_{2,2}, d_{2,2})$, i.e.

$$n_{2,2} = \inf\{n | a_n > b_1, b_n < M\},$$

and set $(c_{2,2}, d_{2,2}) = (a_{n_{2,2}}, b_{n_{2,2}})$. Suppose after step k we have chosen

$$O_1 = (c_{1,1}, d_{1,1}),$$

$$O_2 = (c_{2,1}, d_{2,1}) \cup (c_{2,2}, d_{2,2}),$$

$$\dots$$

$$O_k = \bigcup_{j=1}^{2^{k-1}} (c_{k,j}, d_{k,j}).$$

Now reorder component intervals in $\bigcup_{i=1}^{k} O_i$ by position; between each adjacent pair choose one component interval of E^c with minimal original index. This gives 2^k intervals

$$(c_{k+1,j}, d_{k+1,j}), j = 1, 2, \cdots, 2^k,$$

and define

$$O_{k+1} = \bigcup_{j=1}^{2^k} (c_{k+1,j}, d_{k+1,j}), k = 0, 1, 2, \cdots.$$

We claim $E^c = (-\infty, m) \cup (M, +\infty) \bigcup_{i=1}^{\infty} O_i$. Clearly $\bigcup_{i=1}^{k} O_i \subset E^c$. Conversely, every (a_n, b_n) is eventually chosen: if $(a_1, b_1), \cdots, (a_n, b_n)$ are chosen by step k but (a_{n+1}, b_{n+1}) is not, then it lies between two component intervals of $\bigcup_{i=1}^{k} O_i$, so it must be chosen at step $k+1$ by construction.

Step 2. Construct $\varphi : [m, M] \to [m, M]$. Let $O_0 = \{m, M\}$ and define on O_0:

$$\varphi_0(x) = \begin{cases} m & x = m \\ M & x = M \end{cases},$$

so jump size between adjacent components is $\frac{M-m}{2^0}$. On $O_0 \cup O_1$, define

$$\varphi_1(x) = \begin{cases} \frac{m+M}{2}, & x \in (c_{1,1}, d_{1,1}) \\ \varphi_0(x) & x \in O_0 \end{cases},$$

then jump size is $\frac{M-m}{2^1}$. On $O_0 \cup O_1 \cup O_2$, define

$$\varphi_2(x) = \begin{cases} \frac{m+\varphi_1|_{(c_{1,1}, d_{1,1})}}{2} & x \in (c_{2,1}, d_{2,1}) \\ \frac{M+\frac{m+M}{2}}{2} & x \in (c_{2,2}, d_{2,2}) \\ \varphi_1(x) & x \in O_1 \end{cases},$$

then jump size is $\frac{M-m}{2^2}$. Continuing this process, define φ_{k+1} from φ_k. On each $(c_{k+1,j}, d_{k+1,j})$, set the value to the average of neighboring component values. Then the jump size between adjacent components is

$$\frac{M-m}{2^k}.$$

Proceeding inductively yields a monotone increasing function ψ on $\{m, M\} \cup \left(\bigcup_{i=1}^{\infty} O_i\right)$, constant on each $(c_{k,j}, d_{k,j})$, and with jump size $\frac{M-m}{2^k}$ between adjacent components of $\bigcup_{i=1}^{k} O_i$.

Now define $\varphi|_{O_n} = \psi|_{O_n}$ and

$$\varphi(x) = \sup\Big\{\psi(y) | y \leqslant x, y \in \{m, M\} \cup \Big(\bigcup_{i=1}^{\infty} O_i\Big)\Big\},$$

so φ is monotone increasing.

Step 3. Prove continuity of φ. For any $\varepsilon > 0$ and fixed $x \in [m, M]$, choose k large so that $\frac{M-m}{2^k} < \varepsilon$. If $x \in \bigcup_{i=1}^{k} O_i$, continuity at x is clear. If $x \in [m, M] \backslash \bigcup_{i=1}^{k} O_i$, then x lies between two neighboring components of $\bigcup_{i=1}^{k} O_i$, say left $(c_{i,j}, d_{i,j})$ and right $(c'_{i,j}, d'_{i,j})$. Let $\delta = \max\{x - c_{i,j}, d'_{i,j} - x\}$. For $y \in (x - \delta, x + \delta)$,

$$|\varphi(y) - \varphi(x)| \leqslant \psi\Big(\frac{c'_{i,j} + d'_{i,j}}{2}\Big) - \psi\Big(\frac{c_{i,j} + d_{i,j}}{2}\Big) < \frac{M-m}{2^k} < \varepsilon.$$

Hence φ is continuous. □

Remark 1.3.8. *This proposition shows that, up to ordering of component intervals, complements of one-dimensional nowhere dense perfect sets have the same structure as the complement of the Cantor set.*

1.4 Borel Sets and the Category Theorem

We now introduce Borel sets, which are fundamental in real analysis and probability theory. First we define the σ-ring mentioned in the overview.

Definition 1.4.1 (σ-Ring)**.** *Let X be nonempty and $\Gamma \subset \mathcal{P}(X)$. We call Γ a σ-ring on X if*

(1) *for all $A, B \in \Gamma$, $A \backslash B \in \Gamma$;*

(2) *for $\{A_k\}_{k=1}^{\infty} \subset \Gamma$, $\bigcup\limits_{k=1}^{\infty} A_k \in \Gamma$.*

Remark 1.4.1. *1. From the definition,*

$$A \cap B = (A \cup B) \backslash \big((A \backslash B) \cup (B \backslash A)\big) \in \Gamma.$$

Also from $A_1 \backslash \Big(\bigcap\limits_{k=1}^{\infty} A_k \Big) = \bigcup\limits_{k=1}^{\infty} (A_1 \backslash A_k) \in \Gamma$, we get

$$\bigcap_{k=1}^{\infty} A_k = A_1 \backslash \Big(A_1 \backslash (\bigcap_{k=1}^{\infty} A_k) \Big) \in \Gamma.$$

2. The intersection of any family of σ-rings is still a σ-ring.

Definition 1.4.2 (Generated σ-Ring)**.** *Let X be nonempty and $\Sigma \subset \mathcal{P}(X)$. Define*

$$U = \{\Gamma' \subseteq \mathcal{P}(X) \mid \Sigma \subseteq \Gamma' \text{ and } \Gamma' \text{ is a } \sigma\text{-ring}\}.$$

Then

$$\Gamma(\Sigma) = \bigcap_{\Gamma' \in U} \Gamma'$$

is called the σ-ring generated by Σ.

Remark 1.4.2. *In fact, $\Gamma(\Sigma)$ is the smallest σ-ring containing Σ.*

Now we introduce an important generated σ-ring: the Borel ring.

Definition 1.4.3 (Borel Ring)**.** *Let $\mathcal{B}(\mathbb{R}^n)$ be the σ-ring generated by all open and closed subsets of $\mathbb{R}^n$. This is called the Borel ring on $\mathbb{R}^n$.*

Remark 1.4.3. *1. Clearly $\mathcal{B}(\mathbb{R}^n) \subset \mathcal{P}(\mathbb{R}^n)$, so $\overline{\overline{\mathcal{B}(\mathbb{R}^n)}} \leqslant 2^{\aleph}$. Is $\overline{\overline{\mathcal{B}(\mathbb{R}^n)}} = \aleph$? Intuitively yes, but the proof is nontrivial and requires transfinite induction, which we omit.*

2. From the definition, $(a, b] = \bigcap_{n=1}^{\infty} (a, b + 1/n)$ and $\mathbb{Q} = \bigcup_{n=1}^{\infty} \{r_n\}$ are both in $\mathcal{B}(\mathbb{R}^n)$.

We can further classify elements of $\mathcal{B}(\mathbb{R}^n)$; the simplest are G_δ and F_σ sets.

Definition 1.4.4 (G_δ, F_σ Sets)**.** *$A \subset \mathbb{R}^n$ is called a G_δ set if there exists a sequence of open sets $\{G_k\}_{k=1}^{\infty}$ such that*

$$A = \bigcap_{k=1}^{\infty} G_k;$$

$A \subset \mathbb{R}^n$ is called an F_σ set if there exists a sequence of closed sets $\{F_k\}_{k=1}^{\infty}$ such that

$$A = \bigcup_{k=1}^{\infty} F_k.$$

Remark 1.4.4. *A G_δ set can be small: for example, a singleton $\{x\}$ is a G_δ set in $\mathbb{R}^n$. What makes G_δ sets important in Baire category theory is that in complete metric spaces, countable intersections of* dense open *sets remain dense (Baire category theorem).*

Let us look at examples.

Example 1.4.1. *Any closed set $F \subset \mathbb{R}^n$ is a G_δ set. Indeed, define*

$$G_n = \{x \in \mathbb{R}^n | \text{dist}(x, F) < \frac{1}{n}\},$$

then

$$F = \bigcap_{n=1}^{\infty} G_n.$$

Example 1.4.2. *$\mathbb{R}\backslash\mathbb{Q}$ is not an F_σ set, and $\mathbb{Q}$ is not a G_δ set.*

To prove Example 1.4.2, we need the famous Baire category theorem.

Theorem 1.4.1 (Baire Category Theorem)**.** *$\mathbb{R}^n$ cannot be expressed as a countable union of nowhere dense sets*[9]*.*

Proof. Assume $\mathbb{R}^n = \bigcup_{n=1}^{\infty} A_n$, where each A_n is nowhere dense. Choose $x_0 \in \mathbb{R}^n$, let $B_1 = \overline{B(x_0, 1)}$. Since A_1 is nowhere dense, it is not dense in $B(x_0, 1)$, so there exist $x_1 \in B(x_0, 1)\backslash\overline{A_1}$ and $\delta_1 (< 1/2)$ with

$$\overline{B(x_1, \delta_1)} \cap \overline{A_1} = \varnothing, \overline{B(x_1, \delta_1)} \subset B_1.$$

Let $B_2 = \overline{B(x_1, \delta_1)}$. Since A_2 is nowhere dense, there exist $x_2 \in B(x_1, \delta_1)\backslash\overline{A_2}$ and $\delta_2 (< 1/3)$ such that

$$\overline{B(x_2, \delta_2)} \cap \overline{A_2} = \varnothing, \overline{B(x_2, \delta_2)} \subset B_2.$$

Continue inductively to obtain $\{x_n\} \subset \mathbb{R}^n$ and $\delta_n (< 1/n)$ such that

$$\overline{B(x_{k+1}, \delta_{k+1})} \cap \overline{\bigcup_{i=1}^{k} A_i} = \varnothing,$$

$$\overline{B(x_{k+1}, \delta_{k+1})} \subset \overline{B(x_k, \delta_k)} \subset \cdots \subset \overline{B(x_0, 1)}.$$

Since these are closed balls and $\delta_k \to 0$, $\{x_n\}$ is Cauchy. Let $x^* \in \mathbb{R}^n$ be its limit. Then

$$x^* \in \bigcap_{k=1}^{\infty} \overline{B(x_k, \delta_k)}. \tag{1.1}$$

By assumption $\mathbb{R}^n = \bigcup_{n=1}^{\infty} A_n$, so $x^* \in A_{k_0}$ for some k_0. From (1.1),

$$x^* \in \overline{B(x_{k_0+1}, \delta_{k_0+1})},$$

contradicting $\overline{B(x_{k_0+1}, \delta_{k_0+1})} \cap A_{k_0} = \varnothing$. □

[9]Equivalently, $\mathbb{R}^n$ is of second category. More generally: a nonempty complete metric space cannot be written as a countable union of nowhere dense sets.

Now reconsider Example 1.4.2. Suppose $\mathbb{R}\backslash\mathbb{Q} = \bigcup_{n=1}^{\infty} F_n$ with each F_n closed. Then

$\mathbb{Q} \cap F_n = \varnothing \Rightarrow F_n$ is nowhere dense

$$\Rightarrow \mathbb{R} = \Big(\bigcup_{n=1}^{\infty} F_n \Big) \cup \Big(\bigcup_{r \in \mathbb{Q}} \{r\} \Big)$$

$\Rightarrow \mathbb{R}$ can be written as a union of countably many nowhere dense closed sets.

This contradicts Theorem 1.4.1.

Exercises 1.4

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Chapter 2

Lebesgue Measurable Sets

2.1 Measures on Rings

2.1.1 The Ring $\mathcal{R}_0^n$

Definition 2.1.1. *Let $\mathcal{R}^n$ be a family of subsets of $\mathbb{R}^n$. If whenever $A, B \in \mathcal{R}^n$ we have $A \cup B, A \backslash B \in \mathcal{R}^n$, then $\mathcal{R}^n$ is called a ring on $\mathbb{R}^n$. In particular, if $\mathbb{R}^n \in \mathcal{R}^n$, then $\mathcal{R}^n$ is called an algebra on $\mathbb{R}^n$.*

Remark 2.1.1. *Note the difference from the σ-ring defined in Section 1.4.*

Now let us examine a concrete example. The ring in this example and its higher-dimensional generalization are crucial in what follows.

Example 2.1.1. *Let $\mathcal{R}_0^1$ be the family of all finite unions of bounded left-open right-closed intervals in $\mathbb{R}$:*

$$\mathcal{R}_0^1 = \Big\{ \bigcup_{k=1}^{m} (a_k, b_k] \mid a_k < b_k,\ m \in \mathbb{N} \Big\}.$$

Then $\mathcal{R}_0^1$ is a ring on $\mathbb{R}$.

Proof. Closure under finite unions is immediate.

For set difference, it suffices to consider the difference of two half-open intervals. Let $(a, b]$ and $(c, d]$ be bounded left-open right-closed intervals. Then

$$(a,b] \setminus (c,d] = \begin{cases} (a,b], & d \le a \text{ or } b \le c, \\ (a,c], & a < c < b \le d, \\ (d,b], & c \le a < d < b, \\ (a,c] \cup (d,b], & a < c \le d < b, \\ \varnothing, & c \le a < b \le d. \end{cases}$$

In particular, $(a, b] \setminus (c, d]$ is always a finite union of bounded half-open intervals of the same type.

Now let

$$A = \bigcup_{k=1}^{m} (a_k, b_k], \qquad B = \bigcup_{j=1}^{l} (c_j, d_j].$$

Using distributivity of set difference over finite unions and the closure of this class under finite intersections,

$$A \setminus B = \bigcup_{k=1}^{m} \Big((a_k, b_k] \setminus B \Big), \qquad (a_k, b_k] \setminus B = \bigcap_{j=1}^{l} \Big((a_k, b_k] \setminus (c_j, d_j] \Big),$$

and each intermediate set is a finite union of bounded half-open intervals. Hence $A \setminus B \in \mathcal{R}_0^1$. Therefore $\mathcal{R}_0^1$ is closed under union and difference, so it is a ring. □

We now give a canonical decomposition for elements in $\mathcal{R}_0^1$.

Lemma 2.1.1 (Canonical Decomposition)**.** *If $A \in \mathcal{R}_0^1$, then A can be written as a finite union of pairwise disjoint bounded left-open right-closed intervals.*

Proof. If $A = \varnothing$ there is nothing to prove. Otherwise write

$$A = \bigcup_{k=1}^{m} (a_k, b_k],$$

and let $T = \{a_k, b_k : 1 \le k \le m\}$. List the distinct points of T in increasing order:

$$t_0 < t_1 < \cdots < t_N.$$

For each $i = 1, \ldots, N$, set $J_i = (t_{i-1}, t_i]$. The intervals J_i are pairwise disjoint.

We claim that for each i, either $J_i \subset A$ or $J_i \cap A = \varnothing$. Indeed, no endpoint of any generating interval lies in (t_{i-1}, t_i), so membership in A cannot change inside J_i.

Consequently,

$$A = \bigcup_{\{i : J_i \subset A\}} J_i,$$

which is a finite union of pairwise disjoint half-open intervals. □

Next we generalize Example 2.1.1 to higher dimensions. We first need a lemma: the Cartesian product of rings is still a ring.

Lemma 2.1.2 (Product of Rings)**.** *Let $\mathcal{R}^n$ and $\mathcal{R}^m$ be rings of subsets of $\mathbb{R}^n$ and $\mathbb{R}^m$, respectively. Define*

$$\mathcal{R}^n \times \mathcal{R}^m = \Big\{ \bigcup_{k=1}^{r} (A_k \times B_k) \mid A_k \in \mathcal{R}^n,\ B_k \in \mathcal{R}^m,\ r \in \mathbb{N} \Big\}.$$

Then $\mathcal{R}^n \times \mathcal{R}^m$ is a ring on $\mathbb{R}^{n+m}$.

Proof. Closure under finite unions is immediate.

For set difference, it suffices to handle a single rectangle. Let $A, C \in \mathcal{R}^n$ and $B, D \in \mathcal{R}^m$. Then

$$(A \times B) \setminus (C \times D) = x\big((A \setminus C) \times B\big) \ \cup \ \big((A \cap C) \times (B \setminus D)\big).$$

Since rings are closed under difference and (finite) intersection, both factors lie in $\mathcal{R}^n$ and $\mathcal{R}^m$, so the right-hand side belongs to $\mathcal{R}^n \times \mathcal{R}^m$.

Now for general

$$E = \bigcup_{k=1}^{r} E_k, \qquad F = \bigcup_{j=1}^{s} F_j,$$

with each E_k, F_j a rectangle, we have

$$E \setminus F = \bigcup_{k=1}^{r} (E_k \setminus F), \qquad E_k \setminus F = \bigcap_{j=1}^{s} (E_k \setminus F_j),$$

and each intermediate set is a finite union of rectangles of the product form. Hence $E \setminus F \in \mathcal{R}^n \times \mathcal{R}^m$, as required. □

Now define the higher-dimensional analog.

Definition 2.1.2 (The Ring $\mathcal{R}_0^n$)**.** *Let $\mathcal{R}_0^n$ be the family of all finite unions of bounded left-open right-closed boxes in $\mathbb{R}^n$:*

$$\mathcal{R}_0^n = \Big\{ \bigcup_{k=1}^{l} I_k \mid I_k \text{ is a bounded left-open right-closed box in } \mathbb{R}^n \Big\},$$

where

$$I_k = \{(x_1, \ldots, x_n) \in \mathbb{R}^n \mid a_i < x_i \leq b_i,\ i = 1, \ldots, n\}.$$

Theorem 2.1.1. *With the definition above, $\mathcal{R}_0^n$ is a ring on $\mathbb{R}^n$, and*

$$\mathcal{R}_0^n = \underbrace{\mathcal{R}_0^1 \times \mathcal{R}_0^1 \times \cdots \times \mathcal{R}_0^1}_{n \text{ factors}}.$$

Proof. Each bounded left-open right-closed box in $\mathbb{R}^n$ has the form

$$(a_1, b_1] \times \cdots \times (a_n, b_n],$$

so it is a member of the n-fold product $\mathcal{R}_0^1 \times \cdots \times \mathcal{R}_0^1$. Therefore $\mathcal{R}_0^n \subset \mathcal{R}_0^1 \times \cdots \times \mathcal{R}_0^1$.

Conversely, any element of the product ring is a finite union of sets of the form $E_1 \times \cdots \times E_n$ with $E_i \in \mathcal{R}_0^1$. By Lemma 2.1.1, each E_i is a disjoint finite union of half-open intervals. Expanding the Cartesian product distributes over finite unions, so $E_1 \times \cdots \times E_n$ is a finite union of bounded half-open boxes, hence belongs to $\mathcal{R}_0^n$. Taking a finite union over the generating rectangles shows $\mathcal{R}_0^1 \times \cdots \times \mathcal{R}_0^1 \subset \mathcal{R}_0^n$.

The equality follows. Finally, since the n-fold product is a ring by repeated application of Lemma 2.1.2, $\mathcal{R}_0^n$ is a ring. □

The next lemma is the higher-dimensional version of Lemma 2.1.1.

Lemma 2.1.3 (Canonical Decomposition)**.** *If $A \in \mathcal{R}_0^n$, then A can be written as a finite union of pairwise disjoint left-open right-closed boxes.*

Proof. Write A as a finite union of boxes,

$$A = \bigcup_{k=1}^{m} I_k, \qquad I_k = \prod_{j=1}^{n} (a_{k,j}, b_{k,j}].$$

For each coordinate $j = 1, \ldots, n$, let

$$T_j = \{a_{k,j}, b_{k,j} : 1 \leq k \leq m\},$$

and list the distinct points of T_j in increasing order:

$$t_{j,0} < t_{j,1} < \cdots < t_{j,N_j}.$$

For each multi-index $\alpha = (\alpha_1, \ldots, \alpha_n)$ with $1 \le \alpha_j \le N_j$, define the elementary box

$$Q_\alpha = \prod_{j=1}^{n} (t_{j,\alpha_j - 1}, t_{j,\alpha_j}].$$

The family $\{Q_\alpha\}$ is finite and consists of pairwise disjoint boxes.

By construction, for each k and each Q_α, either $Q_\alpha \subset I_k$ or $Q_\alpha \cap I_k = \varnothing$ (since all boundary hyperplanes of I_k occur among the coordinates $t_{j,\ell}$). Hence each I_k is a union of some of the boxes Q_α, and therefore A is a union of those Q_α that lie in A:

$$A = \bigcup_{\alpha : Q_\alpha \subset A} Q_\alpha.$$

This is a finite union of pairwise disjoint boxes, as claimed. □

2.1.2 Measure on $\mathcal{R}_0^n$

Now we construct a measure on the ring $\mathcal{R}_0^n$.

Definition 2.1.3. *Let $\mathcal{R}_0^n$ be the ring of finite unions of bounded left-open right-closed boxes in $\mathbb{R}^n$. A real-valued function μ on $\mathcal{R}_0^n$ is called a measure on $\mathcal{R}_0^n$ if:*

(1) *$\mu(\varnothing) = 0$;*

(2) *(nonnegativity) $\mu(A) \ge 0$ for all $A \in \mathcal{R}_0^n$;*

(3) *(countable additivity) for any pairwise disjoint sequence $\{A_i\}_{i=1}^{\infty} \subset \mathcal{R}_0^n$, if*

$$\bigcup_{i=1}^{\infty} A_i \in \mathcal{R}_0^n, \tag{2.1}$$

then

$$\mu\Big(\bigcup_{i=1}^{\infty} A_i\Big) = \sum_{i=1}^{\infty} \mu(A_i).$$

Remark 2.1.2. *In (2.1), we require the countable union to lie in $\mathcal{R}_0^n$ because $\mathcal{R}_0^n$ is only closed under finite unions.*

Definition 2.1.4. *For $A \in \mathcal{R}_0^n$, suppose a canonical decomposition is*

$$A = \bigcup_{i=1}^{k} I_i.$$

Define

$$m_0(A) = \sum_{i=1}^{k} |I_i|,$$

where

$$I_i = \{(x_1^i, \ldots, x_n^i) \mid a_j^i < x_j^i \le b_j^i,\ j = 1, \ldots, n\},$$

$$|I_i| = \prod_{j=1}^{n} (b_j^i - a_j^i) = \int_{I_i} \chi_{I_i}(x)\, dx.$$

Clearly, $m_0(A) \ge 0$ and $m_0(\varnothing) = 0$.

Since a given $A \in \mathcal{R}_0^n$ can have multiple canonical decompositions, we must show $m_0(A)$ is independent of the decomposition.

Lemma 2.1.4. *The definition of $m_0(A)$ does not depend on the chosen canonical decomposition of A.*

Proof. Step 1. If I is a single box, then $m_0(I) = |I|$, independent of decomposition.

Suppose

$$I = \bigcup_{i=1}^{k} I_i$$

is a canonical decomposition of I. Then

$$\begin{aligned} |I| &= \int_I \chi_I(x)\,dx = \int_{\bigcup_{i=1}^k I_i} \chi_{\cup I_i}(x)\,dx \\ &= \sum_{i=1}^{k} \int_{I_i} \chi_{I_i}(x)\,dx = \sum_{i=1}^{k} |I_i|. \end{aligned}$$

Step 2. Suppose $A \in \mathcal{R}_0^n$ has two canonical decompositions

$$A = \bigcup_{i=1}^{k} I_i = \bigcup_{j=1}^{l} J_j.$$

Then

$$A = \bigcup_{i=1}^{k} \bigcup_{j=1}^{l} (I_i \cap J_j),$$

$$J_j = \bigcup_{i=1}^{k} (I_i \cap J_j), \quad I_i = \bigcup_{j=1}^{l} (I_i \cap J_j).$$

By Step 1,

$$\begin{aligned} m_0(A) &= \sum_{i=1}^{k} |I_i| = \sum_{i=1}^{k} \sum_{j=1}^{l} |I_i \cap J_j| \\ &= \sum_{j=1}^{l} \sum_{i=1}^{k} |I_i \cap J_j| = \sum_{j=1}^{l} |J_j|. \end{aligned}$$

So $m_0(A)$ is well-defined. □

Lemma 2.1.5. *The set function m_0 on $\mathcal{R}_0^n$ has:*

(1) *(finite additivity) if $A_1, \ldots, A_k \in \mathcal{R}_0^n$ are pairwise disjoint, then*

$$m_0\Big(\bigcup_{i=1}^{k} A_i\Big) = \sum_{i=1}^{k} m_0(A_i);$$

(2) *(monotonicity) if $A_1, \ldots, A_k \in \mathcal{R}_0^n$ are pairwise disjoint and $\bigcup_{i=1}^k A_i \subset A \in \mathcal{R}_0^n$, then*

$$\sum_{i=1}^{k} m_0(A_i) \le m_0(A);$$

(3) *(finite subadditivity under covers) if $A, A_1, \ldots, A_k \in \mathcal{R}_0^n$ and $A \subset \bigcup_{i=1}^k A_i$, then*

$$m_0(A) \le \sum_{i=1}^{k} m_0(A_i).$$

Proof. (1) Let

$$A_i = \bigcup_{j=1}^{m_i} I_j^i, \quad i = 1, \dots, k$$

be canonical decompositions. Then

$$\bigcup_{i=1}^{k} A_i = \bigcup_{i=1}^{k} \bigcup_{j=1}^{m_i} I_j^i$$

is a canonical decomposition, so

$$m_0\Big(\bigcup_{i=1}^{k} A_i\Big) = \sum_{i=1}^{k} \sum_{j=1}^{m_i} |I_j^i| = \sum_{i=1}^{k} m_0(A_i).$$

(2) Let

$$A_{k+1} = A \backslash \bigcup_{i=1}^{k} A_i.$$

Then $A_1, \dots, A_{k+1}$ are pairwise disjoint and

$$m_0(A) = \sum_{i=1}^{k+1} m_0(A_i) \geq \sum_{i=1}^{k} m_0(A_i).$$

(3) By (1) and (2), m_0 is monotone on $\mathcal{R}_0^n$: if $A \subset B$, then $m_0(A) \leq m_0(B)$. For $A \subset \bigcup_{i=1}^{k} A_i$, define

$$B_1 = A_1, \quad B_2 = A_2 \backslash A_1, \quad \dots, \quad B_k = A_k \backslash \Big(\bigcup_{i=1}^{k-1} A_i\Big).$$

Then $B_1, \dots, B_k$ are pairwise disjoint and $\bigcup_{i=1}^{k} A_i = \bigcup_{i=1}^{k} B_i$. Hence

$$m_0(A) \leq m_0\Big(\bigcup_{i=1}^{k} A_i\Big) = m_0\Big(\bigcup_{i=1}^{k} B_i\Big) = \sum_{i=1}^{k} m_0(B_i) \leq \sum_{i=1}^{k} m_0(A_i).$$

□

Theorem 2.1.2. *m_0 is a measure on $\mathcal{R}_0^n$.*

Proof. By Lemma 2.1.5, it remains to prove countable additivity. Let $\{A_k\}$ be pairwise disjoint in $\mathcal{R}_0^n$ and

$$A = \bigcup_{k=1}^{\infty} A_k \in \mathcal{R}_0^n.$$

By monotonicity,

$$\sum_{i=1}^{k} m_0(A_i) \leq m_0(A), \quad \forall k \geq 1.$$

Let $k \to \infty$:

$$\sum_{i=1}^{\infty} m_0(A_i) \leq m_0(A).$$

For the reverse inequality, let

$$A = \bigcup_{i=1}^{l} I_i$$

be a canonical decomposition of A, and

$$A_i = \bigcup_{j=1}^{m_i} I_j^i$$

one for A_i. Write

$$I_i = \{(x_1, \ldots, x_n) \mid a_r < x_r \leq b_r,\ r = 1, \ldots, n\}.$$

Given $\varepsilon > 0$, choose inner closed boxes

$$I_i^{\delta_i} = \{(y_1, \ldots, y_n) \mid a_r + \delta_i \leq y_r \leq b_r,\ r = 1, \ldots, n\}$$

so that

$$|I_i| \geq |I_i^{\delta_i}| \geq |I_i| - \frac{\varepsilon}{2l}.$$

Let

$$A^\delta = \bigcup_{i=1}^{l} I_i^{\delta_i},$$

so A^δ is bounded closed.

For each I_j^i, enlarge slightly to open boxes I_j^{i,δ_j^i} with

$$|I_j^{i,\delta_j^i}| - \frac{\varepsilon}{m_i 2^{i+1}} \leq |I_j^i| \leq |I_j^{i,\delta_j^i}|.$$

Then

$$\{I_j^{i,\delta_j^i} \mid i = 1, 2, \ldots;\ j = 1, 2, \ldots, m_i\}$$

is an open cover of A^δ. By compactness, there exists k_0 such that

$$A^\delta \subset \bigcup_{i=1}^{k_0} \bigcup_{j=1}^{m_i} I_j^{i,\delta_j^i}.$$

Hence

$$\begin{aligned} m_0(A) - \varepsilon &\leq \sum_{i=1}^{l} |I_i^{\delta_i}| \leq \sum_{i=1}^{k_0} \sum_{j=1}^{m_i} |I_j^{i,\delta_j^i}| \\ &\leq \sum_{i=1}^{k_0} \sum_{j=1}^{m_i} \Big(|I_j^i| + \frac{\varepsilon}{m_i 2^{i+1}} \Big) \\ &\leq \sum_{i=1}^{k_0} \Big(m_0(A_i) + \frac{\varepsilon}{2^{i+1}} \Big) \leq \sum_{i=1}^{\infty} m_0(A_i) + \varepsilon. \end{aligned}$$

By arbitrariness of ε,

$$m_0(A) \leq \sum_{i=1}^{\infty} m_0(A_i).$$

□

Corollary 2.1.1. *If $A, A_i \in \mathcal{R}_0^n$ and*

$$A \subset \bigcup_{i=1}^{\infty} A_i,$$

then

$$m_0(A) \leq \sum_{i=1}^{\infty} m_0(A_i).$$

Proof. Since

$$A = \bigcup_{i=1}^{\infty}(A_i \cap A),$$

let $B_i = A_i \cap A$ and

$$C_i = B_i \backslash \Big(\bigcup_{j=1}^{i-1} B_j \Big) \in \mathcal{R}_0^n.$$

Then $C_i \cap C_j = \varnothing$ for $i \neq j$, and

$$A = \bigcup_{i=1}^{\infty} C_i.$$

Thus

$$m_0(A) = \sum_{i=1}^{\infty} m_0(C_i) \leq \sum_{i=1}^{\infty} m_0(A_i).$$

□

Exercises 2.1

1. Construct a measure μ on a ring $\mathcal{R}_0^n$ such that

   (1) for every $r \in \mathbb{Q}$, $\mu^*(\{r\}) = 0$;

   (2) for every closed set $F \subset \mathbb{R}\backslash\mathbb{Q}$, $\mu^*(F) = 0$.

2. Let g be increasing and right-continuous on $\mathbb{R}$. For each box in $\mathbb{R}^n$

   $$I = \{(x_1, \ldots, x_n) \mid a_i \leq x_i \leq b_i,\ i = 1, \ldots, n\},$$

   define

   $$\mu(I) = \prod_{i=1}^{n} (g(b_i) - g(a_i)).$$

   Prove that μ is a measure on $\mathcal{R}_0^n$.

3. Let μ, ν be measures on $\mathcal{R}_0^n$ and $\mathcal{R}_0^m$. For $A \in \mathcal{R}_0^n$, $B \in \mathcal{R}_0^m$ with $\mu(A), \nu(B) < \infty$, define

   $$(\mu * \nu)(A \times B) = \mu(A)\nu(B).$$

   Is $\mu * \nu$ a measure on $\mathcal{R}_0^{n+m}$?

4. Let $\mathcal{S}, \mathcal{R}$ be σ-rings on X, Y. Define

   $$\mathcal{T} = \Big\{ \bigcup_{i=1}^{\infty} A_i \times B_i \mid A_i \in \mathcal{S},\ B_i \in \mathcal{R} \Big\}.$$

   Is $\mathcal{T}$ a σ-ring on $X \times Y$?

2.2 Outer Measure

Definition 2.2.1. *For $A \subset \mathbb{R}^n$, define*

$$m_0^*(A) = \inf\Big\{ \sum_{i=1}^{\infty} m_0(A_i) \mid A_i \in \mathcal{R}_0^n,\ A \subset \bigcup_{i=1}^{\infty} A_i \Big\}.$$

This is the outer measure induced by m_0.

Remark 2.2.1. *In abstract form, for a measure space $(X, \mathcal{R}, \mu)$ with $\mathcal{R}$ a ring and μ a measure on $\mathcal{R}$, define*

$$\mu^*(A) = \inf\Big\{ \sum_{i=1}^{\infty} \mu(A_i) \mid A_i \in \mathcal{R},\ A \subset \bigcup_{i=1}^{\infty} A_i \Big\},$$

called the outer measure induced by μ (see Definition 2.2.2).

Theorem 2.2.1. *The outer measure m_0^* satisfies:*

(1) $m_0^*(\varnothing) = 0$;

(2) *if $A \subset B \subset \mathbb{R}^n$, then $m_0^*(A) \le m_0^*(B)$;*

(3) *if $A \in \mathcal{R}_0^n$, then $m_0^*(A) = m_0(A)$;*

(4)
$$m_0^*\Big(\bigcup_{i=1}^{\infty} A_i \Big) \le \sum_{i=1}^{\infty} m_0^*(A_i).$$

Proof. Proofs of (1) and (2) are omitted.

(3) Set $A_1 = A$, $A_i = \varnothing$ for $i \ge 2$. Then

$$A \subset \bigcup_{i=1}^{\infty} A_i,$$

so

$$m_0^*(A) \le \sum_{i=1}^{\infty} m_0(A_i) = m_0(A).$$

Conversely, for any cover $A \subset \bigcup_{i=1}^{\infty} A_i$ with $A_i \in \mathcal{R}_0^n$,

$$m_0(A) \le \sum_{i=1}^{\infty} m_0(A_i).$$

Taking infimum gives $m_0(A) \le m_0^*(A)$.

(4) Assume without loss of generality that

$$\sum_{i=1}^{\infty} m_0^*(A_i) < \infty.$$

For any $\varepsilon > 0$, by definition of $m_0^*(A_i)$ there exist $A_i^j \in \mathcal{R}_0^n$ such that

$$A_i \subset \bigcup_{j=1}^{\infty} A_i^j, \quad \sum_{j=1}^{\infty} m_0(A_i^j) \le m_0^*(A_i) + \frac{\varepsilon}{2^i}.$$

Hence

$$A := \bigcup_{i=1}^{\infty} A_i \subset \bigcup_{i=1}^{\infty} \bigcup_{j=1}^{\infty} A_i^j,$$

and

$$m_0^*(A) \le \sum_{i=1}^{\infty}\sum_{j=1}^{\infty} m_0(A_i^j) \le \sum_{i=1}^{\infty}\Big(m_0^*(A_i) + \frac{\varepsilon}{2^i}\Big)$$
$$= \sum_{i=1}^{\infty} m_0^*(A_i) + \varepsilon.$$

Let $\varepsilon \downarrow 0$. □

Lemma 2.2.1 (Translation Invariance of m_0^*)**.** *For every $A \subset \mathbb{R}^n$ and every $x_0 \in \mathbb{R}^n$,*

$$m_0^*(A + x_0) = m_0^*(A).$$

Proof. Let $A \subset \mathbb{R}^n$ and $x_0 \in \mathbb{R}^n$. If $A \subset \bigcup_{i=1}^{\infty} A_i$ with $A_i \in \mathcal{R}_0^n$, then

$$A + x_0 \subset \bigcup_{i=1}^{\infty}(A_i + x_0).$$

Since m_0 is translation invariant on $\mathcal{R}_0^n$ (it is defined via Euclidean volume of boxes), we have $m_0(A_i + x_0) = m_0(A_i)$ for all i. Taking infima over all such covers gives $m_0^*(A + x_0) \le m_0^*(A)$. Applying the same argument to $A = (A + x_0) + (-x_0)$ yields the reverse inequality. □

Example 2.2.1. *We have:*

(1) *If $x_0 \in \mathbb{R}^n$, then $m_0^*(\{x_0\}) = 0$;*

(2) *$m_0^*(\mathbb{Q}^n) = 0$;*

(3) *If $\mathcal{C}$ is the middle-third Cantor set, then*

$$m_0^*(\mathcal{C}) \le m_0^*\Big(\bigcup_{i=1}^{2^n} I_i\Big) \le \Big(\frac{2}{3}\Big)^n \to 0 \quad (n \to \infty);$$

(4) *If I is a bounded box, then*

$$m_0^*(\overset{\circ}{I}) = m_0^*(\overline{I}).$$

The measure m_0 on $\mathcal{R}_0^n$ is countably additive, but the outer measure m_0^* is not.

Example 2.2.2 (Outer Measure Is Not Additive)**.** *The outer measure m_0^* is countably subadditive, but it is not additive on disjoint families.*

Let $\sim$ be the equivalence relation on $[0,1)$ defined by $x \sim y$ if $x - y \in \mathbb{Q}$. Choose a set $V \subset [0,1)$ containing exactly one representative from each equivalence class (axiom of choice). Enumerate $\mathbb{Q} \cap [-1,1]$ as $\{q_k\}_{k=1}^{\infty}$ and set

$$V_k = V + q_k.$$

Then $\{V_k\}_{k=1}^{\infty}$ is pairwise disjoint and

$$[0,1) \subset \bigcup_{k=1}^{\infty} V_k \subset (-1,2).$$

By Lemma 2.2.1, $m_0^*(V_k) = m_0^*(V)$ for all k.

If m_0^* were countably additive on disjoint families, we would have

$$1 = m_0^*([0,1)) \le m_0^*\Big(\bigcup_{k=1}^{\infty} V_k\Big) = \sum_{k=1}^{\infty} m_0^*(V_k) = \sum_{k=1}^{\infty} m_0^*(V).$$

The right-hand side is 0 *if* $m_0^*(V) = 0$ *and* $+\infty$ *if* $m_0^*(V) > 0$*, whereas*

$$m_0^*\Big(\bigcup_{k=1}^{\infty} V_k\Big) \leq m_0^*((-1,2)) = 3.$$

This contradiction shows that m_0^* *is not countably additive (and therefore not finitely additive either).*

Theorem 2.2.2. *For any* $A \subset \mathbb{R}^n$*, define*

$$m_{0,\delta}^*(A) = \inf\Big\{\sum_{i=1}^{\infty} m_0(A_i) \mid A_i \in \mathcal{R}_0^n,\ \operatorname{diam}(A_i) < \delta,\ A \subset \bigcup_{i=1}^{\infty} A_i\Big\}.$$

Then

$$m_{0,\delta}^*(A) = m_0^*(A).$$

Proof. Clearly, $m_0^*(A) \leq m_{0,\delta}^*(A)$.

For the opposite inequality, assume $m_0^*(A) < \infty$. Given $\varepsilon > 0$, choose $A_i \in \mathcal{R}_0^n$ such that

$$A \subset \bigcup_{i=1}^{\infty} A_i, \quad \sum_{i=1}^{\infty} m_0(A_i) < m_0^*(A) + \varepsilon.$$

Subdivide each box in each A_i into finitely many boxes of diameter $< \delta$:

$$A_i = \bigcup_{j=1}^{m_i} I_j^i = \bigcup_{j=1}^{m_i} \bigcup_{k=1}^{l_j^i} I_j^{i,k}, \quad \operatorname{diam}(I_j^{i,k}) < \delta.$$

Then

$$m_{0,\delta}^*(A) \leq \sum_{i=1}^{\infty} \sum_{j=1}^{m_i} \sum_{k=1}^{l_j^i} |I_j^{i,k}| < m_0^*(A) + \varepsilon.$$

Let $\varepsilon \downarrow 0$. □

Definition 2.2.2 (Abstract Outer Measure)**.** *Let* X *be a set,* $\mathcal{R} \subset \mathcal{P}(X)$ *a ring, and* μ *a measure on* $\mathcal{R}$*. Define*

$$\Sigma(\mathcal{R}) = \Big\{A \subset X \mid \exists A_i \in \mathcal{R},\ A \subset \bigcup_{i=1}^{\infty} A_i\Big\}.$$

For $A \in \Sigma(\mathcal{R})$*, define*

$$\mu^*(A) = \inf\Big\{\sum_{i=1}^{\infty} \mu(A_i) \mid A \subset \bigcup_{i=1}^{\infty} A_i,\ A_i \in \mathcal{R}\Big\}.$$

This is called the outer measure induced by μ*.*

Remark 2.2.2. *We call* $(X, \mathcal{R}, \mu)$ *a measure space. If additionally* $\mu(X) = 1$*, it is a probability space.*

Theorem 2.2.3. *Let* $(X, \mathcal{R}, \mu)$ *be a measure space and* μ^* *the induced outer measure. Then:*

(1) *if* $A \subset B$*, then* $\mu^*(A) \leq \mu^*(B)$*;*

(2) *if* $A \in \mathcal{R}$*, then* $\mu^*(A) = \mu(A)$*;*

(3) *(subadditivity) for* $A_i \in \Sigma(\mathcal{R})$*,*

$$\mu^*\Big(\bigcup_{i=1}^{\infty} A_i\Big) \leq \sum_{i=1}^{\infty} \mu^*(A_i).$$

Proof. We only prove (2). The inequality $\mu^*(A) \le \mu(A)$ is immediate. For the reverse, assume $\mu^*(A) < \infty$. By definition, for any $\varepsilon > 0$ there exist $A_i \in \mathcal{R}$ such that

$$A \subset \bigcup_{i=1}^{\infty} A_i, \quad \sum_{i=1}^{\infty} \mu(A_i) \le \mu^*(A) + \varepsilon.$$

We may assume $A_i \cap A_j = \varnothing$ for $i \neq j$ (by standard disjointization). Then

$$A = A \cap \Big(\bigcup_{i=1}^{\infty} A_i \Big) = \bigcup_{i=1}^{\infty} (A \cap A_i) \in \mathcal{R},$$

and

$$\mu(A) = \sum_{i=1}^{\infty} \mu(A \cap A_i) \le \sum_{i=1}^{\infty} \mu(A_i) \le \mu^*(A) + \varepsilon.$$

Let $\varepsilon \downarrow 0$. □

Remark 2.2.3. *This proof pattern has already appeared several times and is worth noting.*

2.3 Lebesgue Measure

In this section we define Lebesgue measure. We begin with Lebesgue measurable sets.

Definition 2.3.1 (Lebesgue Measurable Set)**.** *Let $\mathcal{R}_0^n, m_0, m_0^*$ be as above. A set $A \subset \mathbb{R}^n$ is called Lebesgue measurable if for every $E \subset \mathbb{R}^n$,*

$$m_0^*(E) = m_0^*(E \cap A) + m_0^*(E \cap A^c). \tag{2.2}$$

Remark 2.3.1.

1. *The collection of all Lebesgue measurable subsets of $\mathbb{R}^n$ is denoted by $\mathcal{L}^n$. For $A \in \mathcal{L}^n$ we define*

   $$m(A) := m_0^*(A)$$

   and call m the Lebesgue measure*.*

   It is often convenient to write m^* for the outer measure m_0^*; we shall use the two notations interchangeably.

2. *The identity* (2.2) *is called the* Carathéodory condition*.*

3. *If $A \in \mathcal{L}^n$ and $E \in \mathcal{L}^n$, then applying* (2.2) *to the test set E gives*

   $$m(E) = m(E \cap A) + m(E \cap A^c).$$

   Thus, once we restrict to measurable sets, the outer measure behaves additively across measurable partitions.

Now show every set in $\mathcal{R}_0^n$ is Lebesgue measurable.

Theorem 2.3.1. *If $A \in \mathcal{R}_0^n$, then $A \in \mathcal{L}^n$.*

Proof. One side is immediate from subadditivity:

$$m_0^*(E) \le m_0^*(E \cap A) + m_0^*(E \cap A^c).$$

For the reverse, assume $m_0^*(E) < \infty$. For any $\varepsilon > 0$, choose $A_i \in \mathcal{R}_0^n$ with

$$E \subset \bigcup_{i=1}^{\infty} A_i, \quad \sum_{i=1}^{\infty} m_0(A_i) \le m_0^*(E) + \varepsilon.$$

Then

$$E \cap A \subset \bigcup_{i=1}^{\infty} (A_i \cap A), \quad E \cap A^c \subset \bigcup_{i=1}^{\infty} (A_i \cap A^c),$$

and each intersection lies in $\mathcal{R}_0^n$. Hence

$$\begin{aligned} m_0^*(E \cap A) + m_0^*(E \cap A^c) &\le \sum_{i=1}^{\infty} m_0(A_i \cap A) + \sum_{i=1}^{\infty} m_0(A_i \cap A^c) \\ &= \sum_{i=1}^{\infty} m_0(A_i) \le m_0^*(E) + \varepsilon. \end{aligned}$$

Let $\varepsilon \downarrow 0$. □

Remark 2.3.2. *In later proofs of measurability, by subadditivity it is enough to prove*

$$m_0^*(E) \ge m_0^*(E \cap A) + m_0^*(E \cap A^c).$$

The following theorem implies that $\mathcal{L}^n$ is a ring.

Theorem 2.3.2. *The class $\mathcal{L}^n$ has the following closure properties.*

1. *If $m_0^*(A) = 0$, then $A \in \mathcal{L}^n$.*
2. *If $A \in \mathcal{L}^n$, then $A^c \in \mathcal{L}^n$.*
3. *If $A, B \in \mathcal{L}^n$, then $A \cap B$ and $A \cup B$ belong to $\mathcal{L}^n$.*
4. *If $A, B \in \mathcal{L}^n$ are disjoint, then $m(A \cup B) = m(A) + m(B)$.*

Proof. 1. Let $m_0^*(A) = 0$ and $E \subset \mathbb{R}^n$. Subadditivity gives

$$m_0^*(E) \le m_0^*(E \cap A) + m_0^*(E \cap A^c).$$

Conversely, monotonicity and subadditivity give

$$m_0^*(E \cap A) + m_0^*(E \cap A^c) \le m_0^*(A) + m_0^*(E) = m_0^*(E),$$

so equality holds and A satisfies the Carathéodory condition (2.2).

2. If A satisfies (2.2), then the same identity with A replaced by A^c shows that A^c is measurable.

3. Fix $A, B \in \mathcal{L}^n$ and $E \subset \mathbb{R}^n$. First apply (2.2) for A:

$$m_0^*(E) = m_0^*(E \cap A) + m_0^*(E \cap A^c).$$

Next apply (2.2) for B to the test set $E \cap A$:

$$m_0^*(E \cap A) = m_0^*(E \cap A \cap B) + m_0^*(E \cap A \cap B^c).$$

Combining these gives

$$m_0^*(E) = m_0^*(E \cap A \cap B) + \Big(m_0^*(E \cap A \cap B^c) + m_0^*(E \cap A^c)\Big).$$

The two sets $E \cap A \cap B^c$ and $E \cap A^c$ are disjoint and their union equals $E \cap (A \cap B)^c$, so by subadditivity,

$$m_0^*(E) \geq m_0^*(E \cap (A \cap B)) + m_0^*(E \cap (A \cap B)^c).$$

The opposite inequality is always true, hence $A \cap B \in \mathcal{L}^n$. Then

$$A \cup B = (A^c \cap B^c)^c,$$

and the previous item implies $A \cup B \in \mathcal{L}^n$.

4. If $A \cap B = \varnothing$, apply (2.2) with the measurable set A and the test set $E = A \cup B$:

$$m_0^*(A \cup B) = m_0^*((A \cup B) \cap A) + m_0^*((A \cup B) \cap A^c) = m_0^*(A) + m_0^*(B).$$

Since $m = m_0^*$ on $\mathcal{L}^n$, this gives $m(A \cup B) = m(A) + m(B)$.

□

In fact, $\mathcal{L}^n$ is a σ-algebra (it contains $\mathbb{R}^n$ and is closed under countable unions and complements).

Theorem 2.3.3. *Let $\mathcal{L}^n, m_0^*, m$ be as above. If $A_i \in \mathcal{L}^n$, $i = 1, 2, \ldots$, then*

$$\bigcup_{i=1}^{\infty} A_i \in \mathcal{L}^n.$$

Moreover, if $A_i \cap A_j = \varnothing$ for $i \neq j$, then

$$m\Big(\bigcup_{i=1}^{\infty} A_i\Big) = \sum_{i=1}^{\infty} m(A_i).$$

Proof. By disjointization, let

$$B_1 = A_1, \quad B_2 = A_2 \backslash A_1, \quad \ldots, \quad B_k = A_k \backslash \Big(\bigcup_{i=1}^{k-1} A_i\Big), \ \ldots$$

Then each $B_k \in \mathcal{L}^n$ (Theorem 2.3.2),

$$\bigcup_{i=1}^{k} B_i = \bigcup_{i=1}^{k} A_i \quad (k = 1, 2, \ldots, \infty),$$

and $B_i \cap B_j = \varnothing$ for $i \neq j$. So it suffices to treat disjoint $\{B_i\}$.

For any $E \subset \mathbb{R}^n$ and any $k \in \mathbb{N}$,

$$\begin{aligned} m_0^*(E) &= m_0^*\Big(E \cap \big(\bigcup_{i=1}^{k} B_i\big)\Big) + m_0^*\Big(E \cap \big(\bigcup_{i=1}^{k} B_i\big)^c\Big) \\ &= \sum_{i=1}^{k} m_0^*(E \cap B_i) + m_0^*\Big(E \cap \big(\bigcup_{i=1}^{k} B_i\big)^c\Big) \\ &\geq \sum_{i=1}^{k} m_0^*(E \cap B_i) + m_0^*\Big(E \cap \big(\bigcup_{i=1}^{\infty} B_i\big)^c\Big). \end{aligned}$$

Let $k \to \infty$:

$$\begin{aligned} m_0^*(E) &\geq \sum_{i=1}^{\infty} m_0^*(E \cap B_i) + m_0^*\Big(E \cap \big(\bigcup_{i=1}^{\infty} B_i\big)^c\Big) \\ &\geq m_0^*\Big(E \cap \big(\bigcup_{i=1}^{\infty} B_i\big)\Big) + m_0^*\Big(E \cap \big(\bigcup_{i=1}^{\infty} B_i\big)^c\Big). \end{aligned}$$

Hence $\bigcup_{i=1}^{\infty} B_i \in \mathcal{L}^n$. Taking $E = \bigcup_{i=1}^{\infty} B_i$ yields

$$m\Big(\bigcup_{i=1}^{\infty} B_i\Big) = \sum_{i=1}^{\infty} m(B_i). \qquad \square$$

The next two theorems are used frequently.

Theorem 2.3.4 (Measure Limit for Increasing Measurable Sets)**.** *Let $\{E_i\}_{i=1}^{\infty}$ be measurable with*

$$E_1 \subset E_2 \subset \cdots \subset E_n \subset \cdots.$$

Then

$$m\Big(\lim_{k\to\infty} E_k\Big) = \lim_{k\to\infty} m(E_k).$$

Proof. Assume $m(E_k) < \infty$ for all k; otherwise trivial. Define

$$B_1 = E_1, \quad B_2 = E_2 \backslash E_1, \quad \ldots, \quad B_k = E_k \backslash E_{k-1}, \ldots$$

Then $B_k \in \mathcal{L}^n$, pairwise disjoint, and

$$\bigcup_{k=1}^{\infty} B_k = \lim_{k\to\infty} E_k.$$

Hence

$$\begin{aligned} m\Big(\lim_{k\to\infty} E_k\Big) = m\Big(\bigcup_{k=1}^{\infty} B_k\Big) &= \sum_{k=1}^{\infty} (m(E_k) - m(E_{k-1})) \\ &= \lim_{k\to\infty} \sum_{i=1}^{k} (m(E_i) - m(E_{i-1})) = \lim_{k\to\infty} m(E_k), \end{aligned}$$

with $E_0 = \varnothing$. $\square$

With a finiteness condition, we get the decreasing version.

Theorem 2.3.5 (Measure Limit for Decreasing Measurable Sets)**.** *Let* $\{E_i\}_{i=1}^{\infty}$ *be measurable with*

$$E_1 \supset E_2 \supset \cdots \supset E_n \supset \cdots,$$

and

$$m(E_1) < \infty.$$

Then

$$m\Big(\lim_{k\to\infty} E_k\Big) = \lim_{k\to\infty} m(E_k).$$

Proof. Since

$$E_1\backslash E_k \subset E_1\backslash E_{k+1}, \quad k = 1, 2, \ldots,$$

$\{E_1\backslash E_k\}$ is increasing. By Theorem 2.3.4,

$$m\Big(E_1\backslash \lim_{k\to\infty} E_k\Big) = m\Big(\lim_{k\to\infty} (E_1\backslash E_k)\Big) = \lim_{k\to\infty} m(E_1\backslash E_k).$$

Using $m(E_1) < \infty$,

$$m(E_1) - m\Big(\lim_{k\to\infty} E_k\Big) = m(E_1) - \lim_{k\to\infty} m(E_k),$$

so

$$m\Big(\lim_{k\to\infty} E_k\Big) = \lim_{k\to\infty} m(E_k).$$

□

Exercises 2.3

1. P92 1,2,3,4,5,6,7

2.4 Relation Between Lebesgue Measurable Sets and Borel Sets

> *(Real variable theory) does not require as broad a background as is sometimes imagined. Roughly speaking, three principles suffice: every measurable set is close to a finite union of intervals; every measurable function is close to a continuous function; every convergent function sequence is close to a uniformly convergent one. Most results are rather direct applications of these ideas. When students face a real-variable problem, these are the three principles they most need to master. If one principle appears to solve a genuine problem, one naturally asks whether this "closeness" is sufficient; in practice, the answer is generally yes.*

This section concerns Littlewood's first principle.

Theorem 2.4.1. *Every closed set* $F \subset \mathbb{R}^n$ *is Lebesgue measurable, i.e.* $F \in \mathcal{L}^n$.

Proof. Step 1. Suppose F is bounded and closed. Then F is compact. Fix $k \in \mathbb{N}$. For each $x \in F$, define the open cube

$$I_k(x) = \{y = (y_1, \ldots, y_n) \in \mathbb{R}^n \mid |y_i - x_i| < 1/k \ (i = 1, \ldots, n)\}.$$

Then $\{I_k(x) \mid x \in F\}$ is an open cover of F, so by compactness there exists a finite subcover

$$I_k(x^1), \ldots, I_k(x^{l_k}).$$

Let

$$A_k = \bigcup_{j=1}^{l_k} I_k(x^j).$$

Each $I_k(x^j)$ is an open box, and it can be written as a countable union of sets in $\mathcal{R}_0^n$; for example,

$$I_k(x^j) = \bigcup_{m=1}^{\infty} \prod_{i=1}^{n} \Big(x_i^j - \frac{1}{k} + \frac{1}{m},\ x_i^j + \frac{1}{k} - \frac{1}{m}\Big].$$

Since $\mathcal{R}_0^n \subset \mathcal{L}^n$ and $\mathcal{L}^n$ is a σ-algebra, it follows that $A_k \in \mathcal{L}^n$.

We claim that

$$F = \bigcap_{k=1}^{\infty} A_k.$$

Clearly $F \subset A_k$ for every k, so $F \subset \bigcap_k A_k$. Conversely, if $y \in \bigcap_k A_k$, then for each k there exists $x^k \in F$ with

$$|y_i - x_i^k| < 1/k, \quad i = 1, \ldots, n,$$

thus

$$\|y - x^k\| < \frac{\sqrt{n}}{k}.$$

Hence $x^k \to y$, and since F is closed, $y \in F$. Since $\mathcal{L}^n$ is a σ-algebra, it is closed under countable intersections, so $F \in \mathcal{L}^n$.

Step 2. If F is an arbitrary closed subset of $\mathbb{R}^n$, define

$$F_k = F \cap \overline{B(0,k)}.$$

Each F_k is bounded and closed, hence $F_k \in \mathcal{L}^n$ by Step 1. Moreover,

$$F = \bigcup_{k=1}^{\infty} F_k.$$

Closure under countable unions gives $F \in \mathcal{L}^n$. □

By definitions of open and Borel sets, we immediately get:

Corollary 2.4.1. *Every open set and every Borel set in $\mathbb{R}^n$ is Lebesgue measurable.*

Theorem 2.4.2. *If $A \in \mathcal{L}^n$, then for every $\varepsilon > 0$ there exist open G and closed F in $\mathbb{R}^n$ such that*

$$F \subset A \subset G, \quad m(G \backslash A) < \varepsilon, \quad m(A \backslash F) < \varepsilon.$$

Proof. Step 1. If A is bounded, there exists open $G \supset A$ with $m(G \backslash A) < \varepsilon$. By $m(A) = m^*(A)$, for any $\varepsilon > 0$ there exist $A_k \in \mathcal{R}_0^n$ such that

$$A \subset \bigcup_{k=1}^{\infty} A_k, \quad m(A) + \frac{\varepsilon}{2} \geq \sum_{k=1}^{\infty} m(A_k).$$

Write canonical decompositions $A_k = \bigcup_{j=1}^{l_k} I_{k,j}$. Enlarge each box to an open box $I_{k,j}^{\varepsilon}$ with

$$I_{k,j} \subset I_{k,j}^{\varepsilon}, \quad m(I_{k,j}^{\varepsilon}) - m(I_{k,j}) < \frac{\varepsilon}{2^k l_k}.$$

Then

$$\sum_{k=1}^{\infty} \sum_{j=1}^{l_k} m(I_{k,j}^{\varepsilon}) < m(A) + \varepsilon.$$

Let

$$G = \bigcup_{k=1}^{\infty} \bigcup_{j=1}^{l_k} I_{k,j}^{\varepsilon}.$$

Then G is open, $A \subset G$, and

$$m(G \backslash A) = m(G) - m(A) < \varepsilon.$$

Step 2. For a general measurable set A, there exists an open set $G \supset A$ with $m(G \backslash A) < \varepsilon$. For each $k \in \mathbb{N}$, define the truncation

$$A_k = A \cap (-k, k)^n.$$

Then $A_k \in \mathcal{L}^n$ and A_k is bounded. By Step 1, for each k there exists an open set $G_k \supset A_k$ such that

$$m(G_k \backslash A_k) < \frac{\varepsilon}{2^{k+1}}.$$

Replacing G_k by $G_k \cap (-k, k)^n$, we may assume $G_k \subset (-k, k)^n$. Let

$$G = \bigcup_{k=1}^{\infty} G_k.$$

Then G is open and contains A (since every $x \in A$ lies in $(-k, k)^n$ for some k). Moreover, because $G_k \subset (-k, k)^n$ we have $G_k \backslash A = G_k \backslash A_k$, and therefore

$$m(G \backslash A) \leq m\Big(\bigcup_{k=1}^{\infty} (G_k \backslash A) \Big) \leq \sum_{k=1}^{\infty} m(G_k \backslash A) = \sum_{k=1}^{\infty} m(G_k \backslash A_k) < \varepsilon.$$

Step 3. Existence of closed $F \subset A$ with $m(A \backslash F) < \varepsilon$. Apply Step 2 to A^c: there is open $G \supset A^c$ with

$$m(G \backslash A^c) < \varepsilon.$$

Let $F = G^c$. Then F is closed, $F \subset A$, and

$$m(A \backslash F) = m(G \backslash A^c) < \varepsilon. \qquad \square$$

The next theorem is a direct consequence of Theorem 2.4.2.

Theorem 2.4.3. *If $A \in \mathcal{L}^n$, then*

(1) *there exists a G_δ set G^* with $A \subset G^*$ and $m(G^* \backslash A) = 0$;*

(2) *there exists an F_σ set F^* with $F^* \subset A$ and $m(A \backslash F^*) = 0$.*

Remark 2.4.1. *Since both G_δ and F_σ sets are Borel, Lebesgue measurable sets differ from Borel sets only by null sets from both outside and inside.*

Theorem 2.4.4. *Let A be a Borel set in $\mathbb{R}^n$, and let $B \subset \mathbb{R}^n$ with $m^*(B) = 0$. Then $A \cup B$ and $A\backslash B$ are in $\mathcal{L}^n$, and*

$$m(A \cup B) = m(A), \quad m(A\backslash B) = m(A).$$

Proof. Since $B \in \mathcal{L}^n$, both $A \cup B$ and $A\backslash B$ are measurable. Also,

$$m(A) \leq m(A \cup B) \leq m(A) + m(B) = m(A),$$

$$m(A\backslash B) \leq m(A) \leq m(A\backslash B) + m(B) = m(A\backslash B).$$

So equalities hold. □

Theorem 2.4.5 (Translation Invariance)**.** *If $A \in \mathcal{L}^n$ and $x_0 \in \mathbb{R}^n$, then $A + x_0 \in \mathcal{L}^n$ and*

$$m(A + x_0) = m(A).$$

Proof. Fix $E \subset \mathbb{R}^n$. By Lemma 2.2.1,

$$m_0^*(E) = m_0^*(E - x_0).$$

Since $A \in \mathcal{L}^n$, the Carathéodory condition gives

$$m_0^*(E - x_0) = m_0^*((E - x_0) \cap A) + m_0^*((E - x_0) \cap A^c).$$

Translating back and using Lemma 2.2.1 again,

$$m_0^*((E - x_0) \cap A) = m_0^*(E \cap (A + x_0)), \qquad m_0^*((E - x_0) \cap A^c) = m_0^*(E \cap (A + x_0)^c).$$

Therefore

$$m_0^*(E) = m_0^*(E \cap (A + x_0)) + m_0^*(E \cap (A + x_0)^c),$$

so $A + x_0 \in \mathcal{L}^n$. Finally,

$$m(A + x_0) = m_0^*(A + x_0) = m_0^*(A) = m(A),$$

again by Lemma 2.2.1. □

Remark 2.4.2. *Let μ be a measure on $\mathcal{B}^n$, finite on compact sets. If μ is translation invariant, then there exists constant λ such that*

$$\mu(B) = \lambda m(B), \quad \forall B \in \mathcal{B}^n.$$

We omit the proof.

Exercises 2.4

1. P98 1,2,3
2. P102 1,2,3 (replace the conclusion by $m(E) \geq \delta$)

2.5 Completeness of Lebesgue Measure

2.5.1 Existence of Non-Lebesgue-Measurable Sets

Example 2.5.1 (Nonmeasurable Set)**.** *There exists a subset of* $[0,1]^n$ *that is not Lebesgue measurable.*

Let $I = [0,1)^n$ *and define an equivalence relation* $\sim$ *on* I *by*

$$x \sim y \iff x - y \in \mathbb{Q}^n.$$

Using the axiom of choice, select a set $V \subset I$ *containing exactly one representative from each equivalence class. Enumerate* $\mathbb{Q}^n \cap [-1,1]^n$ *as* $\{q_k\}_{k=1}^{\infty}$ *and set* $V_k = V + q_k$*.*

Then the sets $\{V_k\}$ *are pairwise disjoint, and*

$$I \subset \bigcup_{k=1}^{\infty} V_k \subset (-1,2)^n.$$

Assume, for contradiction, that $V \in \mathcal{L}^n$*. By translation invariance, each* V_k *is measurable and* $m(V_k) = m(V)$*. Since the sets* V_k *are pairwise disjoint,*

$$1 = m(I) \le m\Big(\bigcup_{k=1}^{\infty} V_k\Big) = \sum_{k=1}^{\infty} m(V_k) = \sum_{k=1}^{\infty} m(V).$$

If $m(V) = 0$*, the right-hand side equals* 0*, contradicting* $1 \le \cdots$*. If* $m(V) > 0$*, the series diverges to* $+\infty$*, but*

$$m\Big(\bigcup_{k=1}^{\infty} V_k\Big) \le m((-1,2)^n) = 3^n < \infty,$$

another contradiction. Therefore V *is not Lebesgue measurable.*

Remark 2.5.1.

1. *The Vitali-type argument above works for any translation-invariant measure on* $\mathbb{R}^n$ *that assigns a positive finite measure to cubes.*

2. *It also shows, a fortiori, that Lebesgue outer measure* m^* *cannot be countably additive on all subsets of* $\mathbb{R}^n$ *(so it is not finitely additive either).*

2.5.2 Extension of Measures

For any $A \subset \mathbb{R}^n$, define

$$m^{**}(A) = \inf\Big\{\sum_{i=1}^{\infty} m(A_i) \mid A \subset \bigcup_{i=1}^{\infty} A_i,\ A_i \in \mathcal{L}^n\Big\}.$$

If for all $E \subset \mathbb{R}^n$,

$$m^{**}(E) = m^{**}(E \cap A) + m^{**}(E \cap A^c),$$

then A is called second-order Lebesgue measurable.

Clearly every Lebesgue measurable set is second-order measurable. It might appear this extends the measurable class, but the next theorem shows otherwise.

Theorem 2.5.1. *For all $A \subset \mathbb{R}^n$,*

$$m^{**}(A) = m^*(A).$$

Proof. Since $\mathcal{R}_0^n \subset \mathcal{L}^n$,

$$m^{**}(A) \leq m^*(A).$$

For the reverse inequality, assume $m^{**}(A) < \infty$. By definition, for any $\varepsilon > 0$ there exist $A_k \in \mathcal{L}^n$ such that

$$A \subset \bigcup_{k=1}^{\infty} A_k, \quad \sum_{k=1}^{\infty} m(A_k) \leq m^{**}(A) + \varepsilon.$$

For each k, by definition of $m^*(A_k)$ there exist $B_j^k \in \mathcal{R}_0^n$ with

$$A_k \subset \bigcup_{j=1}^{\infty} B_j^k, \quad \sum_{j=1}^{\infty} m_0(B_j^k) \leq m^*(A_k) + \frac{\varepsilon}{2^k}.$$

Hence

$$A \subset \bigcup_{k=1}^{\infty} \bigcup_{j=1}^{\infty} B_j^k,$$

so

$$\begin{aligned} m^*(A) &\leq \sum_{k=1}^{\infty} \sum_{j=1}^{\infty} m_0(B_j^k) \leq \sum_{k=1}^{\infty} \left(m^*(A_k) + \frac{\varepsilon}{2^k} \right) \\ &\leq m^{**}(A) + 2\varepsilon. \end{aligned}$$

Let $\varepsilon \downarrow 0$ to get $m^*(A) \leq m^{**}(A)$. □

Theorem 2.5.2 (Uniqueness of Measure Extension)**.** *Let $\mathcal{R}$ be a σ-ring of subsets of $\mathbb{R}^n$ with $\mathcal{R}_0^n \subset \mathcal{R}$. Let μ be a measure on $\mathcal{R}$ satisfying*

$$\mu|_{\mathcal{R}_0^n} = m|_{\mathcal{R}_0^n},$$

i.e. $\mu(A) = m(A)$ for all $A \in \mathcal{R}_0^n$. Then

$$\mu|_{\mathcal{B}^n} = m|_{\mathcal{B}^n},$$

where $\mathcal{B}^n$ is the smallest σ-ring containing $\mathcal{R}_0^n$.

Proof. Step 1. For every open set $G \subset \mathbb{R}^n$, $\mu(G) = m(G)$.

For each $x \in G$, there is a left-open right-closed box $I(y_x, r_x)$ with rational center y_x and rational side length $r_x > 0$ such that

$$I(y_x, r_x) \subset G.$$

Hence

$$G = \bigcup_{x \in G} I(y_x, r_x).$$

This family is at most countable, so

$$G = \bigcup_{k=1}^{\infty} I(y_{x_k}, r_{x_k}).$$

Disjointize into boxes $\{B_k\}$. By countable additivity of μ and m,

$$\mu(G) = \sum_{k=1}^{\infty} \mu(B_k) = \sum_{k=1}^{\infty} m(B_k) = m(G).$$

Step 2. For every bounded closed set $F \subset \mathbb{R}^n$, $\mu(F) = m(F)$.

Take open $G \supset F$. Then

$$G = (G \backslash F) \cup F,$$

so

$$\mu(G) = m(G) = m(G \backslash F) + m(F) = \mu(G \backslash F) + m(F),$$

hence

$$\mu(F) = \mu(G) - \mu(G \backslash F) = m(F).$$

Step 3. For every closed set $F \subset \mathbb{R}^n$, $\mu(F) = m(F)$.

Let

$$F_k = F \cap \overline{B(0,k)}.$$

By Step 2, $\mu(F_k) = m(F_k)$. Since $F_k \uparrow F$, by continuity from below,

$$\mu(F) = \lim_{k\to\infty} \mu(F_k) = \lim_{k\to\infty} m(F_k) = m(F).$$

Step 4. For every Borel set $E \in \mathcal{B}^n$, $\mu(E) = m(E)$.

Given $\varepsilon > 0$, choose open G_ε and closed F_ε with

$$F_\varepsilon \subset E \subset G_\varepsilon,$$

$$m(E \backslash F_\varepsilon) < \varepsilon, \quad m(G_\varepsilon \backslash E) < \varepsilon.$$

Then

$$m(G_\varepsilon) - m(F_\varepsilon) < 2\varepsilon.$$

Also

$$m(F_\varepsilon) \leq m(E) \leq m(G_\varepsilon),$$

$$\mu(F_\varepsilon) \leq \mu(E) \leq \mu(G_\varepsilon),$$

and by previous steps $\mu(F_\varepsilon) = m(F_\varepsilon)$, $\mu(G_\varepsilon) = m(G_\varepsilon)$. Let $\varepsilon \downarrow 0$ to conclude $\mu(E) = m(E)$. □

Remark 2.5.2. *For measures on $\mathbb{R}^1$ induced by monotone right-continuous functions, all sets may be measurable; this does not contradict the theorem above.*

Finally, we give an example of a Lebesgue measurable set that is not Borel.

Example 2.5.2 (Lebesgue Measurable but Non-Borel Set)**.** *Let $\mathcal{C}$ be the middle-third Cantor set, and let $\{(a_k, b_k)\}$ be component intervals of its complement. Let $\varphi : [0,1] \to [0,1]$ be the Cantor function constructed as in Proposition 1.3.5, so*

$$\varphi|_{(a_k,b_k)} \equiv c_k.$$

Define

$$f(x) = \frac{1}{2}(x + \varphi(x)), \quad x \in [0,1].$$

Then $f(0) = 0$, $f(1) = 1$, and f is strictly increasing; hence bijective. For every k,

$$m(f((a_k, b_k))) = \frac{1}{2}(b_k - a_k).$$

So

$$m\Big(f\Big(\bigcup_{k=1}^{\infty}(a_k, b_k)\Big)\Big) = \frac{1}{2}.$$

Since

$$m\Big(f(\mathcal{C}) \cup f\Big(\bigcup_{k=1}^{\infty}(a_k, b_k)\Big)\Big) = 1,$$

we obtain

$$m(f(\mathcal{C})) = \frac{1}{2}.$$

By the construction in Example 2.5.1, choose a Lebesgue nonmeasurable set $A \subset f(\mathcal{C})$. Let

$$B = f^{-1}(A) \subset \mathcal{C}.$$

Then

$$0 \leq m^*(B) \leq m(\mathcal{C}) = 0,$$

so $B \in \mathcal{L}^1$. But B is not Borel; otherwise $f(B) = A$ would be Borel (see exercise), contradicting nonmeasurability of A.

Exercises 2.5

1. If $f(x)$ is strictly increasing, show that both $f^{-1} : \mathcal{B}^n \to \mathcal{B}^n$ and $f : \mathcal{B}^n \to \mathcal{B}^n$ are bijections.

Chapter 3

Lebesgue Measurable Functions

3.1 Definition and Basic Properties

Throughout this chapter we occasionally work with extended real-valued functions. In particular, whenever the product $0 \cdot \infty$ occurs, we adopt the convention

$$0 \cdot \infty = 0,$$

so that truncation and approximation arguments can be stated without repeated case distinctions.

Definition 3.1.1 (Almost Everywhere)**.** *Let $E \subset \mathbb{R}^n$ be Lebesgue measurable, and let $P(x)$ be a statement about $x \in E$. If there exists a null set $E_0 \subset E$ such that $P(x)$ holds for all $x \in E \backslash E_0$, then we say $P(x)$ holds almost everywhere on E, written as*

$$P(x) \text{ holds for a.e. } x \in E.$$

Example 3.1.1. *By Definition 3.1.1, it makes sense to say that a function is* finite a.e. *on E (meaning $|f(x)| < \infty$ for a.e. x) and* bounded a.e. *on E (meaning $|f(x)| \leq M$ for a.e. x and some M). These are genuinely different notions: a function may be finite a.e. while being unbounded on every set of positive measure.*

Definition 3.1.2 (Lebesgue Measurable Function)**.** *Let $E \subset \mathbb{R}^n$ be Lebesgue measurable, and let*

$$f : E \to \mathbb{R} \cup \{\pm\infty\}.$$

We call f a Lebesgue measurable function if for every $c \in \mathbb{R} \cup \{\pm\infty\}$,

$$E(x|f(x) > c) = \{x \in E | f(x) > c\}$$

is Lebesgue measurable.[1]

Remark 3.1.1. *1. Sometimes one does not assume* a priori *that E is measurable. In that case one typically assumes f is finite-valued*[2] *and then* defines *the measurable domain by*

$$E = \bigcup_{n=1}^{\infty} E(x|f(x) > -n),$$

which is measurable once the superlevel sets are.

[1] Unless otherwise stated, we abbreviate $\{x \in E | f(x) > c\}$ by $E(x|f(x) > c)$.

[2] Finite-valued means the range of f does not contain $\pm\infty$.

2. *For extended real-valued f, one may equivalently require $E(x|f(x) \geq c)$ to be measurable for all $c \in \mathbb{R} \cup \{\pm\infty\}$. In particular, taking $c = -\infty$ forces E to be measurable.*

3. *(Borel measurable functions) If $E \in \mathcal{B}^n$ and $f : E \to \mathbb{R} \cup \{\pm\infty\}$ satisfies*

$$E(x|f(x) > c) \in \mathcal{B}^n \quad \forall c \in \mathbb{R} \cup \{\pm\infty\},$$

then f is called Borel measurable*. Every Borel measurable function is Lebesgue measurable.*

Definition 3.1.3 (Measurable Function on an Abstract Measure Space)**.** *Let $(X, \mathcal{R}, \mu)$ be a measure space, $E \in \mathcal{R}$, and $f : E \to \mathbb{R} \cup \{\pm\infty\}$. If for every $c \in \mathbb{R} \cup \{\pm\infty\}$,*

$$E(x|f(x) > c) \in \mathcal{R},$$

then f is called μ-measurable.

Theorem 3.1.1. *Let $E \in \mathcal{L}^n$, $f : E \to \mathbb{R} \cup \{\pm\infty\}$, and let $D \subset \mathbb{R}$ be a dense subset. If for every $r \in D$,*

$$E(x|f(x) > r) \in \mathcal{L}^n,$$

then f is Lebesgue measurable on E.

Proof. Fix $c \in \mathbb{R}$. Choose a sequence $\{r_j\} \subset D$ such that $r_j \downarrow c$ (possible since D is dense). Then

$$E(x|f(x) > c) = \bigcup_{j=1}^{\infty} E(x|f(x) > r_j),$$

which is measurable as a countable union of measurable sets. By the definition of measurability, f is Lebesgue measurable. □

Theorem 3.1.2. *Let $E \in \mathcal{L}^n$ and $f : E \to \mathbb{R} \cup \{\pm\infty\}$. The following are equivalent:*

(1) *f is Lebesgue measurable on E;*

(2) *for every $c \in \mathbb{R} \cup \{\pm\infty\}$,*

$$E(x|f(x) \leq c) \in \mathcal{L}^n;$$

(3) *for every $c \in \mathbb{R} \cup \{\pm\infty\}$,*

$$E(x|f(x) < c) \in \mathcal{L}^n;$$

(4) *for every $c \in \mathbb{R} \cup \{\pm\infty\}$,*

$$E(x|f(x) \geq c) \in \mathcal{L}^n;$$

(5) *for every $c, d \in \mathbb{R} \cup \{\pm\infty\}$,*

$$E(x|c < f(x) \leq d) \in \mathcal{L}^n.$$

Proof. We sketch the implications (all are elementary set manipulations).

(1) ⇒ (2). For any $c \in \mathbb{R} \cup \{\pm\infty\}$,

$$E(x|f(x) \leq c) = E \setminus E(x|f(x) > c),$$

so measurability of the superlevel set implies measurability of the sublevel set.

(2) ⇒ (3). Note that

$$E(x|f(x) < c) = \bigcup_{m=1}^{\infty} E(x|f(x) \leq c - \tfrac{1}{m}),$$

and the right-hand side is a countable union of measurable sets.

(3) ⇒ (1). Since

$$E(x|f(x) > c) = E \setminus E(x|f(x) \leq c) = E \setminus \bigcap_{m=1}^{\infty} E(x|f(x) < c + \tfrac{1}{m}),$$

measurability of the sets in (3) implies measurability of every superlevel set.

The equivalences between (1)–(4) follow similarly by taking complements and by replacing strict by weak inequalities via rational approximation. Finally,

$$E(x|c < f(x) \leq d) = E(x|f(x) > c) \cap E(x|f(x) \leq d),$$

so (5) is equivalent to (1)–(2). □

Corollary 3.1.1. *Let $E \in \mathcal{L}^n$ and $f : E \to \mathbb{R} \cup \{\pm\infty\}$. If f is Lebesgue measurable, then for every $c \in \mathbb{R} \cup \{\pm\infty\}$,*

$$E(x|f(x) = c) \in \mathcal{L}^n.$$

Proof. This follows directly from Theorem 3.1.2 by writing

$$E(x|f(x) = c) = E(x|f(x) \leq c) \cap E(x|f(x) \geq c).$$ □

Example 3.1.2. *Let $E \in \mathcal{L}^n$. The indicator function χ_E is Lebesgue measurable. In particular, the Dirichlet function is both Lebesgue measurable and Borel measurable.*

Theorem 3.1.3 (Stability Under Algebraic Operations)**.** *Let $E \in \mathcal{L}^n$, and let f, g be Lebesgue measurable on E and finite a.e. Then:*

(1) *for every $\alpha \in \mathbb{R}$, αf is Lebesgue measurable;*

(2) *$f + g$ is Lebesgue measurable;*

(3) *fg is Lebesgue measurable;*

(4) *if $g(x) \neq 0$ for $x \in E$, then f/g is Lebesgue measurable;*

(5) *$\max\{f, g\}$ and $\min\{f, g\}$ are Lebesgue measurable;*

(6) *if $\{f_k\}$ is a sequence of Lebesgue measurable functions on E, then $\sup_k f_k$, $\inf_k f_k$, $\overline{\lim}_{k\to\infty} f_k$, and $\underline{\lim}_{k\to\infty} f_k$ are Lebesgue measurable.*

Proof. Since f and g are finite a.e., we may (if necessary) redefine them on a null set so that they are real-valued everywhere; this does not affect measurability.

(1) follows immediately from the definition.

For (2) (addition), fix $c \in \mathbb{R}$. Using the density of $\mathbb{Q}$,

$$E(x|f(x) + g(x) > c) = \bigcup_{q\in\mathbb{Q}} \Big(E(x|f(x) > q) \cap E(x|g(x) > c - q)\Big),$$

which is measurable as a countable union of measurable sets.

For (3) (multiplication), write $f = f^+ - f^-$ and $g = g^+ - g^-$. Expanding

$$fg = f^+g^+ + f^-g^- - f^+g^- - f^-g^+,$$

reduces the problem to products of nonnegative measurable functions. For nonnegative u, v and any $c > 0$, using the density of $\mathbb{Q}$ we have

$$E(x|u(x)v(x) > c) = \bigcup_{q\in\mathbb{Q},\, q>0} \Big(E(x|u(x) > q) \cap E(x|v(x) > c/q)\Big),$$

which is measurable as a countable union of measurable sets. (If $c \leq 0$, the set is all of E.)

For (5),

$$E\big(x|\max\{f(x), g(x)\} \geq c\big) = E\big(x|f(x) \geq c\big) \cup E\big(x|g(x) \geq c\big).$$

For (4), note that $h(t) = 1/t$ is continuous on $\mathbb{R} \setminus \{0\}$. Since $g \neq 0$ on E, the composition $h \circ g$ is measurable, hence $1/g$ is measurable; multiplying by f gives f/g.

For (6),

$$\overline{\lim_{k\to\infty}} f_k(x) = \inf_m \Big\{ \sup_{k\geq m} f_k(x)\Big\}.$$

The remaining cases follow from closure of measurable sets under countable unions and intersections. □

Example 3.1.3. *Let $E_i \in \mathcal{L}^n$, $i = 1, 2, \ldots, k$. Then*

$$\chi(x) = \sum_{i=1}^{k} \alpha_i \chi_{E_i}(x) \tag{3.1}$$

is Lebesgue measurable.

Remark 3.1.2. *A function of the form* (3.1) *is called a simple function (or step function).*

Example 3.1.4. *If f is Lebesgue measurable on E, and $A \subset E$ with $A \in \mathcal{L}^n$, then the restriction $f|_A$ is Lebesgue measurable.*[3]

Example 3.1.5. *Let $E \in \mathcal{L}^n$. If f is continuous on E, then f is Lebesgue measurable.*

Exercises 3.1

1. P127 1,2,3,4,5,6,7

2. P129 8,9

[3]That is, $f|_A$ is the restriction of f to A.

3.2 Structure of Lebesgue Measurable Functions

Theorem 3.2.1. *Let $E \subset \mathbb{R}^n$ be Lebesgue measurable.*

(1) *If f is a nonnegative Lebesgue measurable function on E and finite a.e., then there exists an increasing*[4] *sequence of simple functions $\{\varphi_k\}_{k=1}^{\infty}$ such that*

$$\varphi_k \to f \quad a.e.\ on\ E.$$

If in addition f is bounded, then

$$\varphi_k \rightrightarrows f.$$

(2) *If f is Lebesgue measurable on E and finite a.e., then there exists a sequence of simple functions $\{\varphi_k\}_{k=1}^{\infty}$ such that*

$$|\varphi_k| \le |f| \quad a.e.\ on\ E,$$

and

$$\varphi_k \to f \quad a.e.\ on\ E.$$

If f is bounded, then

$$\varphi_k \rightrightarrows f.$$

Proof. Step 1. For each natural number k, define

$$E_{k,i} = E\Big(x\Big|\frac{i-1}{2^k} \le f(x) \le \frac{i}{2^k}\Big), \quad i = 1, 2, \ldots, k2^k.$$

Set

$$\varphi_k(x) = \begin{cases} \frac{i-1}{2^k}, & x \in E_{k,i}, \\ k, & x \in E \setminus \bigcup\limits_{i=1}^{k2^k} E_{k,i}. \end{cases}$$

Then $\varphi_k \le \varphi_{k+1}$, and if $x \in E_{k,i}$,

$$0 \le f(x) - \varphi_k(x) \le \frac{1}{2^k}.$$

Since f is finite a.e. on E, the set $E_\infty = \{x \in E | f(x) = +\infty\}$ has measure zero. For $x \in E \setminus E_\infty$ and any $\varepsilon > 0$, choose k_0 so that $f(x) < k_0$ and $2^{-k_0} < \varepsilon$. Then $x \in E_{k_0,i_0}$ for some i_0, hence

$$0 \le f(x) - \varphi_{k_0}(x) \le \frac{1}{2^{k_0}} < \varepsilon.$$

Thus $\varphi_k \to f$ a.e. If $f(x) \le M$ on E, then for $k > M$ the same estimate holds for all $x \in E$, so $\varphi_k \rightrightarrows f$.

Step 2. Let

$$f^+ = \max\{f, 0\}, \quad f^- = \max\{-f, 0\}, \quad f = f^+ - f^-.$$

Apply Step 1 to f^+ and f^- to obtain simple functions $\varphi_k \nearrow f^+$ and $\psi_k \nearrow f^-$ a.e., with $\varphi_k \le f^+$ and $\psi_k \le f^-$. Then

$$\varphi_k - \psi_k \to f^+ - f^- = f,$$

and

$$|\varphi_k - \psi_k| \le \varphi_k + \psi_k \le f^+ + f^- = |f|.$$

So part (2) follows. □

[4] Namely, $\varphi_k \le \varphi_{k+1}$.

Following this theorem, a standard strategy for statements about measurable functions is:

1. prove it for indicator functions;
2. prove it for simple functions;
3. prove it for nonnegative measurable functions;
4. decompose into positive and negative parts, then prove it for general measurable functions.

Lemma 3.2.1. *Let $F_1, F_2, \dots, F_k \subset \mathbb{R}^n$ be pairwise disjoint closed sets, and define*

$$\varphi(x) = \sum_{i=1}^{k} \alpha_i \chi_{F_i}(x).$$

Then $\varphi|_{\cup_{i=1}^k F_i}$ is continuous.

Proof. Fix $x_0 \in \bigcup_{i=1}^k F_i$. Then $x_0 \in F_{i_0}$ for some i_0. Since the F_i are pairwise disjoint and closed, for each $i \neq i_0$,

$$\delta_i = \operatorname{dist}(x_0, F_i) > 0.$$

Set $\delta_0 = \min_{i \neq i_0} \delta_i$. If

$$x \in \Big(\bigcup_{i=1}^{k} F_i \Big) \cap B(x_0, \delta_0),$$

then $x \in F_{i_0}$, hence

$$|\varphi(x) - \varphi(x_0)| = 0 < \varepsilon \quad \forall \varepsilon > 0.$$

So $\varphi|_{\cup_{i=1}^k F_i}$ is continuous. □

The next theorem is Lusin's theorem, corresponding to Littlewood's second principle: every measurable function is close to a continuous one.

Theorem 3.2.2 (Lusin's Theorem)**.** *Let $E \subset \mathbb{R}^n$ be Lebesgue measurable with $m(E) < \infty$, and let f be Lebesgue measurable on E and finite a.e. Then for every $\delta > 0$ there exists a closed set $F \subset E$ such that $f|_F$ is continuous and*

$$m(E \backslash F) < \delta.$$

Proof. Fix $\delta > 0$. For each $k \in \mathbb{N}$, define the measurable sets

$$E_{k,i} = E\Big(x \Big| \frac{i-1}{k} \leq f(x) < \frac{i}{k}\Big), \quad i = 0, \pm 1, \pm 2, \dots$$

so that the $E_{k,i}$ are pairwise disjoint and

$$E = \Big(\bigcup_{i=-\infty}^{\infty} E_{k,i} \Big) \cup E_{-\infty} \cup E_{+\infty},$$

where $E_{\pm\infty} = E(x | f(x) = \pm\infty)$ are null sets.

Since $m(E) < \infty$, for each k we may choose an integer $i_k \geq 1$ such that the "tails" satisfy

$$\Big(\sum_{i=-\infty}^{-i_k-1} + \sum_{i=i_k+1}^{\infty} \Big) m(E_{k,i}) < \frac{\delta}{2^{k+2}}.$$

Define the simple function

$$\varphi_k(x) = \sum_{i=-i_k}^{i_k} \frac{i-1}{k}\,\chi_{E_{k,i}}(x).$$

For each index $i \in \{-i_k, \dots, i_k\}$, use the inner regularity of Lebesgue measure to choose a closed set $F_{k,i} \subset E_{k,i}$ such that

$$m(E_{k,i}\backslash F_{k,i}) < \frac{\delta}{2^{k+2}(2i_k+1)}.$$

Set

$$F_k = \bigcup_{i=-i_k}^{i_k} F_{k,i}, \qquad \psi_k = \varphi_k|_{F_k}.$$

Then F_k is closed, and by Lemma 3.2.1, ψ_k is continuous on F_k. Moreover, for $x \in F_k$ we have $x \in E_{k,i}$ for some $i \in \{-i_k, \dots, i_k\}$, hence

$$|f(x) - \psi_k(x)| = |f(x) - \varphi_k(x)| \le \frac{1}{k}.$$

Finally,

$$\begin{aligned} m(E\backslash F_k) &\le m\Big(E \cap \bigcup_{|i|>i_k} E_{k,i}\Big) + \sum_{i=-i_k}^{i_k} m(E_{k,i}\backslash F_{k,i}) + m(E_{\pm\infty}) \\ &< \frac{\delta}{2^{k+2}} + \frac{\delta}{2^{k+2}} + 0 = \frac{\delta}{2^{k+1}}. \end{aligned}$$

Now define

$$F = \bigcap_{k=1}^{\infty} F_k.$$

Then F is closed and $F \subset E$, and by subadditivity,

$$m(E\backslash F) = m\Big(\bigcup_{k=1}^{\infty}(E\backslash F_k)\Big) \le \sum_{k=1}^{\infty} m(E\backslash F_k) < \sum_{k=1}^{\infty} \frac{\delta}{2^{k+1}} = \delta.$$

On F we have the uniform estimate $|f - \psi_k| \le 1/k$, and each $\psi_k|_F$ is continuous. Hence $f|_F$ is continuous. □

Corollary 3.2.1. *Let $E \subset \mathbb{R}^n$ be Lebesgue measurable with $m(E) < \infty$, and let f be Lebesgue measurable on E and finite a.e. For every $\delta > 0$, there exists a continuous function φ on $\mathbb{R}^n$ such that*

$$m\big(E(x|f(x) \neq \varphi(x))\big) < \delta.$$

Proof. This follows immediately from Theorem 3.2.2 and Theorem 1.3.10. □

Corollary 3.2.2. *Let $E \subset \mathbb{R}^n$ be Lebesgue measurable with $m(E) < \infty$, and let f be Lebesgue measurable on E and finite a.e. Then there exists a sequence of continuous functions $\{\varphi_k\}$ on $\mathbb{R}^n$ such that*

$$\varphi_k \to f \quad \text{a.e. on } E.$$

Proof. Apply the previous corollary with $\delta = 2^{-k}$: choose continuous φ_k on $\mathbb{R}^n$ such that

$$m\big(E(x|f(x) \neq \varphi_k(x))\big) < 2^{-k}.$$

Let $A_k = E(x|f(x) \neq \varphi_k(x))$. Then $\sum_{k=1}^{\infty} m(A_k) < \infty$, hence

$$m\Big(\overline{\lim_{k\to\infty}} A_k\Big) = 0.$$

For $x \in E \setminus \overline{\lim} A_k$, there exists $k_0(x)$ such that $x \notin A_k$ for all $k \geq k_0(x)$, i.e.

$$\varphi_k(x) = f(x) \quad \forall k \geq k_0(x).$$

Therefore $\varphi_k(x) \to f(x)$ for a.e. $x \in E$. □

Theorem 3.2.3. *Let $f : [0,1] \to [0,1]$ be strictly increasing and continuous, with $f([0,1]) = [0,1]$. Then*

$$\mathcal{B} = f^{-1}(\mathcal{B}).$$

Proof. Let

$$\mathcal{A} = \{f^{-1}(B) \mid B \in \mathcal{B},\ B \subset [0,1]\}.$$

We prove in two steps.

Step 1: Prove $\mathcal{B} \subset \mathcal{A}$.

(1) If $A_1, A_2 \in \mathcal{A}$, then $A_1 = f^{-1}(B_1)$ and $A_2 = f^{-1}(B_2)$ for some $B_1, B_2 \in \mathcal{B}$. Thus

$$A_1 \backslash A_2 = f^{-1}(B_1 \backslash B_2) \in \mathcal{A}.$$

(2) If $A_i \in \mathcal{A}$ with $A_i = f^{-1}(B_i)$, then

$$\bigcup_{i=1}^{\infty} A_i = f^{-1}\Big(\bigcup_{i=1}^{\infty} B_i\Big) \in \mathcal{A}.$$

So $\mathcal{A}$ is a σ-algebra.

(3) Since f is continuous, preimages of open or closed subsets of $[0,1]$ are open or closed in $[0,1]$, hence belong to $\mathcal{A}$. Therefore $\mathcal{B} \subset \mathcal{A}$.

Step 2: Let $g = f^{-1}$. Then g is also strictly increasing and continuous. By Step 1, $\mathcal{B} \subset g^{-1}(\mathcal{B}) = f(\mathcal{B})$, so

$$f^{-1}(\mathcal{B}) \subset f^{-1}(f(\mathcal{B})) = \mathcal{B}.$$

Combining with Step 1 gives $\mathcal{B} = f^{-1}(\mathcal{B})$. □

Remark 3.2.1. *This proof method is typical and is often called the "good-set principle": first collect all sets with the desired property, then show that this family forms an appropriate class.*

3.3 Almost Everywhere Convergence and Convergence in Measure

In this section we introduce two important notions of convergence and their relationship.

3.3.1 Almost Everywhere Convergence

Definition 3.3.1 (Almost Everywhere Convergence)**.** *Let $E \subset \mathbb{R}^n$ be Lebesgue measurable, and*

$$f, f_k : E \to \mathbb{R} \cup \{\pm\infty\}, \quad k = 1, 2, \ldots,$$

be Lebesgue measurable. If

$$f_k(x) \to f(x)$$

for a.e. $x \in E$, then we say $\{f_k\}$ converges to f almost everywhere on E, written

$$f_k \to f \quad \text{a.e. on } E,$$

or

$$f_k \overset{\text{a.e.}}{\to} f \quad x \in E.$$

In Theorem 3.4.1 we will see that measurable functions generalize continuous functions. In real analysis, even pointwise limits of continuous functions need not be continuous. In measure theory, however, a.e. limits preserve Lebesgue measurability.

Theorem 3.3.1. *Let $E \subset \mathbb{R}^n$ be Lebesgue measurable, and let $f_k : E \to \mathbb{R} \cup \{\pm\infty\}$ be Lebesgue measurable for $k = 1, 2, \ldots$. If*

$$f_k \overset{\text{a.e.}}{\to} f \quad x \in E,$$

then f is measurable on E.

Proof. Let $N \subset E$ be a null set such that $f_k(x) \to f(x)$ for all $x \in E \setminus N$. Fix $c \in \mathbb{R}$. Define

$$B_c = \bigcup_{m=1}^{\infty} \bigcup_{j=1}^{\infty} \bigcap_{k \geq j} E(x | f_k(x) > c + \tfrac{1}{m}).$$

Each set on the right is measurable (it is built from superlevel sets of the measurable functions f_k using countable unions and intersections), hence B_c is measurable.

We claim that on $E \setminus N$,

$$E(x | f(x) > c) = B_c.$$

Indeed, if $x \in E \setminus N$ and $f(x) > c$, choose m such that $f(x) > c + 1/m$; by pointwise convergence there exists j with $f_k(x) > c + 1/m$ for all $k \geq j$, hence $x \in B_c$. Conversely, if $x \in E \setminus N$ belongs to B_c, then for some m and j we have $f_k(x) > c + 1/m$ for all $k \geq j$, so letting $k \to \infty$ gives $f(x) \geq c + 1/m > c$.

Therefore

$$E(x | f(x) > c) = (B_c \cap (E \setminus N)) \cup (E(x | f(x) > c) \cap N).$$

The first set is measurable as an intersection, and the second is a subset of the null set N, hence measurable (Lebesgue measure is complete). Thus $E(x | f(x) > c)$ is measurable for every $c \in \mathbb{R}$, and f is Lebesgue measurable. □

Next we present the classical Egorov theorem. We first need a lemma.

Lemma 3.3.1. *Let $E \subset \mathbb{R}^n$ be Lebesgue measurable with*

$$m(E) < \infty.$$

Let f, f_k be measurable extended real-valued functions on E, finite a.e., and suppose

$$f_k \overset{a.e.}{\to} f \quad x \in E.$$

For every $\varepsilon > 0$, define

$$E_k(\varepsilon) = E\big(x \big| |f_k(x) - f(x)| > \varepsilon\big).$$

Then

$$\lim_{j\to\infty} m\Big(\bigcup_{k=j}^{\infty} E_k(\varepsilon)\Big) = 0.$$

Proof. Let

$$E(\varepsilon) = \overline{\lim_{k\to\infty}} E_k(\varepsilon) = \bigcap_{j=1}^{\infty}\bigcup_{k=j}^{\infty} E_k(\varepsilon).$$

The set $E(\varepsilon)$ contains points where the deviation $|f_k - f| > \varepsilon$ happens infinitely often. Since $f_k \to f$ a.e., $m(E(\varepsilon)) = 0$. Then the conclusion follows from continuity of measure from above (using $m(E) < \infty$). □

The Egorov theorem below corresponds to Littlewood's third principle: a.e. convergence is nearly uniform convergence on large subsets.

Theorem 3.3.2 (Egorov)**.** *Let $E \subset \mathbb{R}^n$ be Lebesgue measurable with*

$$m(E) < \infty.$$

Let f, f_k be measurable on E, finite a.e., and assume

$$f_k \overset{a.e.}{\to} f \quad x \in E.$$

Then for every $\delta > 0$, there exists a measurable subset $E_\delta \subset E$ such that $m(E_\delta) < \delta$ and

$$f_k \rightrightarrows f \quad \text{on } E\backslash E_\delta.$$

Proof. Let

$$E_0 = E\big(x \big| \exists k : \ |f_k(x)| = \infty \text{ or } |f(x)| = \infty\big),$$

so $m(E_0) = 0$.

In Lemma 3.3.1, take $\varepsilon = 1/m$ for $m = 1, 2, \ldots$. Then for each m, there exists j_m such that

$$m\Big(\bigcup_{k=j_m}^{\infty} E_k\Big(\frac{1}{m}\Big)\Big) < \frac{\delta}{2^m}.$$

Define

$$E_\delta = E_0 \cup \Big(\bigcup_{m=1}^{\infty}\bigcup_{k=j_m}^{\infty} E_k\Big(\frac{1}{m}\Big)\Big).$$

Then

$$m(E_\delta) \le \sum_{m=1}^{\infty} \frac{\delta}{2^m} = \delta.$$

Now fix $\varepsilon > 0$, choose m_0 with $1/m_0 < \varepsilon$. If $x \in E \backslash E_\delta$, then for all $k \geq j_{m_0}$,

$$|f_k(x) - f(x)| \leq \frac{1}{m_0} < \varepsilon.$$

Hence $f_k \rightrightarrows f$ on $E \backslash E_\delta$. □

Remark 3.3.1. *The finite measure assumption in Egorov's theorem cannot be removed.*

Example 3.3.1. *Let $f_k(x) = \chi_{(0,k)}(x)$. Then*

$$f_k \overset{a.e.}{\to} \chi_{(0,\infty)} \quad x \in \mathbb{R}.$$

But for any set E_δ with $m(E_\delta) < \infty$, uniform convergence on $\mathbb{R} \backslash E_\delta$ fails.

3.3.2 Convergence in Measure

Definition 3.3.2 (Convergence in Measure)**.** *Let $E \subset \mathbb{R}^n$ be Lebesgue measurable with $m(E) < \infty$, and let f, f_k be measurable and finite a.e. on E. If for every $\varepsilon > 0$,*

$$\lim_{k \to \infty} m(E(x||f_k(x) - f(x)| > \varepsilon)) = 0,$$

then $\{f_k\}$ is said to converge to f in measure on E, written

$$f_k \overset{m}{\to} f \quad x \in E,$$

or

$$f_k \Rightarrow f \quad x \in E.$$

Convergence in measure is fundamental in probability theory. It is weaker than uniform convergence and typically weaker than almost everywhere convergence.

Definition 3.3.3 (True in Measure)**.** *Let $P_n(x)$ be a sequence of statements about $x \in E$. If*

$$\lim_{n \to \infty} m(E(x|P_n(x) \text{ is false})) = 0,$$

then we say this sequence of statements is true in measure.

The next theorem gives a basic implication between the two convergences.

Theorem 3.3.3. *Let $E \subset \mathbb{R}^n$ be Lebesgue measurable with $m(E) < \infty$. Suppose f, f_k are measurable and finite a.e. on E, and*

$$f_k \overset{a.e.}{\to} f \quad x \in E.$$

Then

$$f_k \overset{m}{\to} f \quad x \in E.$$

Proof. Fix $\varepsilon > 0$ and $\delta > 0$. By Theorem 3.3.2, there exists $E_\delta \subset E$ with $m(E_\delta) < \delta$ such that $f_k \rightrightarrows f$ on $E \backslash E_\delta$. Hence there is k_0 such that for $k > k_0$,

$$|f_k(x) - f(x)| < \varepsilon \quad \forall x \in E \backslash E_\delta.$$

Therefore

$$E\big(x\big||f_k(x)-f(x)|\geq\varepsilon\big)\subset E_\delta,$$

so

$$m\big(E(x||f_k(x)-f(x)|\geq\varepsilon)\big)<\delta\quad(k>k_0).$$

This is exactly $f_k\overset{m}{\to}f$. □

Remark 3.3.2. *The condition $m(E)<\infty$ in Theorem 3.3.3 is essential. For example,*

$$f_k(x)=\begin{cases}1, & |x|\geq k,\\ 0, & |x|<k,\end{cases}$$

satisfies $f_k\overset{a.e.}{\to}0$ on $\mathbb{R}$, but $f_k\not\Rightarrow 0$ on $\mathbb{R}$.

Theorem 3.3.3 shows that a.e. convergence implies convergence in measure on finite-measure sets. The converse generally fails.

Example 3.3.2. *Construct a sequence on $[0,1]$ by blocks:*

$$f_{1,1}(x)=1,\quad x\in[0,1],$$

$$f_{2,1}(x)=\begin{cases}1, & x\in[0,\frac12),\\ 0, & x\in[\frac12,1],\end{cases}\qquad f_{2,2}(x)=\begin{cases}0, & x\in[0,\frac12),\\ 1, & x\in[\frac12,1],\end{cases},$$

and in general,

$$f_{k,i}(x)=\begin{cases}1, & x\in[\frac{i}{2^{k-1}},\frac{i+1}{2^{k-1}}),\\ 0, & \textit{otherwise},\end{cases}\qquad k=1,2,\dots,\ i=1,2,\dots,2^{k-1}.$$

Enumerate these functions into a single sequence $\{f_n\}$ by listing the kth block

$$f_{k,1},f_{k,2},\dots,f_{k,2^{k-1}}$$

after the $(k-1)$st block.

For any fixed $\varepsilon\in(0,1)$, each f_n is an indicator function and

$$m\big(\{x\in[0,1]:|f_n(x)-0|>\varepsilon\}\big)=m\big(\{x\in[0,1]:f_n(x)=1\}\big)=2^{-(k-1)},$$

where k is the block index of f_n. Since $k\to\infty$ as $n\to\infty$, this tends to 0, so $f_n\overset{m}{\to}0$ on $[0,1]$.

However, for every $x\in[0,1]$ and every k, there exists an index $i=i(k,x)$ with $x\in[i/2^{k-1},(i+1)/2^{k-1})$, hence $f_{k,i}(x)=1$. Along the corresponding subsequence of $\{f_n\}$, the values stay equal to 1, so $f_n(x)$ does not converge to 0. Thus $f_n\not\to 0$ a.e.

Hence convergence in measure is strictly weaker than almost everywhere convergence.

Theorem 3.3.4 (Riesz)**.** *Let $E\subset\mathbb{R}^n$ be Lebesgue measurable with $m(E)<\infty$. Let f,f_k be measurable and finite a.e. on E. If*

$$f_k\overset{m}{\to}f\quad x\in E,$$

then there exists a subsequence $\{f_{k_j}\}$ such that

$$f_{k_j}\overset{a.e.}{\to}f\quad x\in E.$$

Before proving Theorem 3.3.4, we record a corollary.

Corollary 3.3.1. *Let $E \subset \mathbb{R}^n$ be Lebesgue measurable with $m(E) < \infty$, and let f, f_k be measurable and finite a.e. on E. Then $f_k \xrightarrow{m} f$ on E if and only if every subsequence $\{f_{k_i}\}$ has a further subsequence $\{f_{k_{i_j}}\}$ such that*

$$f_{k_{i_j}} \xrightarrow{a.e.} f \quad x \in E.$$

Proof. Necessity follows from Theorem 3.3.4.

For sufficiency, argue by contradiction. If $f_k \not\xrightarrow{m} f$, then there exist $\varepsilon_0, \delta_0 > 0$ and a subsequence $\{f_{k_i}\}$ such that

$$m(E(x||f_{k_i}(x) - f(x)| > \varepsilon_0)) \geq \delta_0 \quad \forall i.$$

By the assumption on subsequences, $\{f_{k_i}\}$ has a further subsequence $\{f_{k_{i_j}}\}$ with $f_{k_{i_j}} \xrightarrow{a.e.} f$. By Theorem 3.3.3, this implies $f_{k_{i_j}} \xrightarrow{m} f$, contradiction. □

Proof of Theorem 3.3.4. Since $f_k \xrightarrow{m} f$, for each $j \in \mathbb{N}$ we can choose k_j (strictly increasing) such that

$$m\Big(E\Big(x\Big||f_{k_j}(x) - f(x)| \geq \frac{1}{2^j}\Big)\Big) < \frac{1}{2^j}.$$

Define

$$E_j = E\Big(x\Big||f_{k_j}(x) - f(x)| \geq \frac{1}{2^j}\Big).$$

Then

$$\sum_{j=1}^{\infty} m(E_j) < \infty,$$

so

$$m\Big(\overline{\lim_{j\to\infty}} E_j\Big) = 0.$$

For $x \notin \overline{\lim} E_j$, there exists $j_0(x)$ such that for all $j \geq j_0(x)$,

$$|f_{k_j}(x) - f(x)| < \frac{1}{2^j},$$

which implies $f_{k_j}(x) \to f(x)$. Hence $f_{k_j} \xrightarrow{a.e.} f$ on E. □

When introducing a new convergence notion, one naturally asks about uniqueness of limits. To address this, we introduce Cauchy sequences in measure.

Definition 3.3.4 (Cauchy in Measure)**.** *Let $E \subset \mathbb{R}^n$ be Lebesgue measurable with $m(E) < \infty$. A sequence of measurable functions $\{f_k\}$ on E is called Cauchy in measure if for every $\varepsilon > 0$ and $\delta > 0$, there exists K such that whenever $i, j > K$,*

$$m(E(x||f_i(x) - f_j(x)| > \varepsilon)) < \delta.$$

Remark 3.3.3. *A Cauchy sequence in measure is sometimes also called a measure-Cauchy sequence.*

Theorem 3.3.5. *Let $E \subset \mathbb{R}^n$ be Lebesgue measurable with $m(E) < \infty$. Then $\{f_k\}$ is Cauchy in measure if and only if there exists a Lebesgue measurable function f on E such that*

$$f_k \xrightarrow{m} f \quad x \in E.$$

Proof. Sufficiency is immediate from the definition of convergence in measure.

For necessity, choose inductively a subsequence $\{f_{k_j}\}$ such that

$$m\Big(E\Big(x\Big||f_{k_{j+1}}(x) - f_{k_j}(x)| \geq \frac{1}{2^j}\Big)\Big) < \frac{1}{2^j}.$$

Set

$$A_j = E\Big(x\Big||f_{k_{j+1}}(x) - f_{k_j}(x)| \geq \frac{1}{2^j}\Big).$$

Then $\sum_j m(A_j) < \infty$, so $m(\overline{\lim} A_j) = 0$. For $x \notin \overline{\lim} A_j$, the series of successive differences is summable, hence $\{f_{k_j}(x)\}$ is Cauchy in $\mathbb{R}$ and converges pointwise. Define

$$f(x) = \lim_{j\to\infty} f_{k_j}(x)$$

on this full-measure set and extend on the null set arbitrarily; then f is measurable and

$$f_{k_j} \overset{a.e.}{\to} f \quad \text{on } E.$$

Therefore $f_{k_j} \overset{m}{\to} f$ by Theorem 3.3.3.

Finally, use the Cauchy-in-measure property of the full sequence. Given $\varepsilon, \delta > 0$, choose K so that for all $i, j > K$,

$$m(E(x||f_i(x) - f_j(x)| > \varepsilon/2)) < \delta/2.$$

Next choose j so large that $k_j > K$ and

$$m(E(x||f_{k_j}(x) - f(x)| > \varepsilon/2)) < \delta/2.$$

Then for every $k > K$,

$$\begin{aligned} m(E(x||f_k(x) - f(x)| > \varepsilon)) &\leq m(E(x||f_k(x) - f_{k_j}(x)| > \varepsilon/2)) + m(E(x||f_{k_j}(x) - f(x)| > \varepsilon/2)) \\ &< \delta. \end{aligned}$$

Hence $f_k \overset{m}{\to} f$. □

Exercises 3.3

1. P140 1,3,4,5,6
2. P144 1,2
3. P149 10,11,13,15

3.4 Characterizations of Measurability and Composition Measurability

We now present a necessary and sufficient condition for function measurability.

Theorem 3.4.1 (Characterization of Measurable Functions)**.** *Let $E \subset \mathbb{R}^n$ be Lebesgue measurable. Then $f : E \to \mathbb{R}$ is Lebesgue measurable if and only if for every Borel set $B \subset \mathbb{R}$,*

$$E(x|f(x) \in B)$$

is Lebesgue measurable.

Proof. Sufficiency is immediate: for each $t \in \mathbb{R}$, $(t, \infty) \in \mathcal{B}$, so $E(x|f(x) > t)$ is measurable.

For necessity, use the good-set principle again. Define

$$\mathcal{S} = \{B \subset \mathbb{R} \mid f^{-1}(B) \text{ is Lebesgue measurable in } E\}.$$

(1) If $B_k \in \mathcal{S}$, then $f^{-1}(B_k)$ is measurable, so

$$f^{-1}\Big(\bigcup_{k=1}^{\infty} B_k\Big) = \bigcup_{k=1}^{\infty} f^{-1}(B_k)$$

is measurable. Hence $\mathcal{S}$ is closed under countable unions.

(2) If $B_1, B_2 \in \mathcal{S}$, then

$$f^{-1}(B_1 \cap B_2) = f^{-1}(B_1) \cap f^{-1}(B_2)$$

is measurable. So $\mathcal{S}$ is closed under finite intersections.

(3) If $B \in \mathcal{S}$, then

$$E = f^{-1}(\mathbb{R}) = f^{-1}(B \cup B^c) = f^{-1}(B) \cup f^{-1}(B^c),$$

so $f^{-1}(B^c)$ is measurable. Thus $\mathcal{S}$ is closed under complements in $\mathbb{R}$, hence under differences.

So $\mathcal{S}$ is a σ-algebra. Also, for every interval $(c, d) \subset \mathbb{R}$,

$$f^{-1}((c, d)) = E(x|f(x) > c) \cap E(x|f(x) < d) \in \mathcal{S}.$$

Therefore all Borel sets belong to $\mathcal{S}$, and necessity follows. □

Remark 3.4.1. *Recall: a real-valued function on $\mathbb{R}$ is continuous iff the preimage of every open set is open. In this sense, measurable functions extend continuity.*

Now consider

$$E \xrightarrow{f} \mathbb{R} \xrightarrow{g} \mathbb{R},$$

where both f and g are Lebesgue measurable. Is $g \circ f : E \to \mathbb{R}$ always measurable? We have

$$\begin{aligned} E(x|g \circ f(x) > t) &= E(x|g \circ f(x) \in (t, \infty)) \\ &= E(x|f(x) \in g^{-1}((t, \infty))). \end{aligned}$$

The set $g^{-1}((t, \infty))$ is Lebesgue measurable, but it need not be Borel. Therefore Theorem 3.4.1 does not automatically apply.

On the other hand, if g is Borel measurable *(in particular, if g is continuous), then $g^{-1}((t, \infty))$ is a Borel set for every $t \in \mathbb{R}$, and the argument above shows that $x \mapsto g(f(x))$ is Lebesgue measurable.*

The obstruction arises only when g is Lebesgue measurable but not Borel measurable; in that case, the composition can indeed fail to be measurable.

Exercises 3.4

1. Give a direct example showing that the composition of two Lebesgue measurable functions need not be measurable.

Chapter 4

Lebesgue Integral

Lebesgue integration is designed to make limit operations behave the way analysis needs them to behave. In elementary calculus one quickly discovers that pointwise limiting processes can destroy continuity and Riemann integrability; nonetheless, the integrals of approximating functions often converge. The Lebesgue integral repairs this mismatch by tying integration to measure and by building approximation and closure principles directly into the definition.

This chapter develops the integral in three stages. First, we define and study the integral for bounded nonnegative measurable functions on sets of finite measure. Second, we extend the definition to general measurable functions on arbitrary measurable sets. Finally, we prove the foundational limit theorems (monotone convergence, Fatou, dominated convergence), establish density of continuous functions in L^1, and prove Tonelli–Fubini theorems.

4.1 Integrals of Nonnegative Measurable Functions on Finite-Measure Sets

Throughout this section, $E \subset \mathbb{R}^n$ is Lebesgue measurable with $m(E) < \infty$.

4.1.1 Simple functions and the integral

Definition 4.1.1 (Integral of a nonnegative simple function)**.** *A* nonnegative simple function *on E is a finite linear combination*

$$\varphi(x) = \sum_{i=1}^{k} a_i \, \chi_{A_i}(x), \qquad a_i \geq 0, \ A_i \in \mathcal{L}^n, \ A_i \subset E,$$

where the sets A_i are pairwise disjoint. We define its integral over E by

$$\int_E \varphi(x)\, dx \ \stackrel{\text{def}}{=} \ \sum_{i=1}^{k} a_i \, m(A_i).$$

Lemma 4.1.1 (Well-definedness)**.** *If a nonnegative simple function φ admits two representations as above, then the value of $\int_E \varphi \, dx$ is the same.*

Proof. Refine the two partitions to a common partition of E by taking all possible intersections of the sets involved. On each atom of the common partition the function φ is constant, and the sum $\sum a_i m(A_i)$ reduces to the same constant-weighted sum over the common atoms. □

4.1.2 The Lebesgue integral for bounded nonnegative functions

The next definition is the standard one: integrate by approximation from below.

Definition 4.1.2 (Lebesgue integral of a bounded nonnegative measurable function)**.** *Let $f : E \to [0,\infty)$ be bounded and Lebesgue measurable. Define*

$$\int_E f\,dx \stackrel{\text{def}}{=} \sup\Big\{ \int_E \varphi\,dx \;\Big|\; \varphi \text{ is a nonnegative simple function on } E \text{ and } \varphi \le f \Big\}. \tag{4.1}$$

When the context is clear, we omit the qualifier "Lebesgue".

Remark 4.1.1. *One can also define $\int_E f$ using partitions of the range of f (a Darboux–type definition). The definition* (4.1) *is equivalent and interacts more transparently with monotone limits.*

Example 4.1.1. *If $E \in \mathcal{L}^n$ and $m(E) = 0$, then every bounded measurable function on E is integrable and*

$$\int_E f(x)\,dx = 0.$$

Example 4.1.2 (Dirichlet function)**.** *The Dirichlet function (see Example 5) is Lebesgue integrable on $[0,1]$, and*

$$\int_{[0,1]} D(x)\,dx = 0.$$

4.1.3 Basic properties on finite-measure sets

Theorem 4.1.1. *Let $E \subset \mathbb{R}^n$ be Lebesgue measurable with $m(E) < \infty$. If f is bounded, nonnegative, and Lebesgue measurable on E, then $\int_E f\,dx < \infty$.*

Proof. Since f is bounded, there exists $M > 0$ such that $0 \le f \le M$ on E. By monotonicity of the supremum definition,

$$0 \le \int_E f\,dx \le \int_E M\,dx = M\,m(E) < \infty.$$

□

Theorem 4.1.2 (Finite additivity over the region)**.** *Let $E_1, \dots, E_m \in \mathcal{L}^n$ be pairwise disjoint and set $E = \bigcup_{i=1}^m E_i$. If $f : E \to [0,\infty)$ is bounded and measurable, then*

$$\int_E f\,dx = \sum_{i=1}^m \int_{E_i} f\,dx.$$

Proof. For simple functions the formula is immediate from the definition and finite additivity of measure. For general bounded nonnegative f, approximate from below by simple functions $\varphi \le f$. Then $\varphi\chi_{E_i} \le f\chi_{E_i}$ are simple functions on E_i and

$$\int_E \varphi\,dx = \sum_{i=1}^m \int_{E_i} \varphi\,dx \le \sum_{i=1}^m \int_{E_i} f\,dx.$$

Taking the supremum over $\varphi \le f$ gives $\int_E f \le \sum_i \int_{E_i} f$. For the reverse inequality, fix i and approximate $f\chi_{E_i}$ from below by simple functions on E supported in E_i; summing these approximations yields the opposite inequality. □

Theorem 4.1.3 (Linearity for nonnegative functions)**.** *Let $f, g : E \to [0, \infty)$ be bounded and measurable and let $\alpha, \beta \geq 0$. Then $\alpha f + \beta g$ is integrable and*

$$\int_E (\alpha f + \beta g)\, dx = \alpha \int_E f\, dx + \beta \int_E g\, dx.$$

Proof. The statement is immediate for simple functions. For bounded nonnegative functions, approximate f and g from below by simple functions and use the supremum definition (4.1) together with monotonicity. □

Theorem 4.1.4 (Monotonicity)**.** *If $f, g : E \to [0, \infty)$ are bounded measurable and $f \leq g$ a.e. on E, then*

$$\int_E f\, dx \leq \int_E g\, dx.$$

Proof. Every simple function $\varphi \leq f$ also satisfies $\varphi \leq g$ a.e. Taking suprema in (4.1) yields the claim. □

Theorem 4.1.5. *Let $f : E \to [0, \infty)$ be bounded and measurable. If $\int_E f\, dx = 0$, then $f = 0$ a.e. on E.*

Proof. For each $k \in \mathbb{N}$, let $A_k = \{x \in E : f(x) \geq 1/k\}$. Then $\chi_{A_k}/k \leq f$, hence

$$\frac{1}{k} m(A_k) = \int_E \frac{1}{k} \chi_{A_k}\, dx \leq \int_E f\, dx = 0,$$

so $m(A_k) = 0$. Since $\{f > 0\} = \bigcup_{k \geq 1} A_k$, we have $m(\{f > 0\}) = 0$ and thus $f = 0$ a.e. □

4.1.4 Compatibility with the Riemann integral

Theorem 4.1.6. *If f is Riemann integrable on $[a, b] \subset \mathbb{R}$, then f is Lebesgue integrable on $[a, b]$ and*

$$(L) \int_{[a,b]} f(x)\, dx = (R) \int_a^b f(x)\, dx.$$

Proof. A Riemann integrable function is bounded and is continuous almost everywhere on $[a, b]$. In particular, it is Lebesgue measurable. To relate the two integrals, consider a sequence of partitions with mesh tending to 0 and the corresponding lower and upper step functions $\varphi_k \leq f \leq \psi_k$. Riemann integrability implies

$$\int_{[a,b]} \varphi_k\, dx \uparrow (R) \int_a^b f, \qquad \int_{[a,b]} \psi_k\, dx \downarrow (R) \int_a^b f.$$

Since φ_k are simple and increase to f pointwise a.e. at continuity points, the supremum definition (4.1) yields

$$\int_{[a,b]} f\, dx \geq \sup_k \int \varphi_k\, dx = (R) \int_a^b f.$$

Similarly $\int f \leq \inf_k \int \psi_k\, dx = (R) \int_a^b f$. Therefore the two integrals agree. □

Exercises 4.1

1. If $E \subset \mathbb{R}^n$ has finite measure and $f \in L^\infty(E)$, prove that

$$I(x) = \int_{B(0,\|x\|)} f(y)\,dy$$

defines a continuous function of $x \in \mathbb{R}^n$.

2. Prove that a bounded function on $[a, b]$ is Riemann integrable if and only if it is continuous almost everywhere.

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4.2 Integrals of General Measurable Functions on General Measurable Sets

We now remove the restrictions "bounded" and "$m(E) < \infty$," and we also allow sign changes.

4.2.1 Nonnegative measurable functions

Definition 4.2.1 (Lebesgue integral of a nonnegative measurable function)**.** *Let $E \in \mathcal{L}^n$ and let $f : E \to [0, \infty]$ be measurable. Define*

$$\int_E f\,dx \stackrel{\text{def}}{=} \sup\Big\{ \int_E \varphi\,dx \;\Big|\; \varphi \text{ is a nonnegative simple function on } E \text{ and } \varphi \le f \Big\} \in [0,\infty]. \tag{4.2}$$

We say f is (Lebesgue) integrable *on E if $\int_E f\,dx < \infty$.*

Remark 4.2.1. *If $m(E) < \infty$ and f is bounded, this definition reduces to Definition 4.1.2.*

Theorem 4.2.1 (Computation by truncation and exhaustion)**.** *Let $E \in \mathcal{L}^n$ and $f : E \to [0, \infty]$ be measurable. Choose an increasing exhaustion $E_k \uparrow E$ with $m(E_k) < \infty$ and define truncations $f_k = \min\{f, k\}$. Then*

$$\int_E f\,dx = \lim_{k\to\infty} \int_{E_k} f_k\,dx.$$

The limit exists in $[0, \infty]$ and is independent of the chosen exhaustion.

Proof. For each k, the function $f_k\chi_{E_k}$ is bounded, nonnegative, and measurable on the finite-measure set E_k, hence integrable. Since $f_k\chi_{E_k} \le f$, monotonicity of the integral gives

$$\int_{E_k} f_k\,dx = \int_E f_k\chi_{E_k}\,dx \le \int_E f\,dx,$$

so $\sup_k \int_{E_k} f_k\,dx \le \int_E f\,dx$.

For the reverse inequality, let φ be a nonnegative simple function on E with $\varphi \le f$. Write $\varphi = \sum_{i=1}^N a_i\chi_{A_i}$ with $a_i \ge 0$ and pairwise disjoint measurable $A_i \subset E$, and set $M = \max_i a_i$. Fix $k \ge M$ and define $\varphi_k = \varphi\chi_{E_k}$. Then φ_k is simple and $\varphi_k \uparrow \varphi$ pointwise. Moreover, since $k \ge M$ and $\varphi \le f$, we have on E_k the pointwise bound $\varphi_k \le f_k$. Therefore

$$\int_E \varphi_k\,dx \le \int_{E_k} f_k\,dx \le \sup_j \int_{E_j} f_j\,dx.$$

Letting $k \to \infty$ and using continuity from below for the integral of simple functions yields

$$\int_E \varphi \, dx \le \sup_j \int_{E_j} f_j \, dx.$$

Taking the supremum over all simple $\varphi \le f$ (Definition 4.2.1) gives

$$\int_E f \, dx \le \sup_j \int_{E_j} f_j \, dx.$$

Combining with the first inequality, we conclude

$$\int_E f \, dx = \sup_k \int_{E_k} f_k \, dx.$$

Since the sequence $\int_{E_k} f_k \, dx$ is increasing in k, the supremum equals the limit. This identity also shows that the limit is independent of the chosen exhaustion. □

4.2.2 Signed functions and L^1

Definition 4.2.2 (Positive and negative parts)**.** *For a measurable function $f : E \to \mathbb{R} \cup \{\pm\infty\}$, define*

$$f^+ = \max\{f, 0\}, \qquad f^- = \max\{-f, 0\}.$$

Then $f = f^+ - f^-$ and $|f| = f^+ + f^-$.

Definition 4.2.3. *Let $E \in \mathcal{L}^n$ and $f : E \to \mathbb{R} \cup \{\pm\infty\}$ be measurable. If both $\int_E f^+ \, dx$ and $\int_E f^- \, dx$ are finite, we say that f is* Lebesgue integrable *on E and define*

$$\int_E f \, dx = \int_E f^+ \, dx - \int_E f^- \, dx.$$

We write $f \in L^1(E)$ when $\int_E |f| \, dx < \infty$.

Theorem 4.2.2. *If $f \in L^1(E)$, then f is finite almost everywhere on E.*

Proof. For each $k \ge 1$ set $A_k = \{x \in E : |f(x)| \ge k\}$. Then

$$k \, m(A_k) \le \int_{A_k} |f| \, dx \le \int_E |f| \, dx < \infty,$$

so $m(A_k) \le k^{-1} \int_E |f| \, dx \to 0$. Since $\{|f| = \infty\} = \bigcap_{k \ge 1} A_k$, we conclude $m(\{|f| = \infty\}) = 0$. □

4.2.3 Structural properties in L^1

Theorem 4.2.3 (Finite additivity in the region)**.** *Let $E_1, E_2 \in \mathcal{L}^n$ be disjoint and set $E = E_1 \cup E_2$. If $f \in L^1(E)$, then $f \in L^1(E_1)$ and $f \in L^1(E_2)$ and*

$$\int_E f \, dx = \int_{E_1} f \, dx + \int_{E_2} f \, dx.$$

Proof. Apply the claim to f^+ and f^- separately (both are nonnegative and integrable), and subtract. □

Theorem 4.2.4 (Domination)**.** *Let $E \in \mathcal{L}^n$ and let f be measurable on E. If $|f| \le F$ a.e. on E for some $F \in L^1(E)$, then $f \in L^1(E)$.*

Proof. From $0 \le f^+ \le |f| \le F$ and $0 \le f^- \le |f| \le F$ we obtain $\int_E f^+ \le \int_E F < \infty$ and $\int_E f^- \le \int_E F < \infty$. □

Theorem 4.2.5 (Linearity)**.** *Let $E \in \mathcal{L}^n$, $f, g \in L^1(E)$, and $\alpha, \beta \in \mathbb{R}$. Then $\alpha f + \beta g \in L^1(E)$ and*

$$\int_E (\alpha f + \beta g)\, dx = \alpha \int_E f\, dx + \beta \int_E g\, dx.$$

Proof. Use the triangle inequality to see $|\alpha f + \beta g| \le |\alpha||f| + |\beta||g| \in L^1(E)$. Linearity of the integral follows from the corresponding property for nonnegative functions applied to positive/negative parts. □

Theorem 4.2.6 (Monotonicity)**.** *Let $f, g \in L^1(E)$. If $f \le g$ a.e. on E, then*

$$\int_E f\, dx \le \int_E g\, dx.$$

Proof. Since $g - f \ge 0$ a.e.,

$$\int_E g\, dx - \int_E f\, dx = \int_E (g - f)\, dx \ge 0.$$

□

Corollary 4.2.1 (Triangle inequality for the integral)**.** *If $f \in L^1(E)$, then*

$$\Big| \int_E f\, dx \Big| \le \int_E |f|\, dx.$$

Proof. Apply Theorem 4.2.6 to $-|f| \le f \le |f|$. □

Theorem 4.2.7. *Let $f \in L^1(E)$ with $f \ge 0$ a.e. If $\int_E f\, dx = 0$, then $f = 0$ a.e. on E.*

Proof. For $\alpha > 0$ define $A_\alpha = \{x \in E : f(x) \ge \alpha\}$. Then $\alpha \chi_{A_\alpha} \le f$, hence

$$\alpha\, m(A_\alpha) = \int_E \alpha \chi_{A_\alpha}\, dx \le \int_E f\, dx = 0.$$

Thus $m(A_\alpha) = 0$ for every $\alpha > 0$. Since $\{f > 0\} = \bigcup_{k \ge 1} A_{1/k}$, we obtain $m(\{f > 0\}) = 0$. □

Example 4.2.1. *The improper Riemann integral may exist while the Lebesgue integral fails due to lack of absolute integrability. For instance, $\sin x/x$ has a convergent improper Riemann integral on $[1, \infty)$, but $\int_1^\infty |\sin x/x|\, dx = \infty$, hence $\sin x/x \notin L^1([1, \infty))$.*

Theorem 4.2.8 (Absolute continuity of the integral)**.** *Let $f \in L^1(E)$. For every $\varepsilon > 0$ there exists $\delta > 0$ such that for every measurable $e \subset E$,*

$$m(e) < \delta \implies \Big| \int_e f\, dx \Big| < \varepsilon.$$

Proof. Fix $\varepsilon > 0$. Choose $M > 0$ so that

$$\int_{\{|f| > M\}} |f|\, dx < \frac{\varepsilon}{2}.$$

(This is possible since $|f| \in L^1(E)$.) Set $\delta = \varepsilon/(2M)$. If $m(e) < \delta$, then

$$\int_e |f|\, dx = \int_{e \cap \{|f| \le M\}} |f|\, dx + \int_{e \cap \{|f| > M\}} |f|\, dx \le M m(e) + \int_{\{|f| > M\}} |f|\, dx < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Therefore $|\int_e f| \le \int_e |f| < \varepsilon$. □

Theorem 4.2.9 (Countable additivity in the region)**.** *Let $E \in \mathcal{L}^n$ and let $\{E_k\}_{k\geq 1} \subset \mathcal{L}^n$ be pairwise disjoint with $E = \bigcup_{k\geq 1} E_k$. Then $f \in L^1(E)$ if and only if*

1. *$f \in L^1(E_k)$ for each k, and*
2. *$\sum_{k=1}^{\infty} \int_{E_k} |f|\,dx < \infty$.*

Moreover, when $f \in L^1(E)$,

$$\int_E f\,dx = \sum_{k=1}^{\infty} \int_{E_k} f\,dx.$$

Proof. Assume $f \in L^1(E)$. Then $\int_{E_k} |f| \leq \int_E |f| < \infty$, so $f \in L^1(E_k)$. Also, by monotone convergence applied to $\sum_{k=1}^{m} \chi_{E_k}|f| = \chi_{\cup_{k\leq m} E_k}|f|$,

$$\sum_{k=1}^{m} \int_{E_k} |f|\,dx = \int_{\cup_{k\leq m} E_k} |f|\,dx \uparrow \int_E |f|\,dx < \infty,$$

so the series converges. Finally, apply the same argument to f^+ and f^- to obtain the series representation for $\int_E f$.

Conversely, assume (1)–(2). Then

$$\int_E |f|\,dx = \sum_{k=1}^{\infty} \int_{E_k} |f|\,dx < \infty,$$

so $f \in L^1(E)$, and the identity for $\int_E f$ follows from the decomposition $f = f^+ - f^-$. □

Corollary 4.2.2 (Measures with density)**.** *Let $f \geq 0$ and $f \in L^1(\mathbb{R}^n)$. Define $\varphi(E) = \int_E f\,dx$ for $E \in \mathcal{L}^n$. Then φ is a measure on $(\mathbb{R}^n, \mathcal{L}^n)$.*

Proof. We have $\varphi(\varnothing) = 0$ and monotonicity is immediate. Countable additivity follows from Theorem 4.2.9 applied to f. □

Theorem 4.2.10. *If $f \in L^1(E)$, then*

$$\lim_{k\to\infty} k\, m(\{x \in E : |f(x)| \geq k\}) = 0.$$

Proof. Let $A_k = \{|f| \geq k\}$. Then

$$k\, m(A_k) \leq \int_{A_k} |f|\,dx.$$

Since $m(A_k) \to 0$ and the integral is absolutely continuous in the region (Theorem 4.2.8), the right-hand side tends to 0. □

Theorem 4.2.11 (Characterization by simple functions)**.** *Let $E \in \mathcal{L}^n$ and let $f \geq 0$ be measurable on E. Then $f \in L^1(E)$ if and only if*

$$\sup\Big\{ \int_E h\,dx \;\Big|\; 0 \leq h \leq f \text{ a.e. and } h \text{ is simple} \Big\} < \infty.$$

In this case, the supremum equals $\int_E f\,dx$.

Proof. This is exactly the definition (4.2). □

Exercises 4.2

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4.3 Limit Theorems for the Lebesgue Integral

The power of Lebesgue integration lies in its compatibility with limits. We begin with the monotone convergence theorem, then derive Fatou's lemma and the dominated convergence theorem.

Theorem 4.3.1 (Monotone convergence theorem (Levi)). *Let $E \in \mathcal{L}^n$ and let $\{f_n\}$ be measurable functions on E such that*

$$0 \le f_1 \le f_2 \le \cdots, \qquad f_n(x) \to f(x) \text{ a.e. on } E.$$

Then

$$\int_E f\,dx = \lim_{n\to\infty} \int_E f_n\,dx \in [0,\infty].$$

Proof. Set

$$\ell = \lim_{n\to\infty} \int_E f_n\,dx \in [0,\infty].$$

Monotonicity implies $\ell \le \int_E f\,dx$. We prove the reverse inequality $\int_E f\,dx \le \ell$.

Let φ be a nonnegative simple function on E with $\varphi \le f$. Fix $m \in \mathbb{N}$ with $m \ge 2$ and set

$$\varphi^{(m)} = (1 - 1/m)\,\varphi.$$

Then $0 \le \varphi^{(m)} \le \varphi \le f$.

Write $\varphi = \sum_{i=1}^{k} a_i \chi_{A_i}$ with $a_i \ge 0$ and pairwise disjoint measurable A_i. For each i and n define

$$A_{i,n}^{(m)} = A_i \cap \{x \in E : f_n(x) \ge (1-1/m)a_i\}.$$

Since $f_n \uparrow f$ a.e. and $f \ge a_i$ on A_i (because $\varphi \le f$), we have $f > (1-1/m)a_i$ on A_i whenever $a_i > 0$. Therefore $A_{i,n}^{(m)} \uparrow A_i$ up to a null set, and $m(A_{i,n}^{(m)}) \uparrow m(A_i)$. Define

$$\varphi_n^{(m)} = \sum_{i=1}^{k} (1-1/m) a_i\, \chi_{A_{i,n}^{(m)}}.$$

Then $\varphi_n^{(m)} \le f_n$ a.e., so

$$\int_E \varphi_n^{(m)}\,dx \le \int_E f_n\,dx \qquad \forall n.$$

Moreover,

$$\int_E \varphi_n^{(m)}\,dx = \sum_{i=1}^{k} (1-1/m) a_i\, m(A_{i,n}^{(m)}) \uparrow \sum_{i=1}^{k} (1-1/m) a_i\, m(A_i) = \int_E \varphi^{(m)}\,dx.$$

Passing to the limit in n yields $\int_E \varphi^{(m)}\,dx \le \ell$. Letting $m \to \infty$ gives $\int_E \varphi\,dx \le \ell$. Taking the supremum over all simple $\varphi \le f$ completes the proof. □

Theorem 4.3.2 (Fatou's lemma). *Let $E \in \mathcal{L}^n$ and let $\{f_n\}$ be nonnegative measurable functions on E. Then*

$$\int_E \liminf_{n\to\infty} f_n\,dx \le \liminf_{n\to\infty} \int_E f_n\,dx.$$

More generally, if there exists $h \in L^1(E)$ with $h \le f_n$ a.e. for all n, then

$$\int_E \liminf_{n\to\infty} f_n\,dx \le \liminf_{n\to\infty} \int_E f_n\,dx.$$

Proof. For the nonnegative case, define $\varphi_k(x) = \inf\{f_k(x), f_{k+1}(x), \dots\}$. Then φ_k is measurable, $\varphi_k \uparrow \liminf_n f_n$, and $\varphi_k \le f_n$ for all $n \ge k$. Hence

$$\int_E \varphi_k \, dx \le \inf_{n \ge k} \int_E f_n \, dx.$$

Letting $k \to \infty$ and applying Levi's theorem (Theorem 4.3.1) gives the desired inequality.

For the general case, apply the nonnegative case to $f_n - h \ge 0$ and note that $\liminf(f_n - h) = \liminf f_n - h$. □

Theorem 4.3.3 (Lebesgue dominated convergence theorem)**.** *Let $E \in \mathcal{L}^n$ and let f_n, f be measurable on E. Assume:*

1. *$f_n \to f$ a.e. on E,* or *(when $m(E) < \infty$) $f_n \xrightarrow{m} f$ on E;*
2. *there exists $F \in L^1(E)$ such that $|f_n| \le F$ a.e. on E for all n.*

Then $f \in L^1(E)$ and

$$\lim_{n\to\infty} \int_E f_n \, dx = \int_E f \, dx.$$

Proof. Since $|f| \le F$ a.e., we have $f \in L^1(E)$. Assume first that $f_n \to f$ a.e. Apply Fatou's lemma (Theorem 4.3.2) to the nonnegative functions $F \pm f_n$:

$$\int_E (F + f) \, dx \le \liminf_n \int_E (F + f_n) \, dx = \int_E F \, dx + \liminf_n \int_E f_n \, dx,$$

which implies

$$\int_E f \, dx \le \liminf_n \int_E f_n \, dx.$$

Similarly, applying Fatou to $F - f_n$ yields

$$\int_E (F - f) \, dx \le \liminf_n \int_E (F - f_n) \, dx = \int_E F \, dx - \limsup_n \int_E f_n \, dx,$$

so

$$\limsup_n \int_E f_n \, dx \le \int_E f \, dx.$$

Combining the two inequalities shows $\int_E f_n \to \int_E f$.

If instead $m(E) < \infty$ and $f_n \xrightarrow{m} f$, then every subsequence has a further subsequence converging to f almost everywhere (Theorem 3.3.4). Applying the a.e. version to that sub-subsequence shows that every subsequence of $\{\int_E f_n\}$ has a further subsequence converging to $\int_E f$. Therefore the whole sequence converges to $\int_E f$. □

Example 4.3.1. *Without an integrable dominating function, the conclusion can fail even under uniform convergence. Let*

$$f_k(x) = \begin{cases} 1/k, & x \in [0, k), \\ 0, & x \in [k, \infty), \end{cases} \qquad x \in [0, \infty).$$

Then $f_k \rightrightarrows 0$ on $[0, \infty)$, but

$$\int_{[0,\infty)} f_k(x) \, dx = 1 \not\to 0.$$

Example 4.3.2. *Pointwise almost everywhere convergence alone is not enough. On* $[0,1]$ *let*

$$f_k(x) = \begin{cases} k, & x \in [0, 1/k), \\ 0, & x \in [1/k, 1]. \end{cases}$$

Then $f_k \overset{a.e.}{\to} 0$ *but* $\int_{[0,1]} f_k \, dx = 1$ *for all* k.

Theorem 4.3.4 (Vitali convergence theorem)**.** *Let* $E \in \mathcal{L}^n$ *and* $\{f_k\} \subset L^1(E)$. *Assume:*

1. *(Uniform integrability) for every* $\varepsilon > 0$ *there exists* $\delta > 0$ *such that for every measurable* $e \subset E$,
$$m(e) < \delta \implies \sup_k \int_e |f_k| \, dx < \varepsilon;$$
2. *(Tightness) for every* $\varepsilon > 0$ *there exists a measurable set* $F \subset E$ *with* $m(F) < \infty$ *such that*
$$\sup_k \int_{E \setminus F} |f_k| \, dx < \varepsilon;$$
3. $f_k \overset{m}{\to} f$ *on* E *(or* $f_k \overset{a.e.}{\to} f$*).*

Then $f \in L^1(E)$ *and* $\int_E f_k \to \int_E f$.

Proof. Condition (2) reduces the problem to the finite-measure set F up to an arbitrarily small error. On F, uniform integrability (1) and convergence in measure (3) imply $\int_F |f_k - f| \to 0$ (a standard consequence of the dominated convergence theorem after truncation). The tail estimate on $E \setminus F$ then shows $\int_E |f_k - f| \to 0$, hence $\int_E f_k \to \int_E f$. □

Theorem 4.3.5 (Term-by-term integration)**.** *Let* $E \in \mathcal{L}^n$ *and let* $\{f_n\} \subset L^1(E)$ *satisfy*

$$\sum_{n=1}^{\infty} \int_E |f_n| \, dx < \infty.$$

Then the series $\sum_{n=1}^{\infty} f_n(x)$ *converges absolutely for almost every* $x \in E$. *If*

$$f(x) = \sum_{n=1}^{\infty} f_n(x),$$

then $f \in L^1(E)$ *and*

$$\int_E f \, dx = \sum_{n=1}^{\infty} \int_E f_n \, dx.$$

Proof. Let $F_m = \sum_{n=1}^{m} |f_n|$. Then $F_m \uparrow F$ and by Levi's theorem,

$$\int_E F \, dx = \lim_m \int_E F_m \, dx = \sum_{n=1}^{\infty} \int_E |f_n| \, dx < \infty.$$

Thus $F < \infty$ a.e., hence the series converges absolutely a.e. For the partial sums $S_m = \sum_{n=1}^{m} f_n$ we have $S_m \overset{a.e.}{\to} f$ and $|S_m| \le F \in L^1(E)$. Apply dominated convergence. □

Theorem 4.3.6 (Continuity of parameter-dependent integrals)**.** *Let* $E \in \mathcal{L}^n$ *and let* $B(y_0, \delta) \subset \mathbb{R}^m$. *Suppose* $f : E \times B(y_0, \delta) \to \mathbb{R}$ *satisfies:*

1. *for each y, the section $f(\cdot, y)$ is measurable on E;*

2. *for almost every x, the map $y \mapsto f(x, y)$ is continuous on $B(y_0, \delta)$;*

3. *there exists $F \in L^1(E)$ such that $|f(x, y)| \leq F(x)$ a.e. for all y.*

Then

$$I(y) = \int_E f(x, y)\, dx$$

defines a continuous function on $B(y_0, \delta)$.

Proof. If $y_k \to y$, then $f(\cdot, y_k) \to f(\cdot, y)$ a.e. Moreover $|f(\cdot, y_k)| \leq F \in L^1(E)$. Apply dominated convergence. □

Theorem 4.3.7 (Differentiation under the integral sign)**.** *Let $E \in \mathcal{L}^n$ and let $f : E \times (a, b) \to \mathbb{R}$ satisfy:*

1. *for each $y \in (a, b)$, the section $f(\cdot, y)$ is measurable;*

2. *for almost every x, the function $y \mapsto f(x, y)$ is differentiable on (a, b);*

3. *there exists $F \in L^1(E)$ such that $|\partial_y f(x, y)| \leq F(x)$ a.e. for all $y \in (a, b)$.*

Then $I(y) = \int_E f(x, y)\, dx$ is differentiable on (a, b) and

$$I'(y) = \int_E \partial_y f(x, y)\, dx.$$

Proof. Fix $y \in (a, b)$. For small h define the difference quotient

$$g_h(x) = \frac{f(x, y + h) - f(x, y)}{h}.$$

By differentiability, $g_h(x) \to \partial_y f(x, y)$ for a.e. x. Also $|g_h(x)| \leq F(x)$ a.e. by the mean value theorem applied pointwise in y. Apply dominated convergence to conclude

$$\lim_{h \to 0} \frac{I(y + h) - I(y)}{h} = \int_E \partial_y f(x, y)\, dx.$$ □

Exercises 4.3

1. Page 185: Problems 1, 5, 6, 7.

2. Page 191: Problem 11.

3. Page 222: Problems 12, 18, 19, 20, 21, 23.

4.4 Relation Between Lebesgue Integrable Functions and Continuous Functions

A key practical message is that L^1 functions, while possibly very irregular pointwise, can be approximated in the integral norm by continuous compactly supported functions.

Theorem 4.4.1 (Density of $C_c(\mathbb{R}^n)$ in $L^1(\mathbb{R}^n)$)**.** *If $f \in L^1(\mathbb{R}^n)$, then for every $\varepsilon > 0$ there exists a continuous, compactly supported function g on $\mathbb{R}^n$ such that*

$$\int_{\mathbb{R}^n} |f(x) - g(x)|\, dx < \varepsilon.$$

Proof. Fix $\varepsilon > 0$.

Step 1: reduce to a bounded function with bounded support. Choose $R > 0$ such that

$$\int_{\mathbb{R}^n \setminus B(0,R)} |f|\, dx < \frac{\varepsilon}{3}.$$

Choose $M > 0$ such that

$$\int_{B(0,R)} (|f| - \min\{|f|, M\})\, dx < \frac{\varepsilon}{3}.$$

Set $f_{R,M} = \chi_{B(0,R)} \operatorname{sgn}(f) \min\{|f|, M\}$. Then $\|f - f_{R,M}\|_{L^1} < 2\varepsilon/3$.

Step 2: approximate by a continuous function on a large closed set. By Lusin's theorem (Theorem 3.2.2), there exists a closed set $F \subset B(0,R)$ such that $m(B(0,R) \setminus F)$ is small and $f_{R,M}$ is continuous on F. By Tietze's extension theorem in $\mathbb{R}^n$ (Theorem 1.3.10), extend $f_{R,M}|_F$ to a continuous function $\tilde{g}$ on $\mathbb{R}^n$. Choose a continuous cutoff function $\eta \in C_c(\mathbb{R}^n)$ with $\eta \equiv 1$ on $B(0,R)$. Set $g = \eta\, \tilde{g} \in C_c(\mathbb{R}^n)$. On F we have $g = f_{R,M}$.

Step 3: control the L^1 error. Since $|f_{R,M}| \le M$ and $|g|$ is bounded on $B(0,R)$, we can bound

$$\int_{\mathbb{R}^n} |f_{R,M} - g|\, dx = \int_{B(0,R)\setminus F} |f_{R,M} - g|\, dx$$

by a constant times $m(B(0,R)\setminus F)$. Choosing F so that this is $< \varepsilon/3$ yields $\|f_{R,M} - g\|_{L^1} < \varepsilon/3$. Combining with Step 1 gives $\|f - g\|_{L^1} < \varepsilon$. □

Corollary 4.4.1 (Continuity of translations in L^1)**.** *If $f \in L^1(\mathbb{R}^n)$, then*

$$\lim_{h \to 0} \int_{\mathbb{R}^n} |f(x+h) - f(x)|\, dx = 0.$$

Proof. Fix $\varepsilon > 0$. Choose $g \in C_c(\mathbb{R}^n)$ with $\|f - g\|_{L^1} < \varepsilon/3$. Then

$$\|f(\cdot + h) - f\|_{L^1} \le \|f(\cdot + h) - g(\cdot + h)\|_{L^1} + \|g(\cdot + h) - g\|_{L^1} + \|g - f\|_{L^1}.$$

The first and third terms equal $\|f - g\|_{L^1} < \varepsilon/3$ by translation invariance of Lebesgue measure. Since g is uniformly continuous and compactly supported, $\|g(\cdot + h) - g\|_{L^1} \to 0$ as $h \to 0$. Thus for $|h|$ small enough, the middle term is $< \varepsilon/3$, proving the claim. □

4.5 Fubini Theorem

We now study multiple integrals and the exchange of integration order. Write $\mathbb{R}^{p+q} = \mathbb{R}^p \times \mathbb{R}^q$ and denote points by (x, y) with $x \in \mathbb{R}^p$ and $y \in \mathbb{R}^q$.

Theorem 4.5.1 (Tonelli)**.** *Let $f : \mathbb{R}^{p+q} \to [0, \infty]$ be measurable. Then:*

1. *for almost every $x \in \mathbb{R}^p$, the function $y \mapsto f(x, y)$ is measurable on $\mathbb{R}^q$;*
2. *the function*
$$F(x) = \int_{\mathbb{R}^q} f(x, y)\, dy \in [0, \infty]$$
is measurable on $\mathbb{R}^p$;
3. *we have the equality (allowing the value $+\infty$)*
$$\int_{\mathbb{R}^{p+q}} f(x, y)\, dx\, dy = \int_{\mathbb{R}^p} \left(\int_{\mathbb{R}^q} f(x, y)\, dy \right) dx.$$

The same conclusions hold with the roles of x and y interchanged.

Proof sketch. For indicator functions of measurable rectangles $A \times B$, the conclusion follows from $m(A \times B) = m(A)m(B)$. Extend to finite linear combinations (simple functions) by linearity. For general $f \geq 0$, choose simple functions $\varphi_n \uparrow f$ a.e. Apply monotone convergence (Theorem 4.3.1) first to the y-integrals and then to the x-integral to pass to the limit. □

Theorem 4.5.2 (Fubini)**.** *Let $f \in L^1(\mathbb{R}^{p+q})$. Then for almost every $x \in \mathbb{R}^p$ the section $f(x, \cdot)$ belongs to $L^1(\mathbb{R}^q)$, the function*
$$F(x) = \int_{\mathbb{R}^q} f(x, y)\, dy$$
is integrable on $\mathbb{R}^p$, and
$$\int_{\mathbb{R}^{p+q}} f(x, y)\, dx\, dy = \int_{\mathbb{R}^p} \left(\int_{\mathbb{R}^q} f(x, y)\, dy \right) dx = \int_{\mathbb{R}^q} \left(\int_{\mathbb{R}^p} f(x, y)\, dx \right) dy.$$

Proof. Apply Tonelli's theorem (Theorem 4.5.1) to f^+ and f^- separately and subtract. □

Definition 4.5.1 (Section)**.** *Let $E \subset \mathbb{R}^{p+q}$. For each $x \in \mathbb{R}^p$, define the* section of E at x *by*
$$E(x) = \{y \in \mathbb{R}^q : (x, y) \in E\}.$$

Remark 4.5.1. *If $E \in \mathcal{L}^{p+q}$, Tonelli's theorem implies that $E(x) \in \mathcal{L}^q$ for almost every x and that*
$$m(E) = \int_{\mathbb{R}^p} m\big(E(x)\big)\, dx.$$

Corollary 4.5.1 (Layer-cake representation)**.** *Let $E \in \mathcal{L}^n$ and let $f : E \to [0, \infty]$ be measurable. Then*
$$\int_E f(x)\, dx = \int_0^\infty m\big(\{x \in E : f(x) > t\}\big)\, dt.$$

Proof. Consider the measurable set
$$D = \{(x, t) \in E \times [0, \infty) : 0 \leq t \leq f(x)\}.$$
Compute $m(D)$ first by integrating χ_D in the t-variable and then by integrating in the x-variable. Tonelli's theorem identifies both computations and yields the formula. □

Corollary 4.5.2 (Measure of products)**.** *Let $A \in \mathcal{L}^p$ and $B \in \mathcal{L}^q$. If at least one of $m(A), m(B)$ is finite, then $A \times B \in \mathcal{L}^{p+q}$ and*

$$m(A \times B) = m(A)\, m(B).$$

Proof. Apply Tonelli to $\chi_{A\times B}(x, y) = \chi_A(x)\chi_B(y)$. □

Chapter 5

Measure Derivatives and the Newton-Leibniz Formula

At the beginning of this chapter we introduce an abstract measure class and then develop differentiation of measures and applications.

Definition 5.0.1 (Radon Measure)**.** *A measure μ on a topological space containing all Borel sets is called a Radon measure if it is inner regular and locally finite. Inner regular means: for every measurable set E,*

$$\mu(E) = \sup\{\mu(F) \mid F \subset E,\ F \text{ closed}\}.$$

Locally finite means: for every x, there is an open neighborhood U of x with $\mu(U) < \infty$.

Our discussion in this chapter is centered around Radon measures.

5.1 Vitali Covering Theorem

Definition 5.1.1 (Vitali Covering)**.** *Let $E \subset \mathbb{R}^n$, and let $\mathcal{F}$ be a family of closed balls in $\mathbb{R}^n$. We call $\mathcal{F}$ a Vitali (fine) cover of E if for every $x \in E$ and every $\varepsilon > 0$, there exists $B_x^\varepsilon \in \mathcal{F}$ such that*

$$x \in B_x^\varepsilon, \quad \operatorname{diam}(B_x^\varepsilon) < \varepsilon.$$

If $\mathcal{F}$ is a fine cover of E, then

$$\mathcal{F}_\delta = \{B \in \mathcal{F} \mid \operatorname{diam}(B) < \delta\}$$

is also a fine cover of E.

If additionally $\mu^*(E) < \infty$ and there exists open $G \supset E$ with $\mu(G) < \infty$, then

$$\mathcal{F}_\delta^G = \{B \in \mathcal{F} \mid B \subset G,\ \operatorname{diam}(B) < \delta\}$$

is still a fine cover of E.

Theorem 5.1.1 (Vitali)**.** *Let $\mathcal{F}$ be a family of nondegenerate closed balls in $\mathbb{R}^n$ with*

$$d^* = \sup\{\operatorname{diam}(B) \mid B \in \mathcal{F}\} < \infty.$$

Then there exists an at most countable disjoint subfamily $\mathcal{G} \subset \mathcal{F}$ such that

$$\bigcup_{B\in\mathcal{F}} B \subset \bigcup_{B\in\mathcal{G}} \widehat{B},$$

where $\widehat{B}$ is the concentric ball with radius 5 times that of B.

Proof. Partition $\mathcal{F}$ by dyadic diameter levels and choose maximal disjoint subfamilies inductively. Any remaining ball at level $k+1$ must intersect a previously chosen ball of level at most $k+1$. The diameter comparison yields containment inside a 5-dilate of the chosen intersecting ball. □

Corollary 5.1.1. *Let $E \subset \mathbb{R}^n$ and let $\mathcal{F}$ be a fine cover of E by closed balls, with bounded diameters. Then there exists a countable disjoint subfamily $\mathcal{G} \subset \mathcal{F}$ such that for every finite subset*

$$\{B_1, \dots, B_k\} \subset \mathcal{G},$$

one has

$$E\setminus \bigcup_{j=1}^{k} B_j \subset \bigcup_{B\in\mathcal{G}\setminus\{B_1,\dots,B_k\}} \widehat{B}.$$

Proof. Apply Theorem 5.1.1 to the remaining part after removing finitely many selected balls. □

Corollary 5.1.2. *Let μ be a nonnegative Radon measure on $\mathbb{R}^n$, let $E \subset \mathbb{R}^n$ with $\mu^*(E) < \infty$, and let $\mathcal{F}$ be a fine cover of E. Then for every $\varepsilon > 0$, there are finitely many pairwise disjoint balls $B_1, \dots, B_k \in \mathcal{F}$ such that*

$$\mu^*\Big(E\setminus \bigcup_{j=1}^{k} B_j\Big) < \varepsilon.$$

Proof. Restrict to balls contained in an open $G \supset E$ with finite measure, use Corollary 5.1.1, and then use countable additivity and tail control of measure. □

Corollary 5.1.3. *Let μ be a nonnegative Radon measure on $\mathbb{R}^n$, let $E \subset \mathbb{R}^n$, and let $\mathcal{F}$ be a fine cover of E. Then there is a countable disjoint family $\{B_j\} \subset \mathcal{F}$ such that*

$$\mu^*\Big(E\setminus \bigcup_{j=1}^{\infty} B_j\Big) = 0.$$

Proof. Localize on bounded annuli, apply Corollary 5.1.2 on each localization with geometric errors, then add a final fine-cover selection on the remainder. □

Remark 5.1.1. *1. These results play the same role as finite covering arguments in classical analysis: local estimates on balls are upgraded to global statements by almost full disjoint-ball coverings.*

2. Compare the strength of the four statements carefully. Theorem 5.1.1 assumes only a bounded-diameter ball family and gives coverage after dilation. Corollary 5.1.3 requires fine covering and yields measure-theoretic almost-full coverage by disjoint original balls.

Exercises 5.1

1. P248 1,2

5.2 Hahn Decomposition Theorem

Definition 5.2.1 (Positive Set, Negative Set)**.** *Let μ be a Radon measure on $\mathbb{R}^n$. A set $A \subset \mathbb{R}^n$ is called positive for μ if for every μ-measurable E,*

$$\mu(A \cap E) \geq 0.$$

A set $B \subset \mathbb{R}^n$ is called negative for μ if for every μ-measurable E,

$$\mu(B \cap E) \leq 0.$$

Remark 5.2.1. *If A is positive, then A is measurable and $\mu(A) \geq 0$. But $\mu(A) \geq 0$ alone does not imply A is positive.*

The goal is to decompose $\mathbb{R}^n$ into one positive and one negative part.

Lemma 5.2.1. *Let μ be a Radon measure on $\mathbb{R}^n$.*

(1) *If A_1, A_2 are positive sets, then $A_1 \cup A_2$, $A_1 \cap A_2$, and $A_1 \backslash A_2$ are positive.*

(2) *If B_1, B_2 are negative sets, then $B_1 \cup B_2$, $B_1 \cap B_2$, and $B_1 \backslash B_2$ are negative.*

(3) *If A is positive and B is negative, then for every measurable E,*

$$\mu(E \cap A \cap B) = 0.$$

Proof. For (1), use set decompositions of intersections and sign constraints term by term. Part (2) follows from (1) by replacing μ with $-\mu$. Part (3) is immediate from both inequalities. □

Theorem 5.2.1 (Hahn Decomposition Theorem)**.** *For any Radon measure μ on $\mathbb{R}^n$, there exist a positive set A and a negative set B such that*

$$A \cap B = \varnothing, \quad \mathbb{R}^n = A \cup B.$$

Proof. Work first on each annulus

$$M_k = \{x \in \mathbb{R}^n \mid k \leq \|x\| < k + 1\}.$$

Take the infimum of μ over negative subsets of M_k, build an increasing sequence of negative sets approaching this infimum, and pass to the union. The complement is then shown to be positive by contradiction using maximality of the infimum and Lemma 5.2.1. Combine annuli. □

Remark 5.2.2. *Define*

$$\mu^+(E) = \mu(E \cap A), \quad \mu^-(E) = -\mu(E \cap B).$$

Then

$$\mu = \mu^+ - \mu^-.$$

This is the Jordan decomposition of a signed Radon measure.

5.3 Derivatives of Radon Measures

We now extend the derivative concept from analysis to Radon measures.

Definition 5.3.1 (Derivative)**.** *Let ν, μ be nonnegative Radon measures on $\mathbb{R}^n$. For each $x \in \mathbb{R}^n$, define*

$$\overline{D}_\mu \nu(x) = \begin{cases} \overline{\lim}\limits_{r \to 0^+} \dfrac{\nu(B(x,r))}{\mu(B(x,r))}, & \mu(B(x,r)) > 0 \ \forall r > 0, \\ \infty, & \exists r > 0 \text{ with } \mu(B(x,r)) = 0, \end{cases}$$

$$\underline{D}_\mu \nu(x) = \begin{cases} \underline{\lim}\limits_{r \to 0^+} \dfrac{\nu(B(x,r))}{\mu(B(x,r))}, & \mu(B(x,r)) > 0 \ \forall r > 0, \\ \infty, & \exists r > 0 \text{ with } \mu(B(x,r)) = 0. \end{cases}$$

If upper and lower derivatives coincide, we denote the common value by

$$D_\mu \nu(x).$$

Example 5.3.1 (Average Value for Continuous Density)**.** *Let $f \geq 0$ be continuous and locally integrable on $\mathbb{R}^n$. Let $\mu = m$ (Lebesgue measure) and*

$$\nu(E) = \int_E f(x)\, dx.$$

Then

$$D_\mu \nu(x) = \lim_{r \to 0} \frac{\int_{B(x,r)} f(y)\, dy}{m(B(x,r))} = f(x).$$

Lemma 5.3.1. *Let ν, μ be nonnegative Radon measures on $\mathbb{R}^n$.*

(1) *If*

$$E \subset \{x \in \mathbb{R}^n \mid \underline{D}_\mu \nu(x) \leq a\},$$

then

$$\nu^*(E) \leq a\, \mu^*(E).$$

(2) *If*

$$E \subset \{x \in \mathbb{R}^n \mid \overline{D}_\mu \nu(x) \geq b\},$$

then

$$\nu^*(E) \geq b\, \mu^*(E).$$

Proof. Use Vitali covering selections built from balls where the local ratio is near the liminf or limsup threshold, then apply Corollary 5.1.3 and countable additivity. □

Theorem 5.3.1. *Let ν, μ be nonnegative Radon measures on $\mathbb{R}^n$.*

(1) *$D_\mu \nu$ exists and is finite μ-a.e. on $\mathbb{R}^n$.*

(2) *$D_\mu \nu$ is μ-measurable.*

Proof. For existence and finiteness, apply Lemma 5.3.1 to sets where $\overline{D}_\mu \nu = \infty$ and where $\underline{D}_\mu \nu < \overline{D}_\mu \nu$. Both have zero μ-measure. For measurability, approximate derivatives through measurable ball-measure maps

$$x \mapsto \nu(B(x,r)), \quad x \mapsto \mu(B(x,r)).$$

Then pass to limsup/liminf operations. □

For a measurable set E with $\mu(E) > 0$, write

$$\fint_E f\,d\mu = \frac{1}{\mu(E)}\int_E f\,d\mu.$$

Theorem 5.3.2 (Average Value Theorem)**.** *If $f \in L^1_{\text{loc}}(\mathbb{R}^n)$, then*

$$\lim_{r\to 0}\fint_{B(x,r)} f(y)\,dy = f(x)$$

for a.e. $x \in \mathbb{R}^n$.

Proof. This is the Lebesgue differentiation theorem. Approximate f in L^1_{loc} by continuous functions and apply Lemma 5.3.1 to the error measure. □

Remark 5.3.1. *A stronger form is*

$$\lim_{r\to 0}\fint_{B(x,r)} |f(y) - f(x)|\,dy = 0$$

a.e. Such points are called Lebesgue points.

Corollary 5.3.1. *Let $E \subset \mathbb{R}^n$ be Lebesgue measurable. Then for a.e. $x \in E$,*

$$\lim_{r\to 0}\frac{m(E \cap B(x,r))}{m(B(x,r))} = 1.$$

Proof. Apply Theorem 5.3.2 to $f = \chi_E$. □

Remark 5.3.2. *For positive-measure nowhere-dense perfect sets, the pointwise ratio is always < 1 for each fixed $r > 0$, yet the limit above is still 1 a.e.*

Exercises 5.3

1. Let $\{x_k\}, \{y_k\} \subset \mathbb{R}^n$ be discrete sequences and define

$$\mu(E) = \sum_{k=1}^{\infty}\frac{1}{2^k}\delta_{x_k}(E) + m(E), \quad \nu(E) = \sum_{k=1}^{\infty}\frac{1}{2^k}\delta_{y_k}(E).$$

   Find $D_\mu\nu(x)$.

5.4 Radon-Nikodym Theorem

For Radon measures we have the analog of the Newton-Leibniz principle.

Definition 5.4.1 (Absolute Continuity)**.** *Let ν, μ be Radon measures on $\mathbb{R}^n$. We say ν is absolutely continuous with respect to μ (written $\nu \ll \mu$) if for every measurable E,*

$$|\mu|(E) = 0 \implies |\nu|(E) = 0,$$

where $|\mu| = \mu^+ + \mu^-$. When $\mu \geq 0$, this is equivalent to $\mu(E) = 0 \Rightarrow \nu(E) = 0$.

Theorem 5.4.1 (Radon-Nikodym)**.** *Let μ be a nonnegative Radon measure on $\mathbb{R}^n$, and let ν be a Radon measure with $\nu \ll \mu$. Then there exists a locally μ-integrable function f such that for every measurable $E \subset \mathbb{R}^n$,*

$$\nu(E) = \int_E f \, d\mu = \int_E D_\mu \nu \, d\mu. \tag{5.1}$$

Proof. Step 1: every μ-measurable set is ν-measurable under $\nu \ll \mu$.

Step 2: for

$$E^\infty = \{D_\mu \nu = \infty\}, \quad E_0 = \{D_\mu \nu = 0\},$$

prove

$$\nu(E^\infty) = \int_{E^\infty} D_\mu \nu \, d\mu, \quad \nu(E_0) = \int_{E_0} D_\mu \nu \, d\mu.$$

Step 3: decompose any measurable E into derivative level sets

$$E_k = \{x \in E \mid D_\mu \nu(x) \in [t^k, t^{k+1})\}, \quad t > 1,$$

and sandwich $\nu(E)$ between $t \int_E D_\mu \nu \, d\mu$ and $t^{-1} \int_E D_\mu \nu \, d\mu$. Let $t \to 1$. □

Corollary 5.4.1. *Let μ be a nonnegative Radon measure on $\mathbb{R}^n$ and let $f \in L^1_{\text{loc}}(\mu)$. Then*

$$\lim_{r \to 0} \fint_{B(x,r)} f(y) \, d\mu = f(x)$$

for μ-a.e. x.

Proof. Apply Theorem 5.4.1 to the signed measure

$$\nu(E) = \int_E f \, d\mu.$$

Then $f = D_\mu \nu$ a.e. □

Remark 5.4.1. *Formally,*

$$D_\mu \left(\int f \, d\mu \right) = f \quad a.e.$$

so measure differentiation is the inverse of integration.

Definition 5.4.2 (Mutual Singularity)**.** *Let ν, μ be nonnegative Radon measures on $\mathbb{R}^n$. We write $\nu \perp \mu$ if there exists measurable $B \subset \mathbb{R}^n$ such that*

$$\mu(\mathbb{R}^n \backslash B) = 0, \quad \nu(B) = 0.$$

Remark 5.4.2. *In general, singularity does not imply absolute continuity.*

Theorem 5.4.2 (Lebesgue Decomposition Theorem)**.** *Let $\mu \geq 0$ be a Radon measure on $\mathbb{R}^n$, and let ν be a Radon measure. Then*

$$\nu = \nu_{\text{AC}} + \nu_s,$$

where $\nu_{\text{AC}} \ll \mu$ and $\nu_s \perp \mu$.

Proof. Define

$$\mathcal{E} = \{A \subset \mathbb{R}^n \mid \mu(\mathbb{R}^n \backslash A) = 0\},$$

choose $B \in \mathcal{E}$ minimizing $\nu(A)$ over $\mathcal{E}$ (via infimizing sequence and intersection), and set

$$\nu_{\mathrm{AC}} = \nu|_B, \quad \nu_s = \nu|_{\mathbb{R}^n \backslash B}.$$

Minimality gives $\nu_{\mathrm{AC}} \ll \mu$, while $\nu_s \perp \mu$ follows from definition of $\mathcal{E}$. □

Remark 5.4.3. *For nonnegative μ,*

$$D_\mu \nu = D_\mu \nu_{\mathrm{AC}} \quad \mu\text{-a.e.}, \qquad D_\mu \nu_s = 0 \quad \mu\text{-a.e.}$$

under the theorem assumptions.

Example 5.4.1. *With $\mu = m$ on $\mathbb{R}$,*

$$\nu(E) = \int_E f(x)\, d\mu + \sum_{i=1}^{\infty} \alpha_i \delta_{x_i}(E)$$

is the sum of an absolutely continuous part and a singular atomic part.

Exercises 5.4

1. Does there exist a non-atomic Radon measure singular to Lebesgue measure?

5.5 Monotone Functions and Functions of Bounded Variation

5.5.1 Monotone Functions

This section parallels the previous one because a monotone function induces a measure.

Theorem 5.5.1. *Let f be monotone increasing on $[a, b]$. Then f is differentiable a.e. on $[a, b]$.*

Example 5.5.1 (Monotone Function Discontinuous on a Dense Set)**.** *Let $\sum(\alpha_n + \beta_n) < \infty$ with $\alpha_n, \beta_n > 0$, let $\{x_n\} = \mathbb{Q}$, and define*

$$f(x) = \sum_{n=1}^{\infty} \alpha_n H_0(x - x_n) + \sum_{n=1}^{\infty} \beta_n H_1(x - x_n),$$

where

$$H_0(x) = \begin{cases} 1, & x > 0, \\ 0, & x \le 0, \end{cases} \qquad H_1(x) = \begin{cases} 1, & x \ge 0, \\ 0, & x < 0. \end{cases}$$

Then f has dense discontinuities but is still differentiable a.e. by Theorem 5.5.1.

Remark 5.5.1. *For background on the Heaviside function, see [4].*

Lemma 5.5.1. *Let f be monotone increasing on $[a,b]$. If*

$$E \subset \{x \in [a,b] \mid \underline{D}f(x) \le \alpha\},$$

then

$$f^*(E) \le \alpha\, m^*(E).$$

If

$$E \subset \{x \in [a,b] \mid \overline{D}f(x) \ge \beta\},$$

then

$$f^*(E) \ge \beta\, m^*(E),$$

where

$$\underline{D}f(x) = \varliminf_{h\to 0} \frac{f(x+h)-f(x)}{h}, \quad \overline{D}f(x) = \varlimsup_{h\to 0} \frac{f(x+h)-f(x)}{h},$$

and $f^*(E) = m^*(f(E))$.

Proof. Construct fine covers by intervals on which difference quotients satisfy the desired bound, apply Vitali covering, and estimate image measure by interval lengths. □

Proof of Theorem 5.5.1. Define

$$E^\infty = \{\overline{D}f = \infty\}.$$

By Lemma 5.5.1, $m(E^\infty) = 0$. For

$$E_\alpha^\beta = \{\underline{D}f \le \alpha < \beta \le \overline{D}f\},$$

Lemma 5.5.1 gives

$$\beta m^*(E_\alpha^\beta) \le f^*(E_\alpha^\beta) \le \alpha m^*(E_\alpha^\beta),$$

so $m(E_\alpha^\beta) = 0$. Take countable rational pairs $\alpha < \beta$. □

Theorem 5.5.2. *If f is monotone increasing on $[a,b]$, then*

$$\int_a^b f'(x)\,dx \le f(b) - f(a).$$

Proof. Use difference quotients

$$g_k(x) = \frac{f(x+1/k)-f(x)}{1/k} \ge 0,$$

then apply Fatou/Levi type passage and monotonicity bounds. □

Theorem 5.5.3 (Fubini Term-by-Term Differentiation)**.** *Let f_k be monotone increasing on $[a,b]$, and suppose $\sum_{k=1}^\infty f_k(x)$ converges a.e. on $[a,b]$. Then*

$$\frac{d}{dx}\Big(\sum_{k=1}^\infty f_k(x)\Big) = \sum_{k=1}^\infty f_k'(x)$$

a.e. on $[a,b]$.

Proof. Set $S = \sum f_k$, so S is monotone and differentiable a.e. Then write

$$S'(x) = \sum_{k=1}^n f_k'(x) + r_n'(x),$$

with $r_n = \sum_{k=n+1}^\infty f_k$. Show $r_n' \downarrow 0$ a.e. via Theorem 5.5.2 and convergence of the tail at endpoints. Let $n \to \infty$. □

5.5.2 Functions of Bounded Variation

Definition 5.5.1 (Function of Bounded Variation)**.** *Let f be finite-valued on $[a,b]$. For a partition*

$$a = x_0 \le x_1 \le \cdots \le x_n = b,$$

define

$$\bigvee_f(\Delta) = \sum_{i=1}^{n} |f(x_i) - f(x_{i-1})|.$$

If

$$\bigvee_a^b(f) = \sup\{\bigvee_f(\Delta) \mid \Delta \text{ partition of } [a,b]\} < \infty,$$

then $f \in BV[a,b]$ and $\bigvee_a^b(f)$ is called total variation.

Example 5.5.2. *Every bounded monotone function on $[a,b]$ is in $BV[a,b]$, and*

$$\bigvee_a^b(f) = |f(b) - f(a)|.$$

Example 5.5.3. *If f is Lipschitz on $[a,b]$ with constant L, then $f \in BV[a,b]$ and*

$$\bigvee_a^b(f) \le L|b-a|.$$

Example 5.5.4 (Continuous but Not of Bounded Variation)**.**

$$f(x) = \begin{cases} x\sin\left(\dfrac{1}{x^2}\right), & x \in (0,1], \\ 0, & x = 0. \end{cases}$$

Theorem 5.5.4. *Functions in $BV[a,b]$ satisfy:*

(1) *boundedness;*

(2) *linear closure and*

$$\bigvee_a^b(\alpha f + \beta g) \le |\alpha| \bigvee_a^b(f) + |\beta| \bigvee_a^b(g);$$

(3) *product closure with standard estimate;*

(4) $\bigvee_a^b(f) = 0 \iff f$ *constant;*

(5) *restriction monotonicity on subintervals;*

(6) *additivity of total variation at a split point c:*

$$\bigvee_a^b(f) = \bigvee_a^c(f) + \bigvee_c^b(f);$$

(7) *if $f_n \in BV[a,b]$, $\sup_n \bigvee_a^b(f_n) \le M$, and $f_n \to f$ pointwise, then $f \in BV[a,b]$ and*

$$\bigvee_a^b(f) \le M.$$

Proof. Only (6) and (7) are nontrivial here. For (6), refine partitions by inserting c and pass to suprema in both directions. For (7), for each fixed partition pass limit inside finite sums:

$$\bigvee_f(\Delta) = \lim_{n\to\infty} \bigvee_{f_n}(\Delta) \le M,$$

then take supremum over Δ. □

Remark 5.5.2. *Property* (6) *implies*

$$x \mapsto \bigvee_a^x(f)$$

is monotone increasing.

Theorem 5.5.5 (Jordan Decomposition Theorem)**.** *If $f \in BV[a,b]$ and*

$$g(x) = \frac{1}{2}\Big(\bigvee_a^x(f) + f(x)\Big), \quad h(x) = \frac{1}{2}\Big(\bigvee_a^x(f) - f(x)\Big),$$

then g, h are increasing on $[a,b]$ and

$$f = g - h.$$

Proof. For $x \le y$,

$$|f(y) - f(x)| \le \bigvee_x^y(f) = \bigvee_a^y(f) - \bigvee_a^x(f),$$

which gives monotonicity of both $\bigvee_a^x(f) \pm f(x)$. □

Corollary 5.5.1. *If $f \in BV[a,b]$, then its discontinuity set is at most countable, f is differentiable a.e., and $f' \in L^1[a,b]$.*

Theorem 5.5.6. *If $f \in BV[a,b]$, then f and $\bigvee_a^x(f)$ have the same continuity/discontinuity points, and at each discontinuity x_0 the one-sided oscillations satisfy*

$$\omega_f^+(x_0) = \lim_{x\to x_0^+} |f(x) - f(x_0)| = \lim_{x\to x_0^+} \bigvee_{x_0}^x(f),$$

$$\omega_f^-(x_0) = \lim_{x\to x_0^-} |f(x) - f(x_0)| = \lim_{x\to x_0^-} \bigvee_{x_0}^x(f).$$

Proof. One inequality is immediate from

$$|f(x) - f(x_0)| \le \bigvee_{x_0}^x(f).$$

The reverse inequality follows by choosing near-optimal partitions and isolating the first subinterval. □

Theorem 5.5.7. *If $f \in BV[a,b]$, then*

$$\frac{d}{dx}\Big(\bigvee_a^x(f)\Big) = |f'(x)|$$

a.e. on $[a,b]$.

Proof. Approximate variation by piecewise monotone envelopes and use the term-by-term differentiation theorem (Theorem 5.5.3) on a suitable positive series of monotone error terms. □

Remark 5.5.3. *Consequently,*

$$\int_a^b |f'(x)|\,dx \le \bigvee_a^b (f).$$

Theorem 5.5.8 (Hally)**.** *Let $\mathcal{F} \subset BV[a,b]$ with*

$$\sup_{f\in\mathcal{F}} \Big(|f(a)| + \bigvee_a^b (f)\Big) < M.$$

Then every sequence in $\mathcal{F}$ has a subsequence converging pointwise to a function in $BV[a,b]$.

Proof. First handle monotone families via diagonal extraction on rational points. Extend limits by monotonicity and adjust values at countably many discontinuities. Then reduce general BV case through Jordan decomposition. □

Exercises 5.5

1. P255 10,11,12
2. P281 11
3. P283 7,8

5.6 Absolutely Continuous Functions

This section gives necessary and sufficient conditions for the Newton-Leibniz formula.

Theorem 5.6.1. *If $f \in L^1[a,b]$ and*

$$F(x) = \int_a^x f(t)\,dt,$$

then F is differentiable a.e. and

$$F'(x) = f(x) \quad a.e.$$

Proof. Write

$$F(x) = \int_a^x f^+(t)\,dt - \int_a^x f^-(t)\,dt.$$

Then $F \in BV[a,b]$, hence differentiable a.e. Use approximation of f by continuous functions in L^1 together with variation estimates to conclude $F' = f$ a.e. □

Definition 5.6.1 (Absolute Continuity)**.** *A real-valued function f on $[a,b]$ is absolutely continuous if for every $\varepsilon > 0$ there exists $\delta > 0$ such that for every finite family of pairwise disjoint intervals*

$$[x_1, y_1], \ldots, [x_n, y_n] \subset [a,b],$$

if

$$\sum_{i=1}^{n}(y_i - x_i) < \delta,$$

then

$$\sum_{i=1}^{n} |f(y_i) - f(x_i)| < \varepsilon.$$

The class is denoted by $AC[a,b]$.

Lemma 5.6.1 (Singular Function)**.** *Suppose* f *is nonconstant on* $[a,b]$*, differentiable a.e., and*

$$f'(x) = 0 \quad \text{a.e.}$$

Then $f \notin AC[a,b]$.

Proof. If f were absolutely continuous, then by definition and Vitali-type interval selection one obtains arbitrarily short disjoint interval families with arbitrarily small total increment, contradicting nonconstancy under $f' = 0$ a.e. □

Theorem 5.6.2. (1) *If* $f \in AC[a,b]$*, then* $f \in C[a,b]$.

(2) *If* $f, g \in AC[a,b]$ *and* $\alpha, \beta \in \mathbb{R}$*, then* $\alpha f + \beta g \in AC[a,b]$.

(3) *If* $f \in AC[a,b]$*, then* $f \in BV[a,b]$.

(4) *If* $f \in L^1[a,b]$ *and*

$$F(x) = \int_a^x f(t)\,dt,$$

then $F \in AC[a,b]$.

Proof. Standard and omitted. □

Now let $f \in AC[a,b]$ and define

$$g(x) = \int_a^x f'(t)\,dt.$$

By part (4) of Theorem 5.6.2, $g \in AC[a,b]$. Hence $h = g - f \in AC[a,b]$. By Theorem 5.6.1,

$$h'(x) = g'(x) - f'(x) = 0 \quad \text{a.e.}$$

and by Lemma 5.6.1, h must be constant. Therefore

$$f(x) = \int_a^x f'(t)\,dt + \text{const.}$$

Evaluating at $x = b$ yields:

Theorem 5.6.3 (Newton-Leibniz)**.** *If* $f \in AC[a,b]$*, then*

$$f(b) - f(a) = \int_a^b f'(t)\,dt.$$

For a.e. differentiable f with $f' \in L^1[a,b]$, define

$$f_s(x) = f(x) - \int_a^x f'(t)\,dt.$$

Then $f_s'(x) = 0$ a.e., so

$$f(x) = \int_a^x f'(t)\,dt + f_s(x) = f_{\mathrm{AC}}(x) + f_s(x),$$

which is the decomposition into absolutely continuous and singular parts.

This parallels the Radon-Nikodym section. Compare the two frameworks.

Appendix

A Another Construction of Lebesgue Measure

Below we reconstruct Lebesgue measure from the viewpoint of abstract measure theory.

Measurable Sets and Measures

Definition A.1 (Measurable Space)**.** *Let X be a nonempty set, and let $\mathcal{M}$ be a σ-algebra on X. Then the pair $(X, \mathcal{M})$ is called a measurable space.*

Definition A.2 (Measure)**.** *Let $(X, \mathcal{M})$ be a measurable space. A function $\mu : \mathcal{M} \to [0, +\infty]$ is called a measure on this measurable space if it is countably additive.*

Remark A.1. *Naturally, we assume that μ is not identically $+\infty$.*

Proposition A.1. *Let $(X, \mathcal{M})$ be a measurable space, and let μ be a measure on it. Then μ has the following properties:*

(1) $\mu(\varnothing) = 0$*;*

(2) *finite additivity;*

(3) *monotonicity;*

(4) *continuity from below: if $A_1 \subset A_2 \subset \cdots$, then*

$$\lim_{n\to\infty} \mu(A_n) = \mu\Big(\bigcup_{n=1}^{\infty} A_n\Big);$$

(5) *continuity from above: if $A_1 \supset A_2 \supset \cdots$ and $\mu(A_1) < +\infty$, then*

$$\lim_{n\to\infty} \mu(A_n) = \mu\Big(\bigcap_{n=1}^{\infty} A_n\Big).$$

Now let us look at some examples of abstract measures.

Example A.1 (Counting Measure)**.** *Take the measurable space $(X, \mathcal{P}(X))$. Define $\mu : \mathcal{P}(X) \to \mathbb{N} \cup \{\infty\}$ by mapping each set A to its cardinality.*

Remark A.2. *In fact, all series theory can be viewed as integration theory for counting measure.*

Example A.2 (Dirac Measure)**.** *If $x \in E$, then $\delta_x(E) = 1$. If $x \notin E$, then $\delta_x(E) = 0$.*

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